

Sequence Models for words and pixels



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Many slides from Steve Seitz's wonderful [5 min Lectures](#)

CS194: Intro to Computer Vision & Comp. Photography
Alexei Efros, UC Berkeley, Fall 2025

Michel Gondry train video

<http://www.youtube.com/watch?v=0S43lwBF0uM>

“Amateur” by Lasse Gjertsen

<http://www.youtube.com/watch?v=JzqumbhfxRo>

Generative AI

Generative models trained on **lots of data** have revolutionized computer science and beyond!



Text

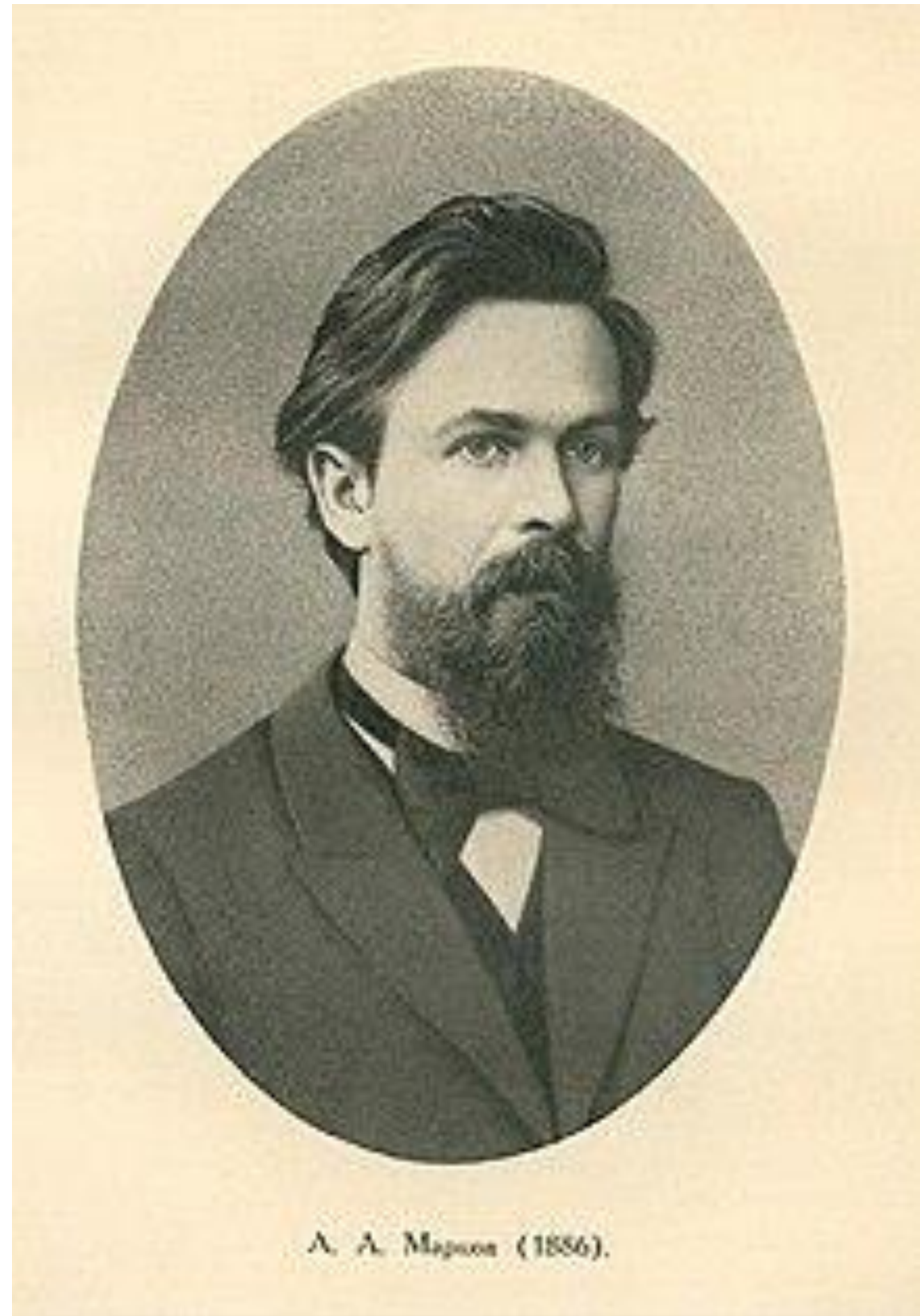


Pixels



Audio

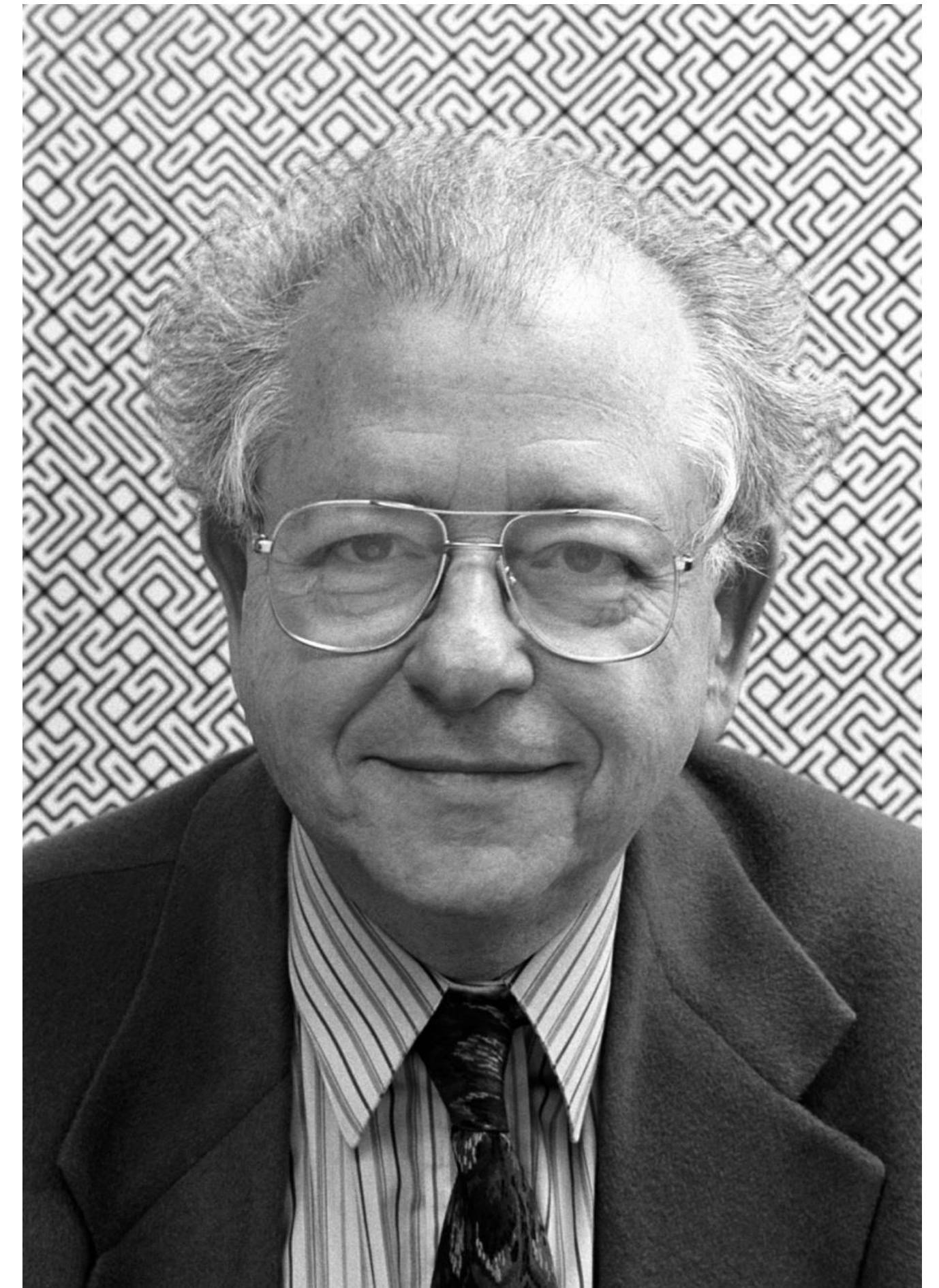
Ideas going back over a hundred years...



Andrey Markov
(1856-1922)



Claude Shannon
(1916-2001)



Bela Julesz
(1928-2003)

Weather Forecasting for Dummies™

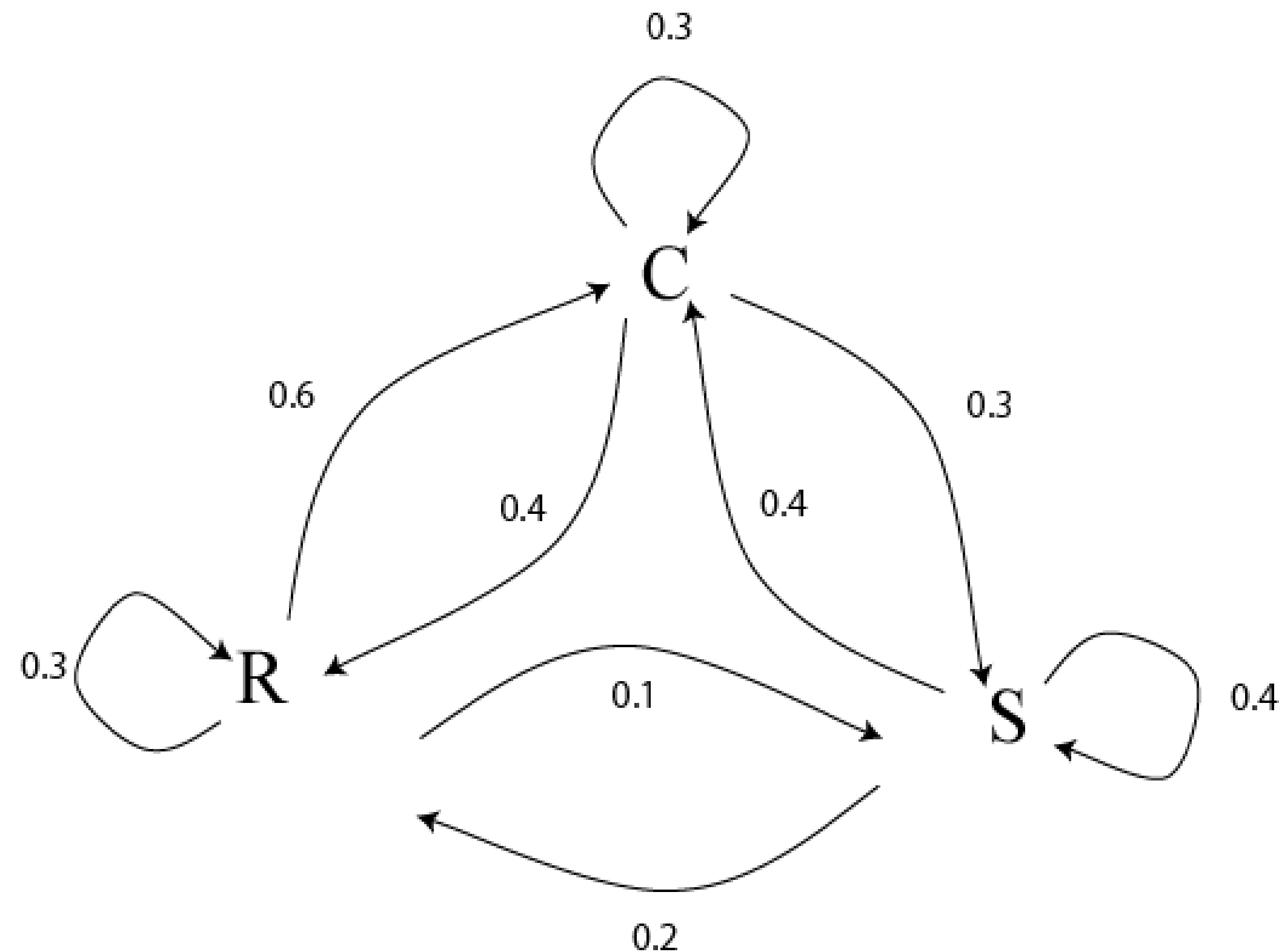
Let's predict weather:

- Given today's weather only, we want to know tomorrow's
- Suppose weather can only be {Sunny, Cloudy, Raining}

The “Weather Channel” algorithm:

- Over a long period of time, record:
 - How often S followed by R
 - How often S followed by S
 - Etc.
- Compute percentages for each state:
 - $P(R|S)$, $P(S|S)$, etc.
- Predict the state with highest probability!
- It's a Markov Chain

Markov Chain



$$\begin{pmatrix} 0.3 & 0.6 & 0.1 \\ 0.4 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.4 \end{pmatrix}$$

What if we know today and yestarday's weather?

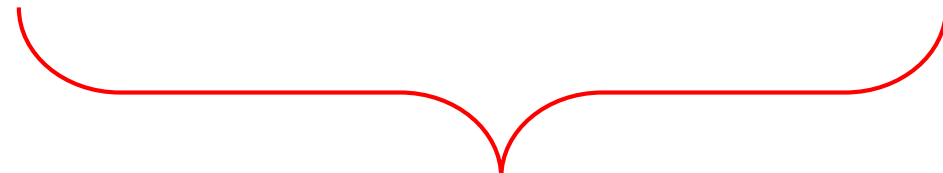
Now, let's apply this to text

[Markov, 1913] statistical analysis of text

[Shannon, 1948] proposed a way to generate English-looking text using N-grams:

- Assume a generalized Markov model of language
- Use a large text to compute prob. distributions of each letter given N-1 previous letters
- Starting from a seed repeatedly sample this Markov chain to generate new letters
- Also works for whole words
- E.g.:

WE NEED TO EAT CAKE



Mark V. Shaney (Bell Labs)

Results (using `alt.singles corpus`):

- *“As I've commented before, really relating to someone involves standing next to impossible.”*
- *“One morning I shot an elephant in my arms and kissed him.”*
- *“I spent an interesting evening recently with a grain of salt”*

Early one morning the sun was shining I was laying in bed
Wondering if she had changed at all if her hair was still red

Bob Dylan, *Tangled up in Blue*

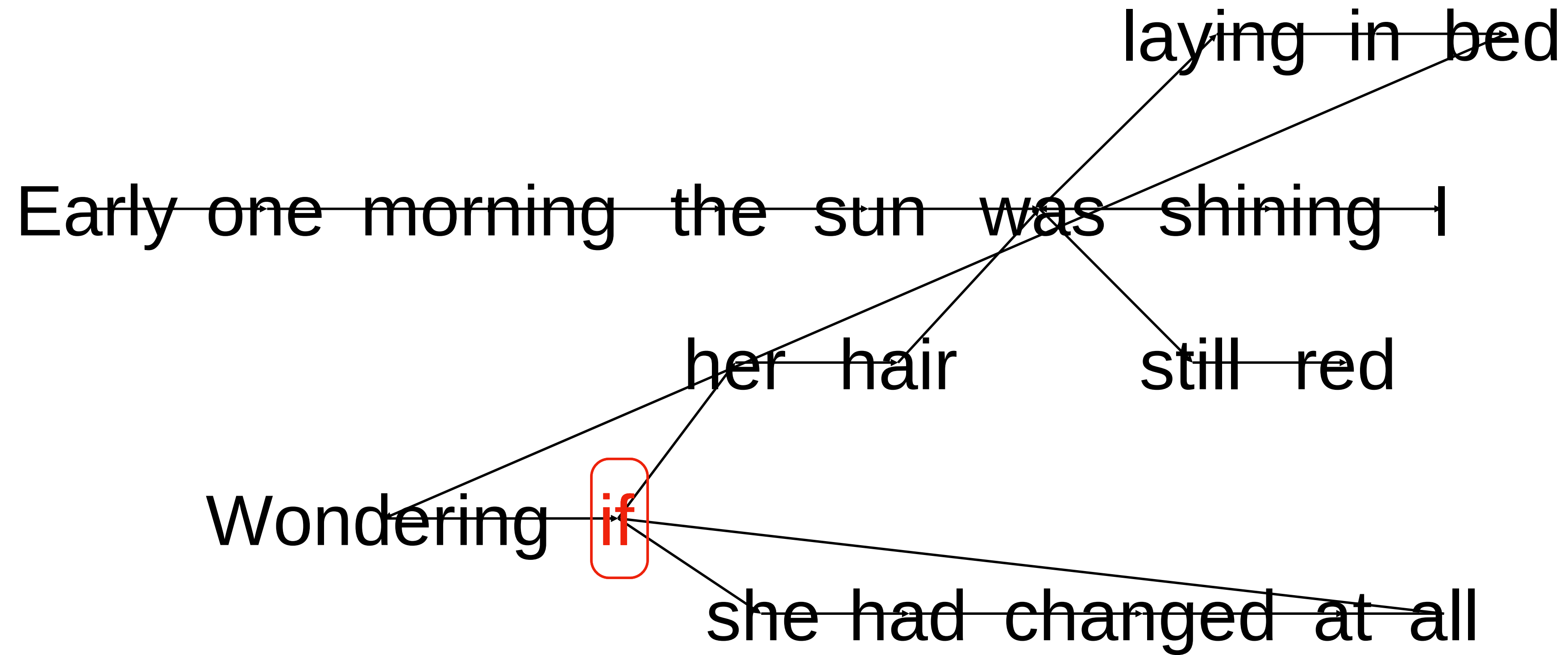
~~Early one morning the sun was shining I was laying in bed
Wondering if she had changed at all if her hair was still red~~

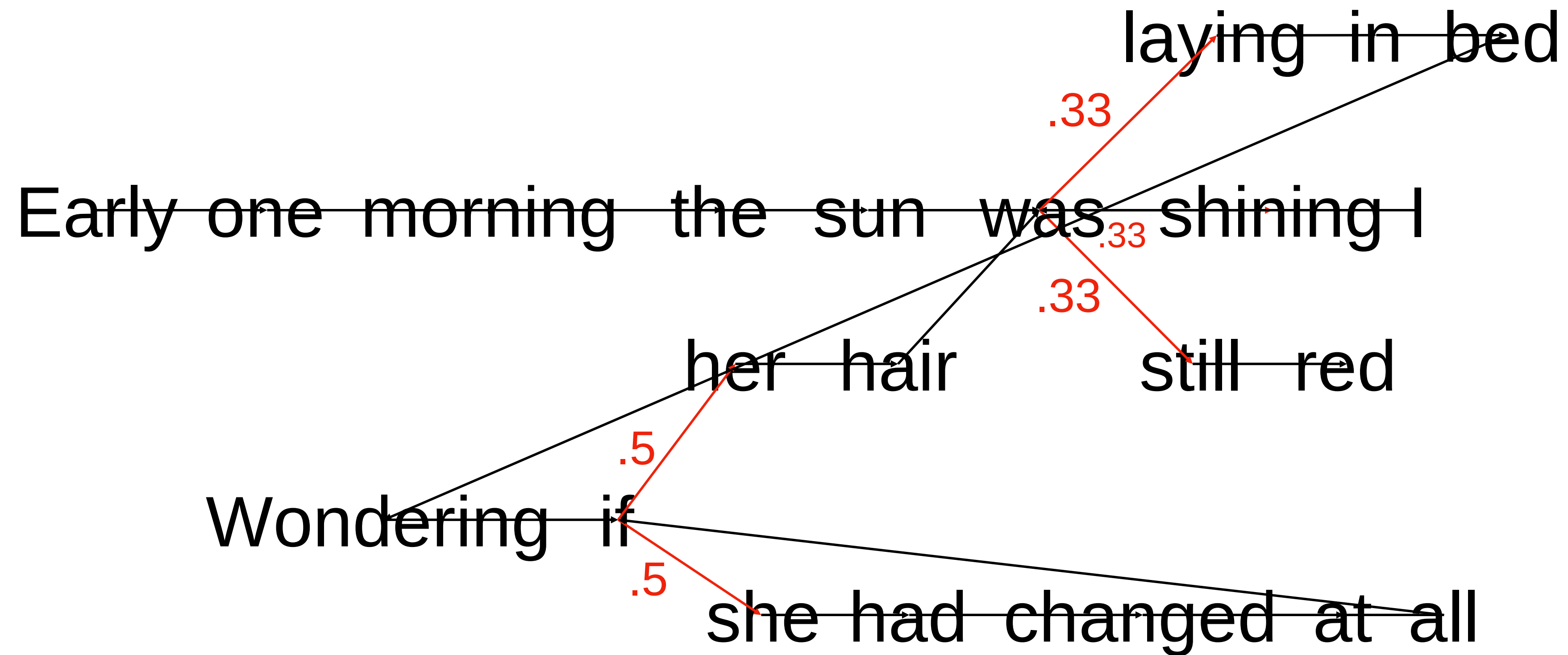
Early one morning the sun **was** shining I **was** laying in bed
Wondering if she had changed at all if her hair **was** still red

Early one morning the sun **was** shining I
laying in bed
her hair still red
Wondering if she had changed at all if

laying in bed
Early one morning the sun was shining I
her hair still red
Wondering **if** she had changed at all **if**

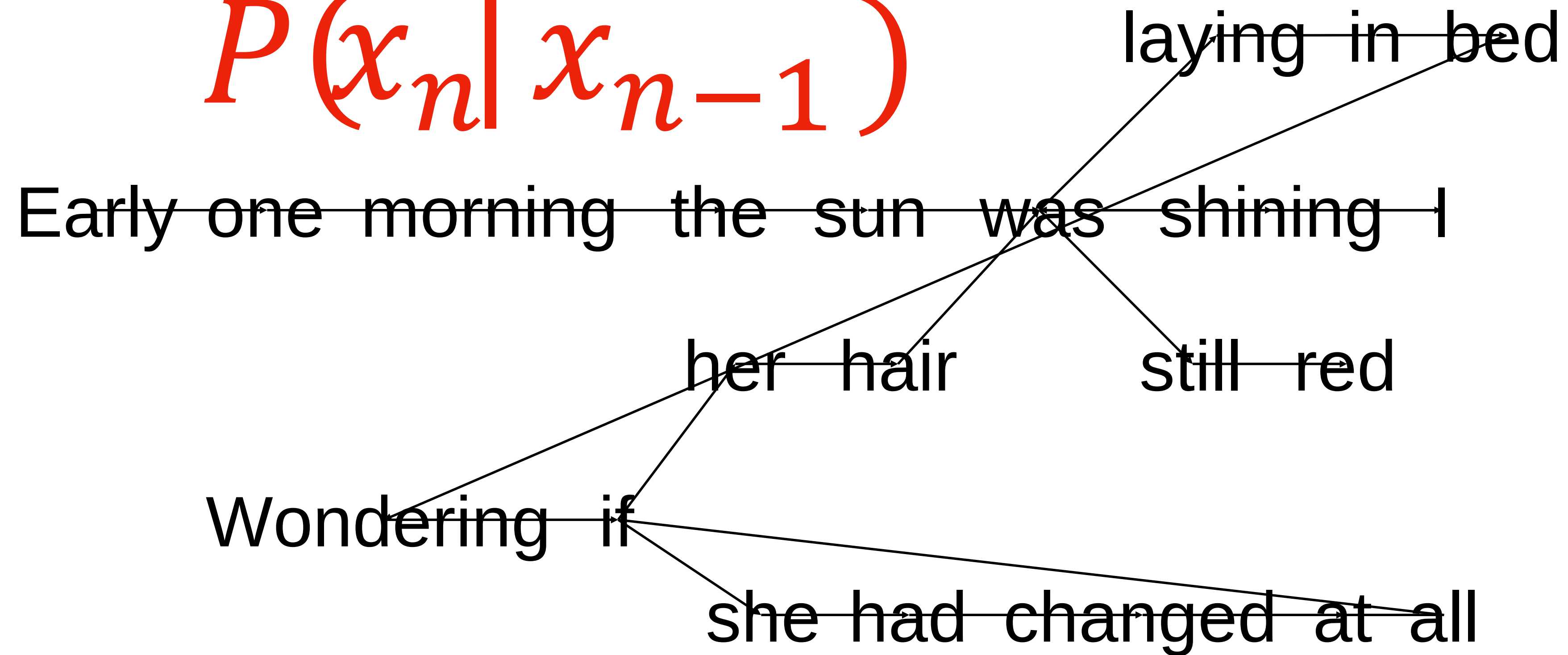
The diagram shows a sentence with several words crossed out. Arrows point from 'was' to 'laying in bed', 'her hair', and 'still red'. Arrows also point from 'if' to 'Wondering' and 'at all'.





Language Model

$$P(x_n | x_{n-1})$$



$$P(x_n | x_{n-1}, x_{n-2})$$

Early one → morning
one morning → the
morning the → sun
the sun → was
sun was → shining
was shining → I
shining I → was
I was → laying

...

Video Textures

Arno Schödl

Richard Szeliski

David Salesin

Irfan Essa

Microsoft Research, Georgia Tech

SIGGRAPH 2000

Still photos



Video clips



Video textures



Problem statement

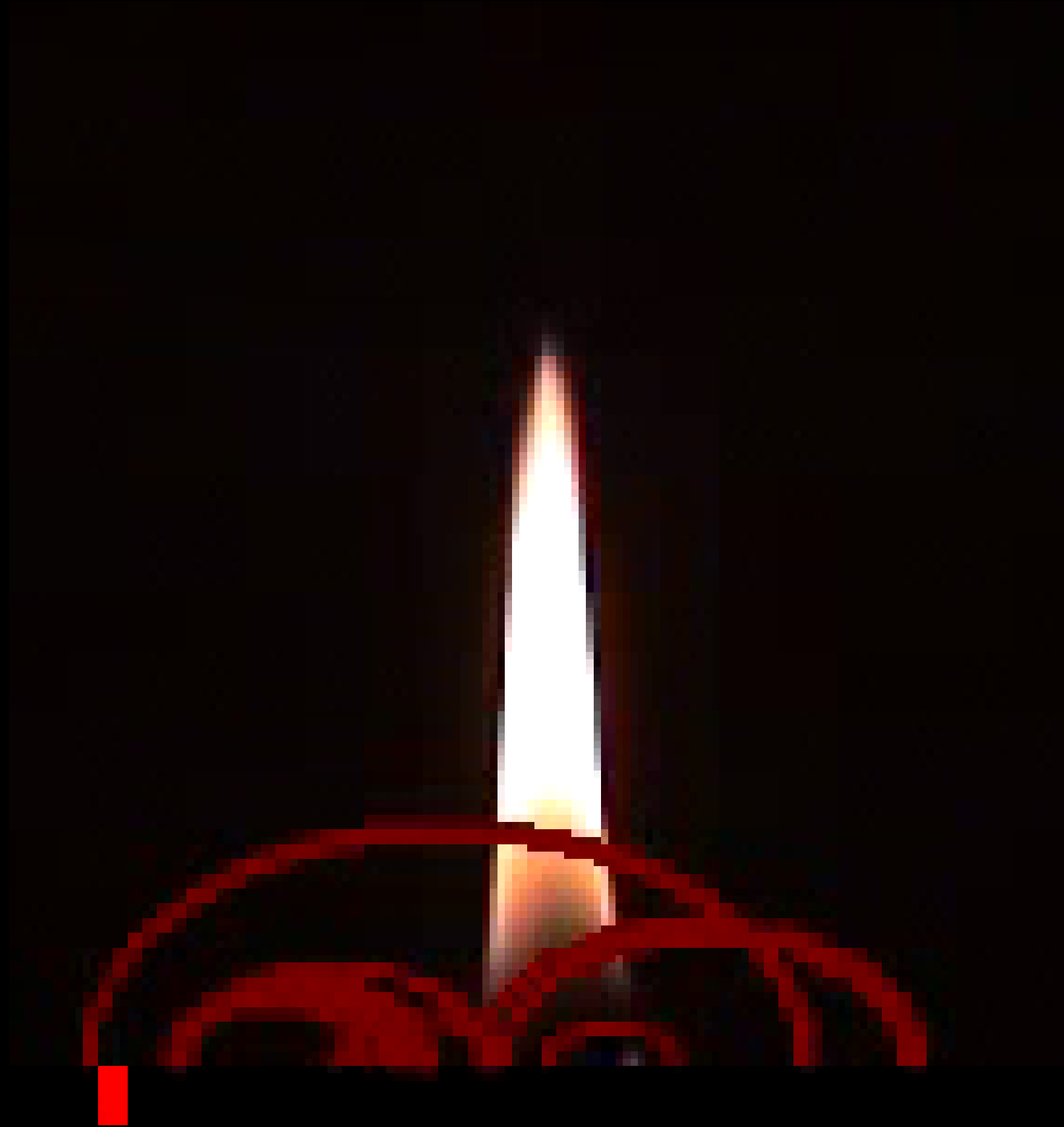


video clip



video texture

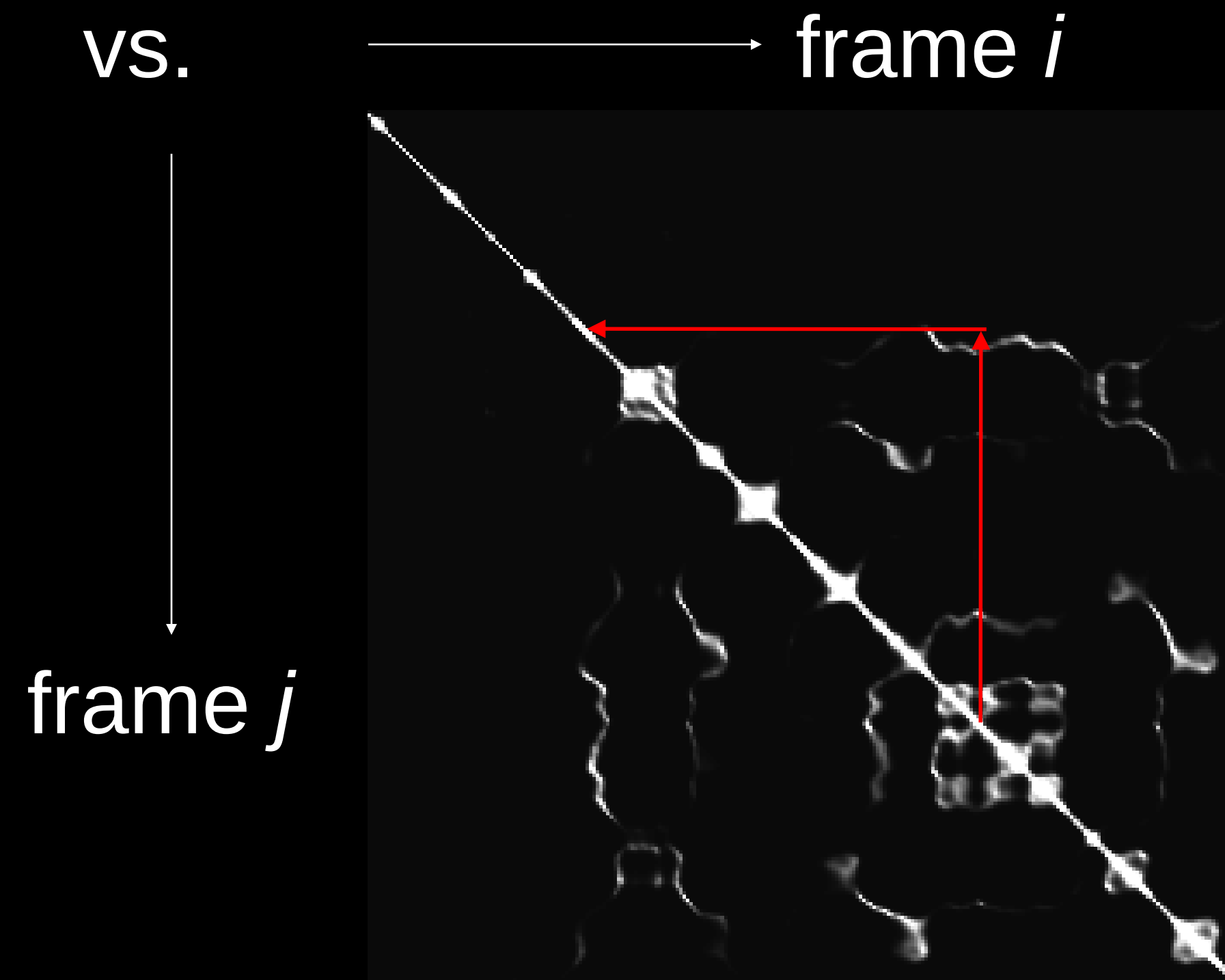
Our approach



- How do we find good transitions?

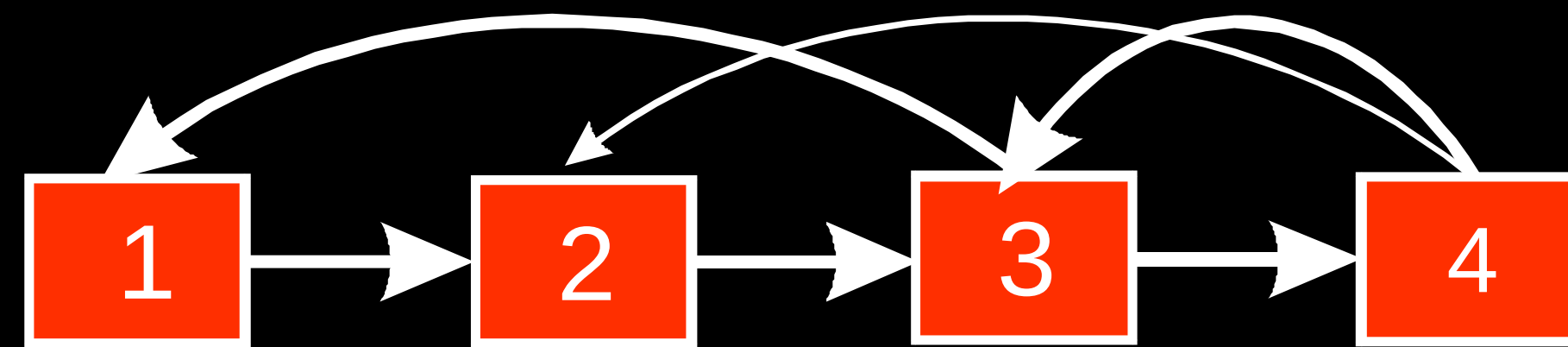
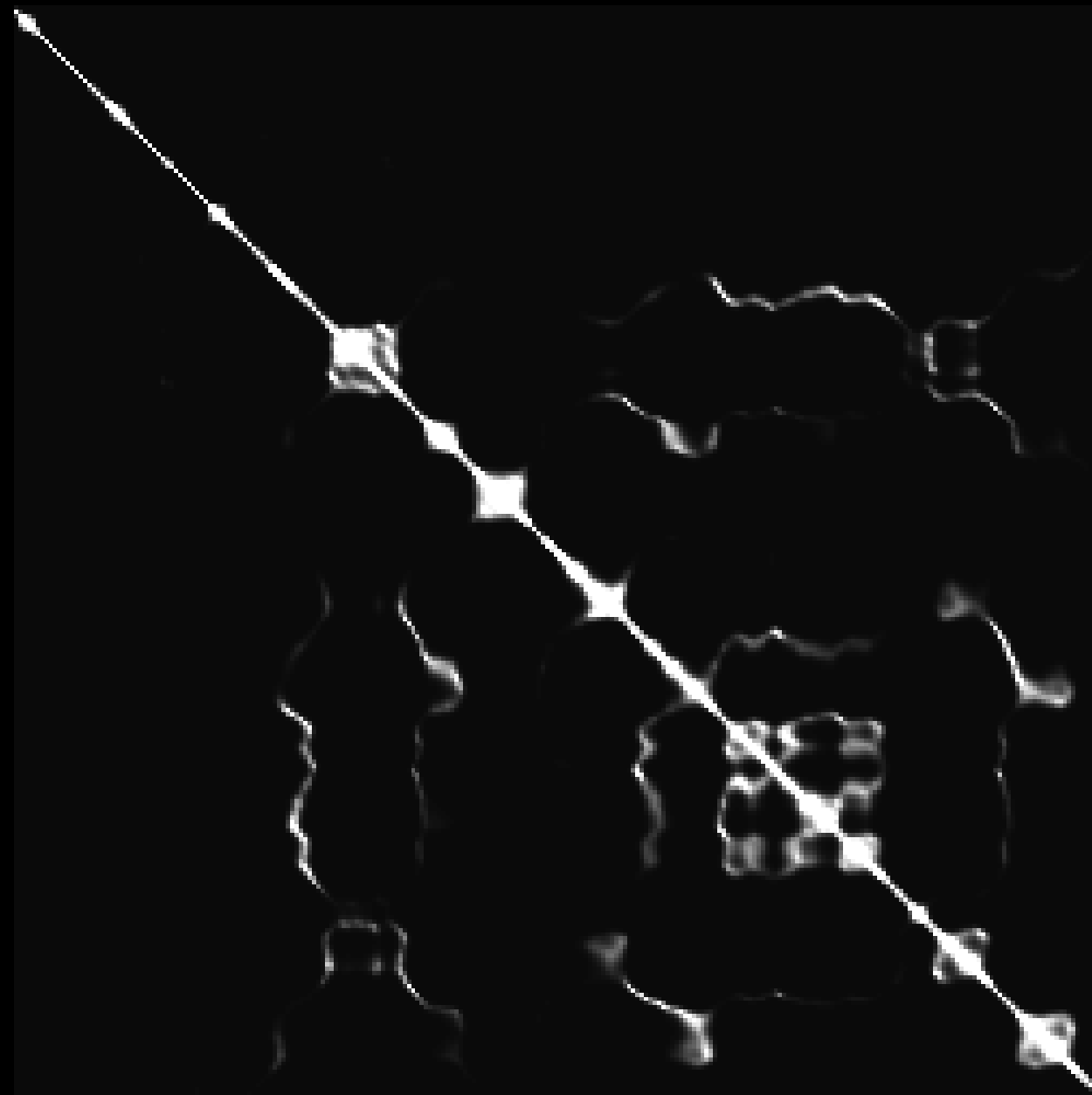
Finding good transitions

- Compute L_2 distance $D_{i,j}$ between all frames



Similar frames make good transitions

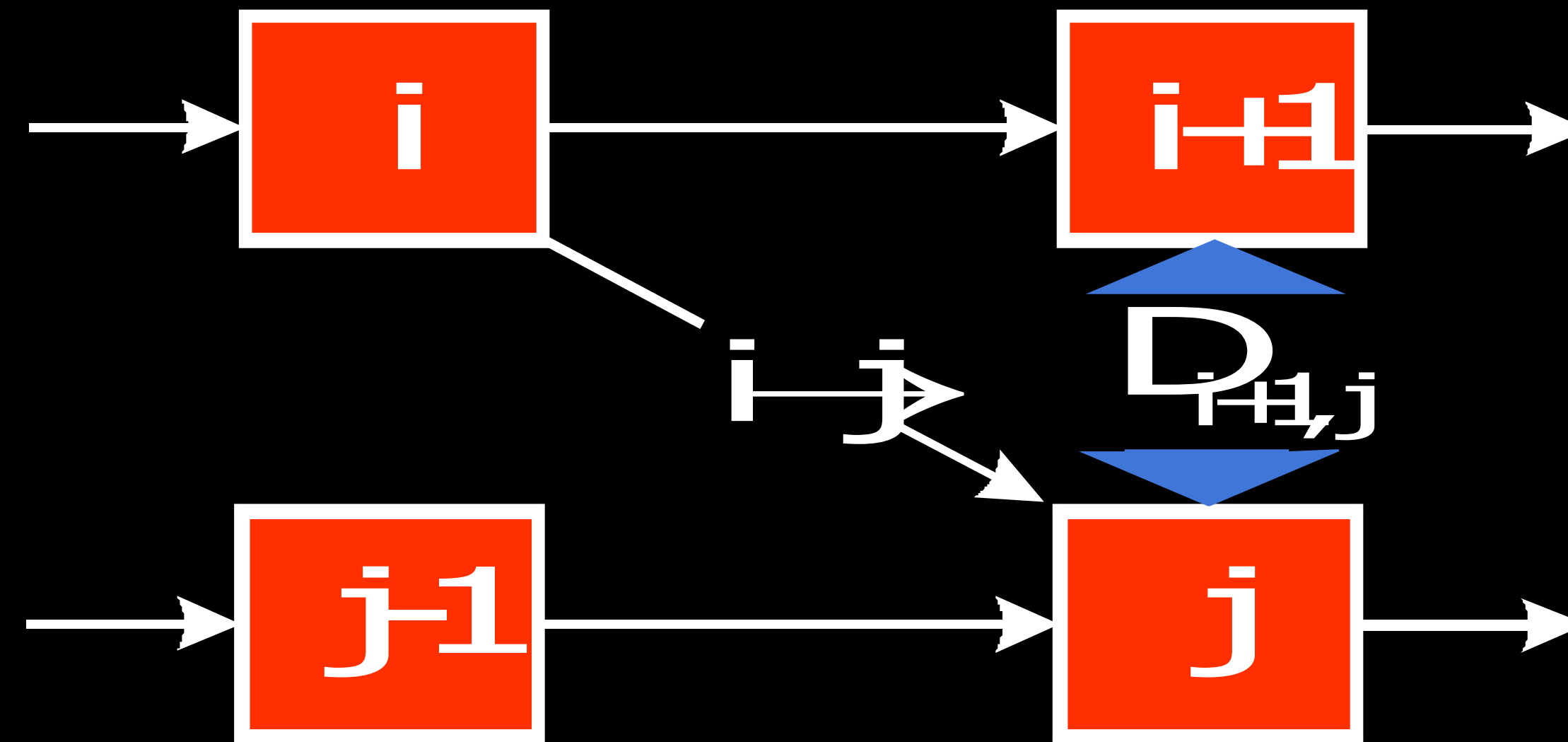
Markov chain representation



Similar frames make good transitions

Transition costs

- Transition from i to j if successor of i is similar to j
 - Cost function: $C_{i \rightarrow j} = D_{i+1, j}$



Transition probabilities

- Probability for transition $P_{i \rightarrow j}$ inversely related to cost:

- $P_{i \rightarrow j} \sim \exp (- C_{i \rightarrow j} / \sigma^2)$



high σ

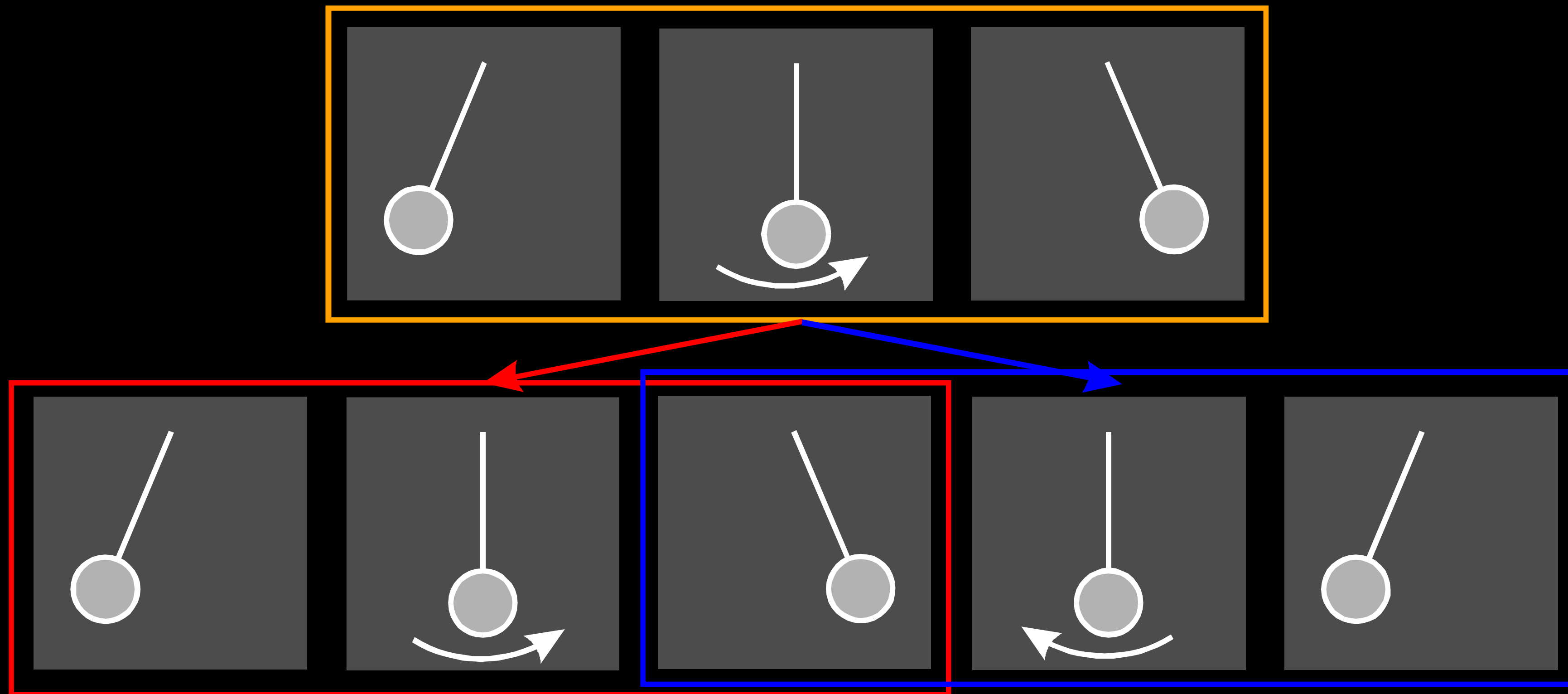


low σ

Preserving dynamics



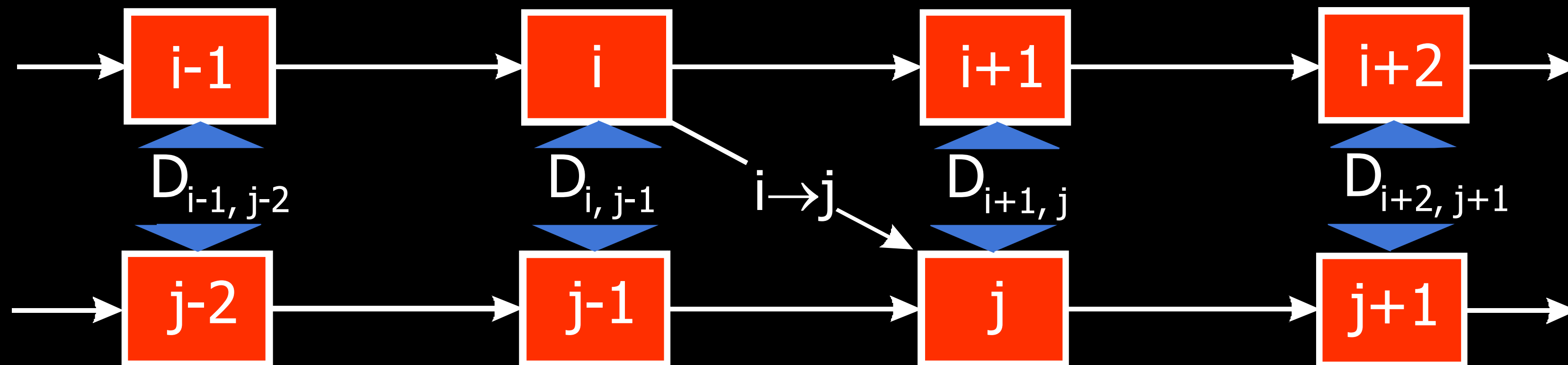
Preserving dynamics



Preserving dynamics

- Cost for transition $i \rightarrow j$

- $$C_{i \rightarrow j} = \sum_{k=-N}^{N-1} w_k D_{i+k+1, j+k}$$



Preserving dynamics – effect

- Cost for transition $i \rightarrow j$
 - $$C_{i \rightarrow j} = \sum_{k=-N}^{N-1} w_k D_{i+k+1, j+k}$$



User-controlled video textures



slow



variable



fast

User selects target frame range

Video-based animation

- Like sprites
computer games
- Extract sprites
from real video
- Interactively control
desired motion



©1985 Nintendo of America Inc.



Video sprite extraction

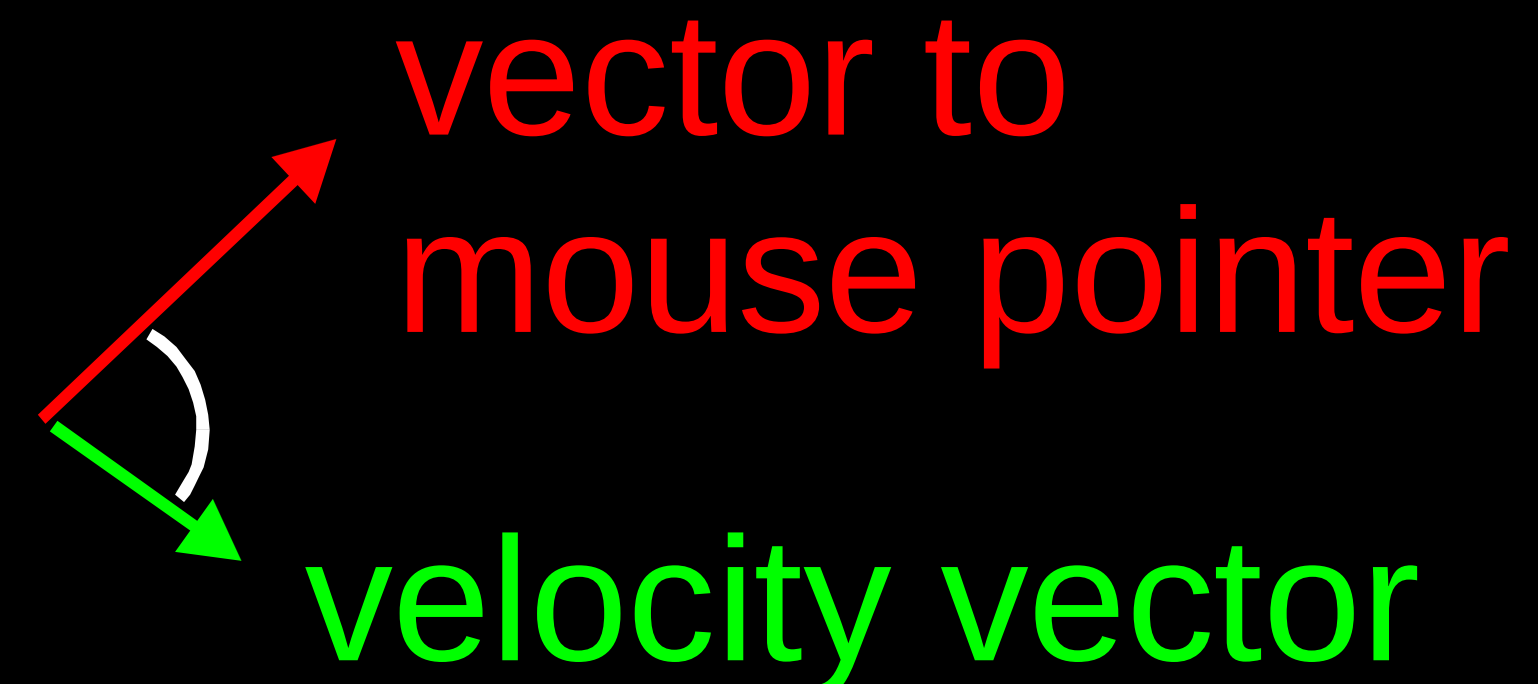


blue screen matting
and velocity estimation



Video sprite control

- Augmented transition cost:

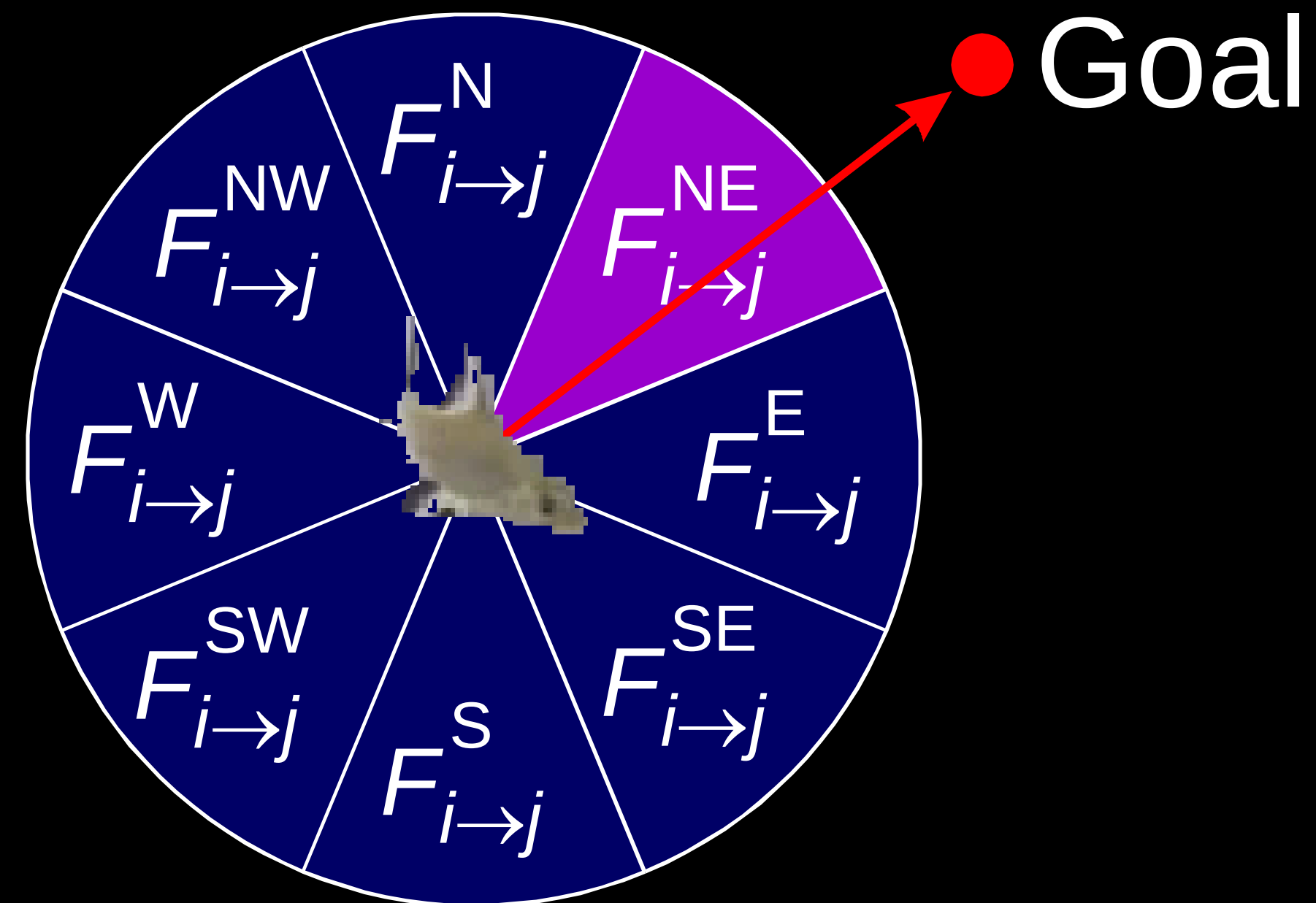
$$C_{i \rightarrow j}^{\text{Animation}} = \alpha \underbrace{C_{i \rightarrow j}}_{\text{Similarity term}} + \beta \underbrace{\text{angle}}_{\text{Control term}}$$


vector to mouse pointer

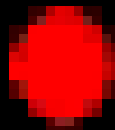
velocity vector

Video sprite control

- Need future cost computation
- Precompute future costs for a few angles.
- Switch between precomputed angles according to user input
- [GIT-GVU-00-11]



Interactive fish



Summary / Discussion

- Some things are relatively easy



Discussion

- Some are hard



Texture

- Texture depicts spatially repeating patterns
- Many natural phenomena are textures



radishes



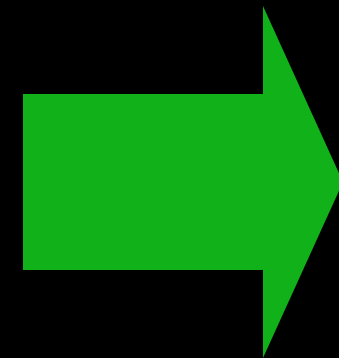
rocks



yogurt

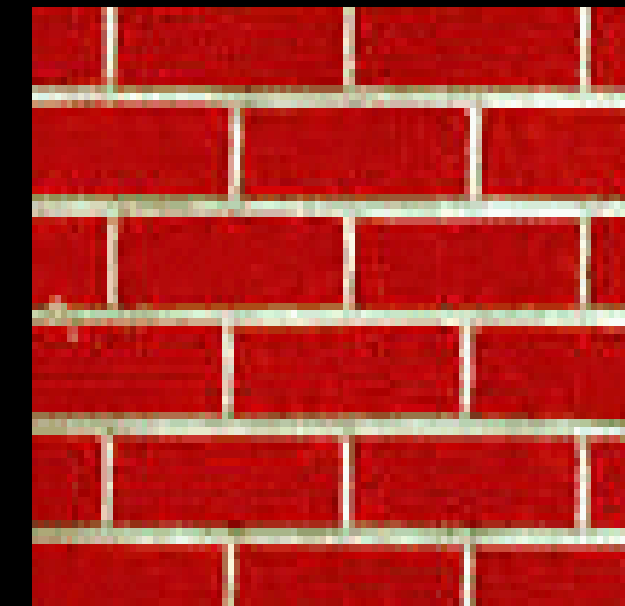
Texture Synthesis

- Goal of Texture Synthesis: create new samples of a given texture
- Many applications: virtual environments, hole-filling, texturing surfaces



The Challenge

- Need to model the whole spectrum: from repeated to stochastic texture



repeated

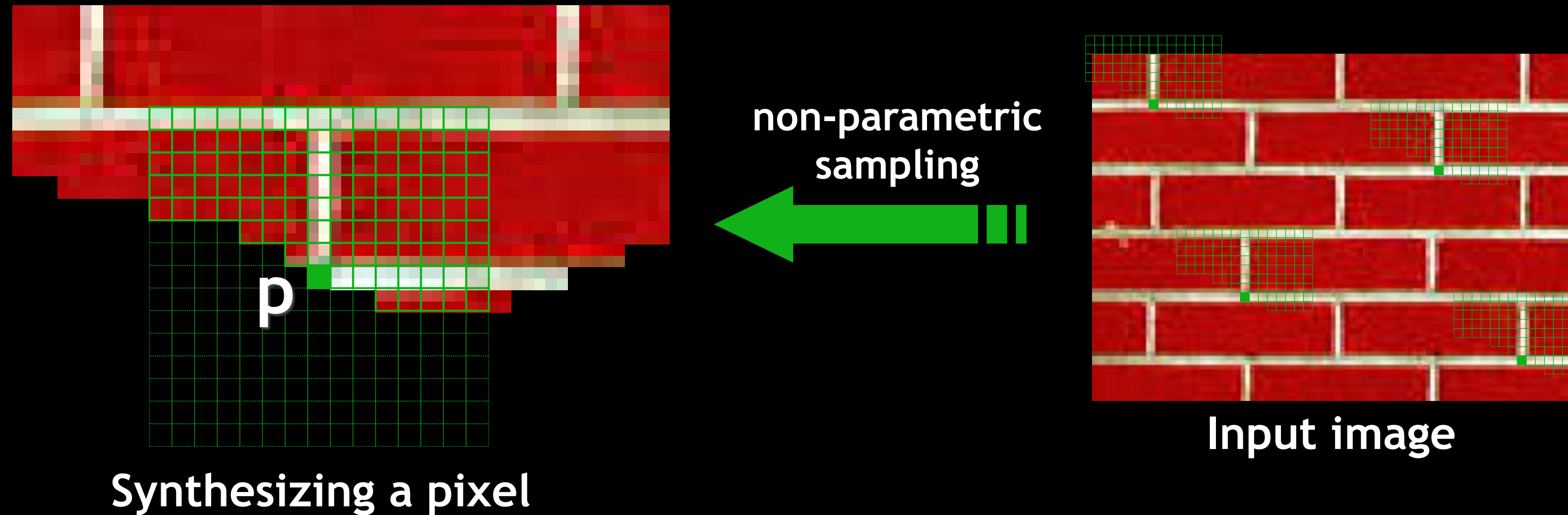


stochastic



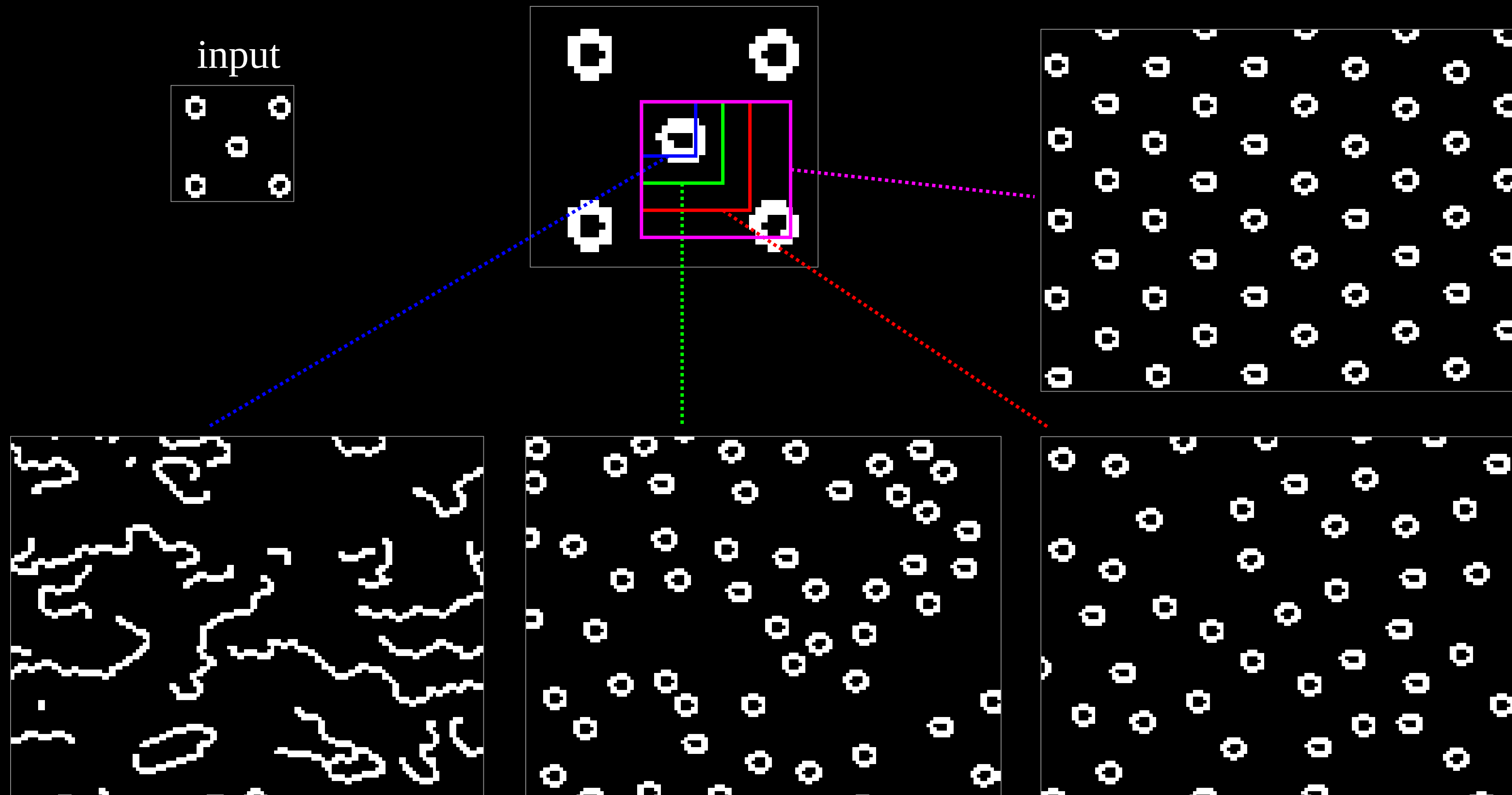
Both?

Efros & Leung Algorithm (ICCV 1999)

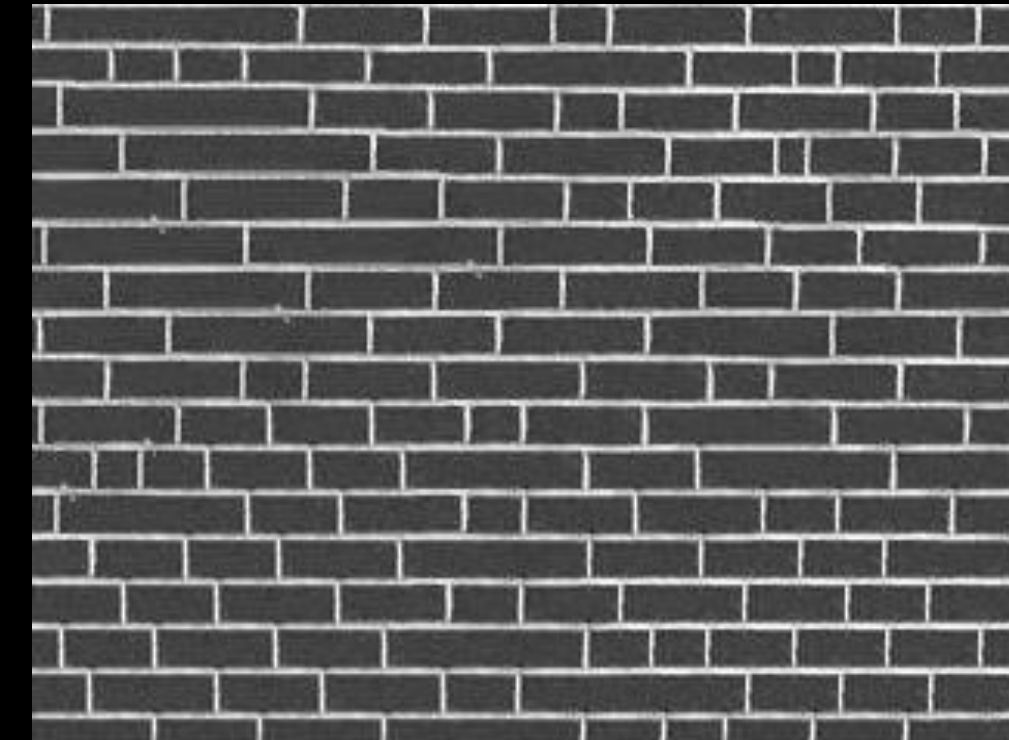
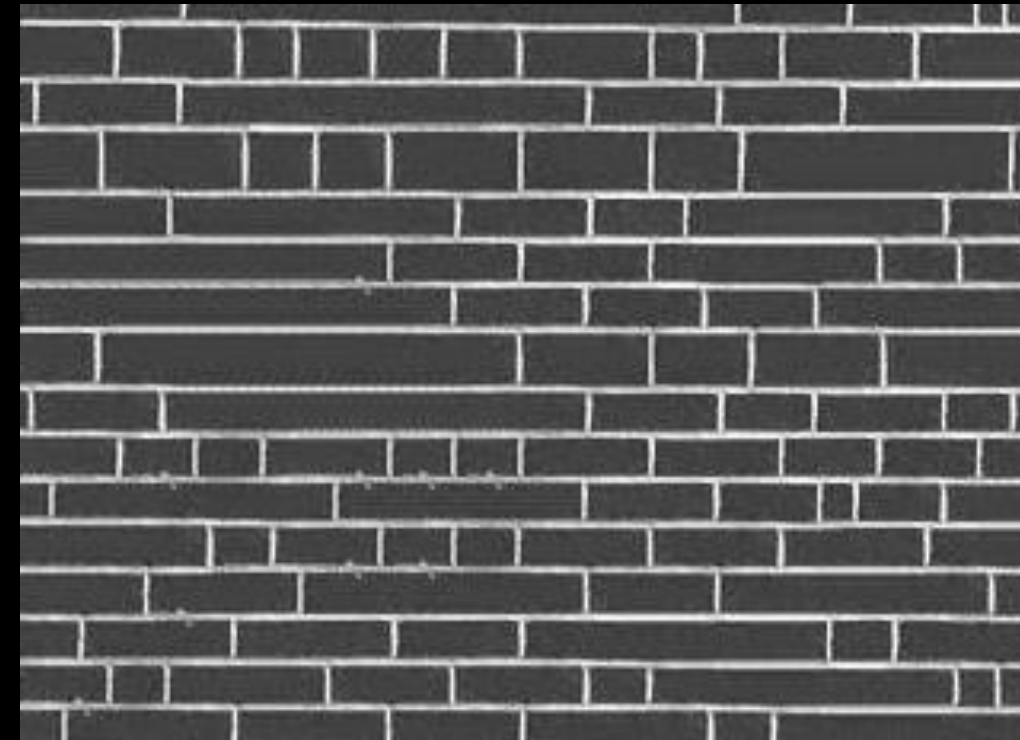
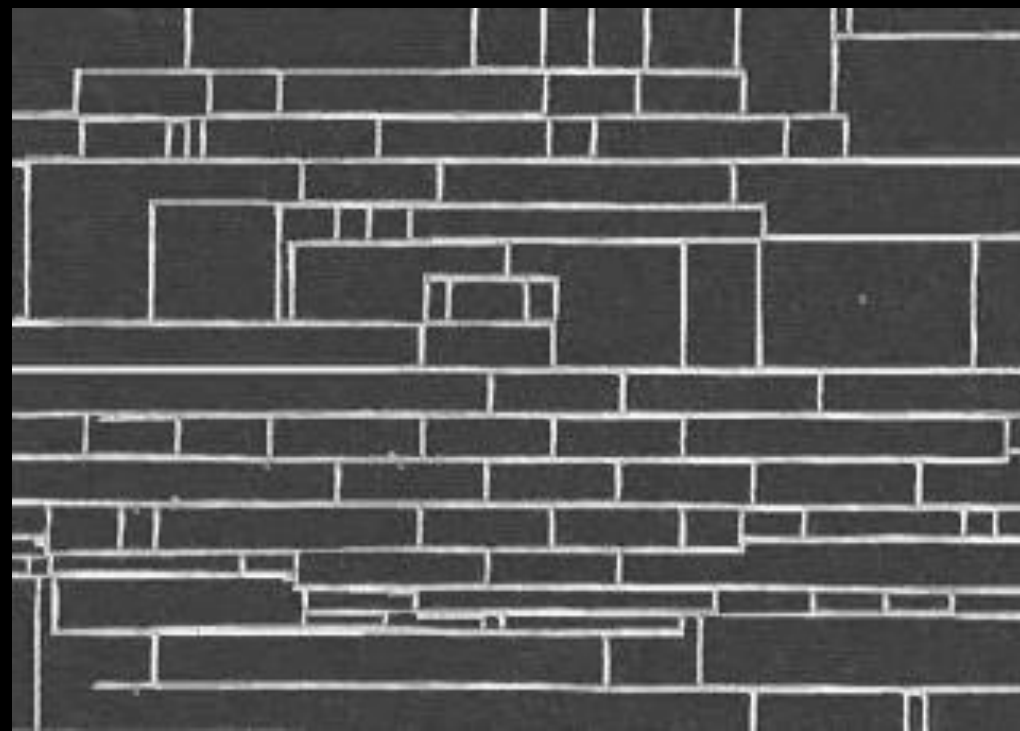
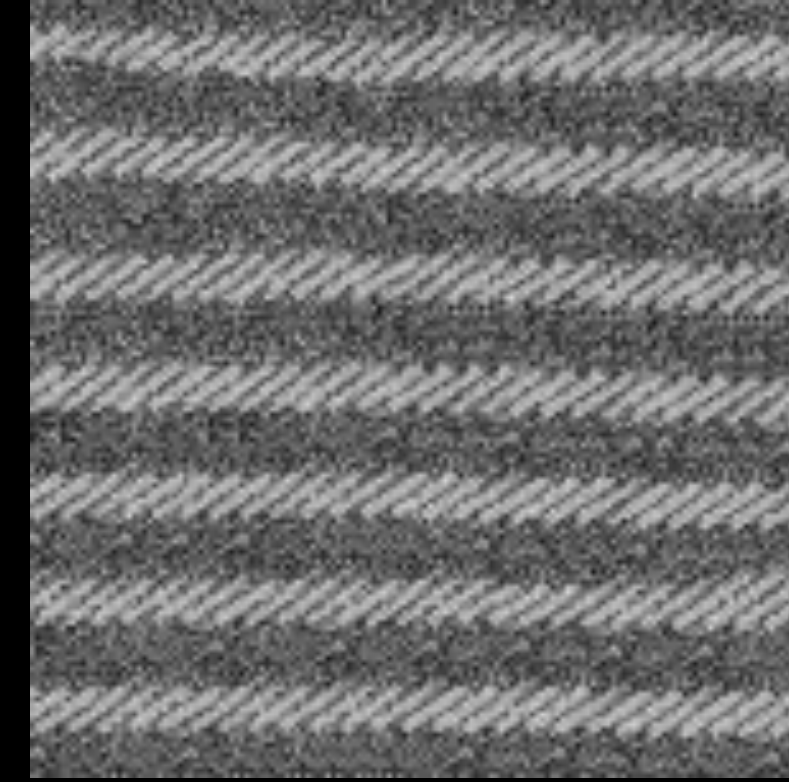
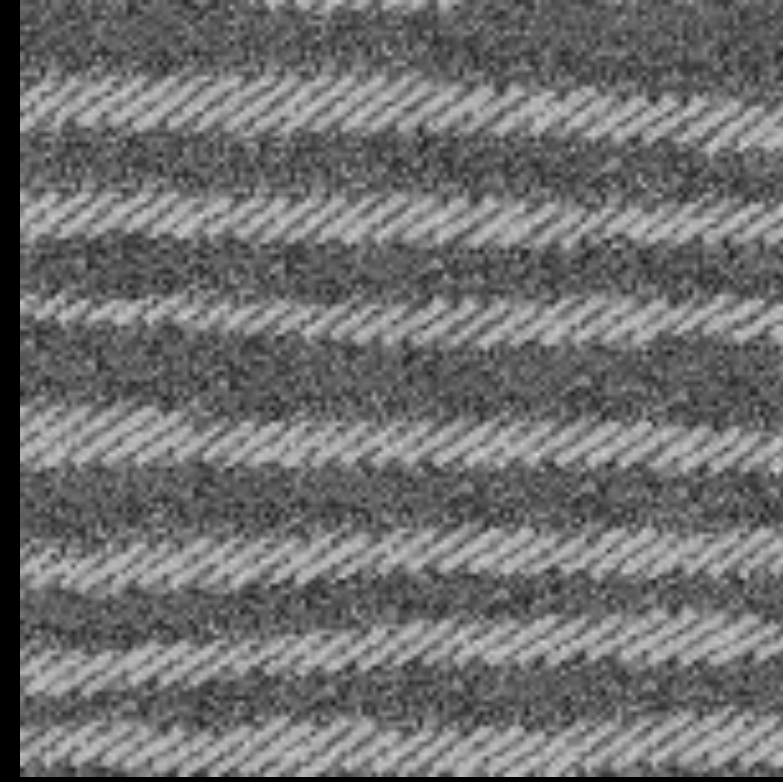
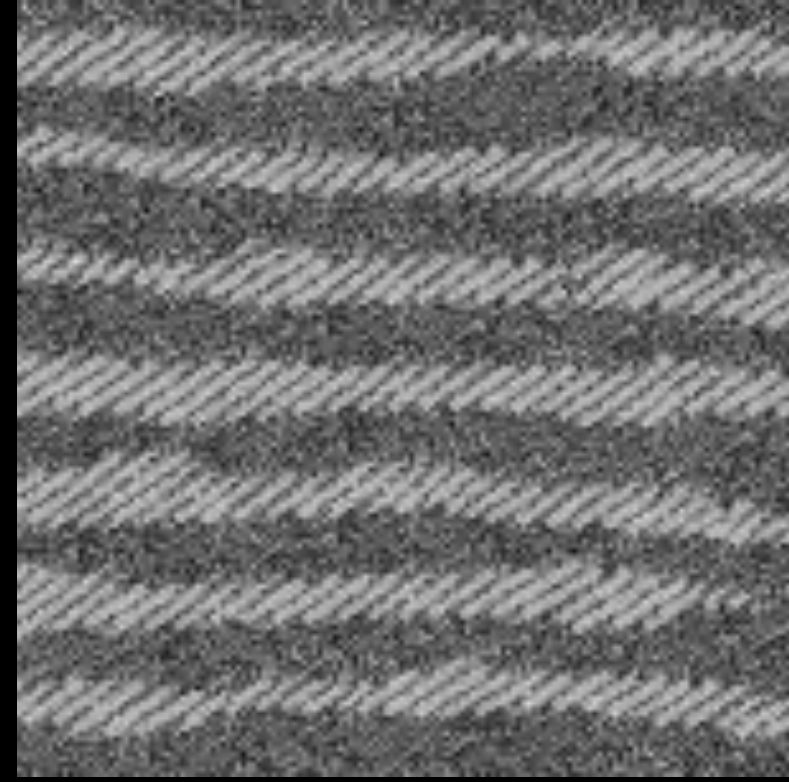
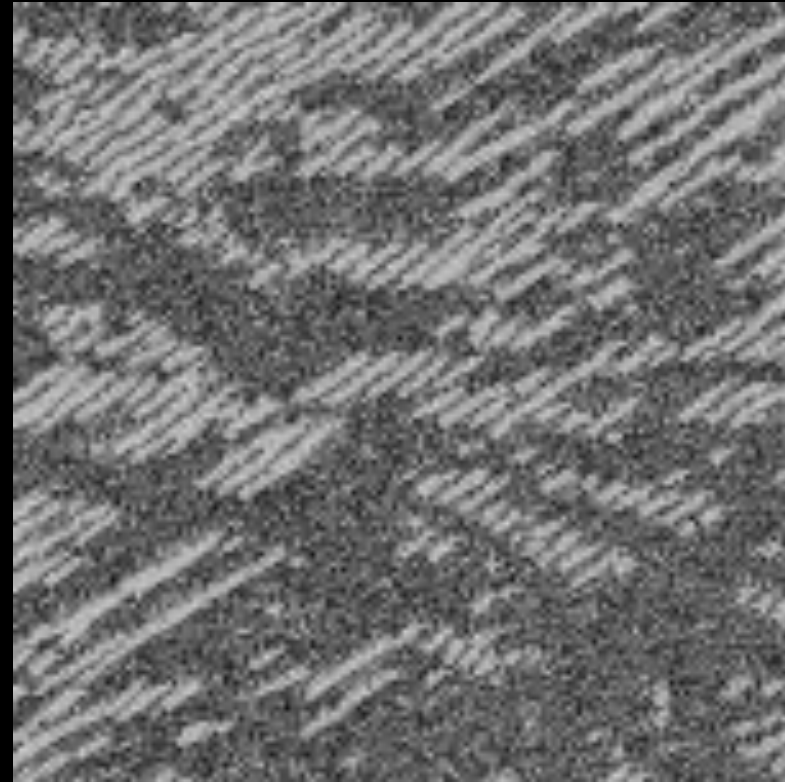
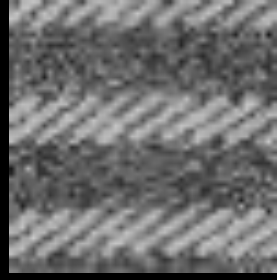


- Assuming Markov property, compute $P(\mathbf{p}|\mathbf{N}(\mathbf{p}))$
 - Building explicit probability tables infeasible
 - Instead, we *search the input image* for all similar neighborhoods — that's our pdf for $\mathbf{p}$
 - To sample from this pdf, just pick one match at random

Neighborhood Window



Varying Window Size

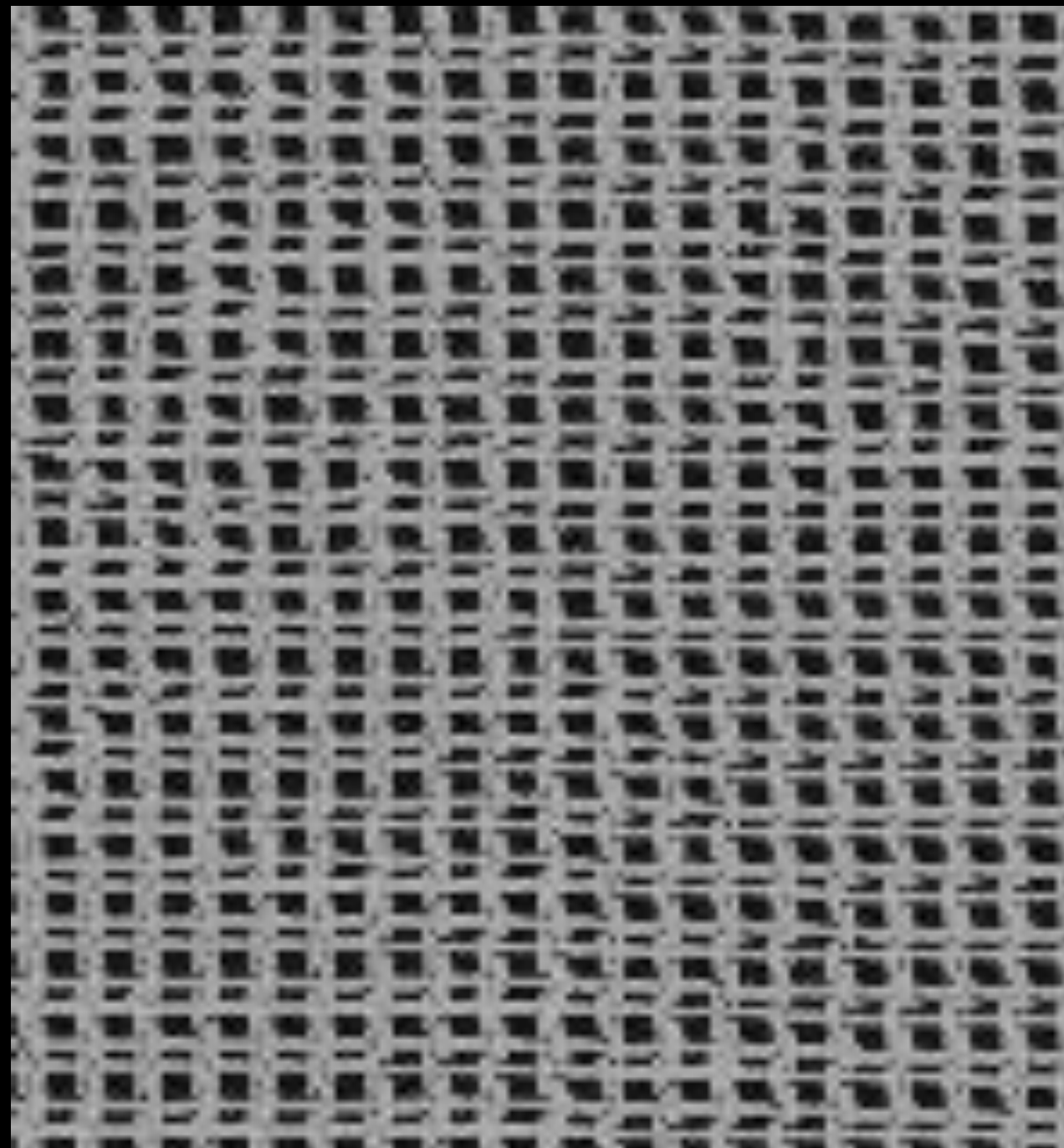
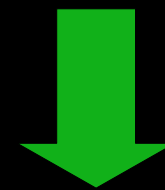
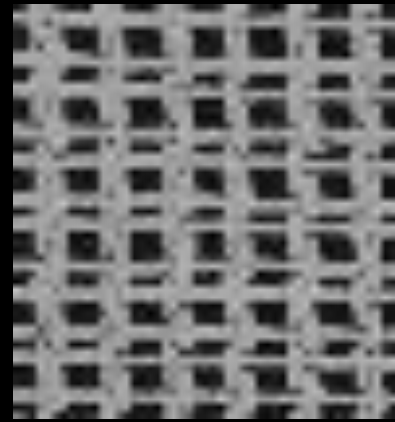


Increasing window size

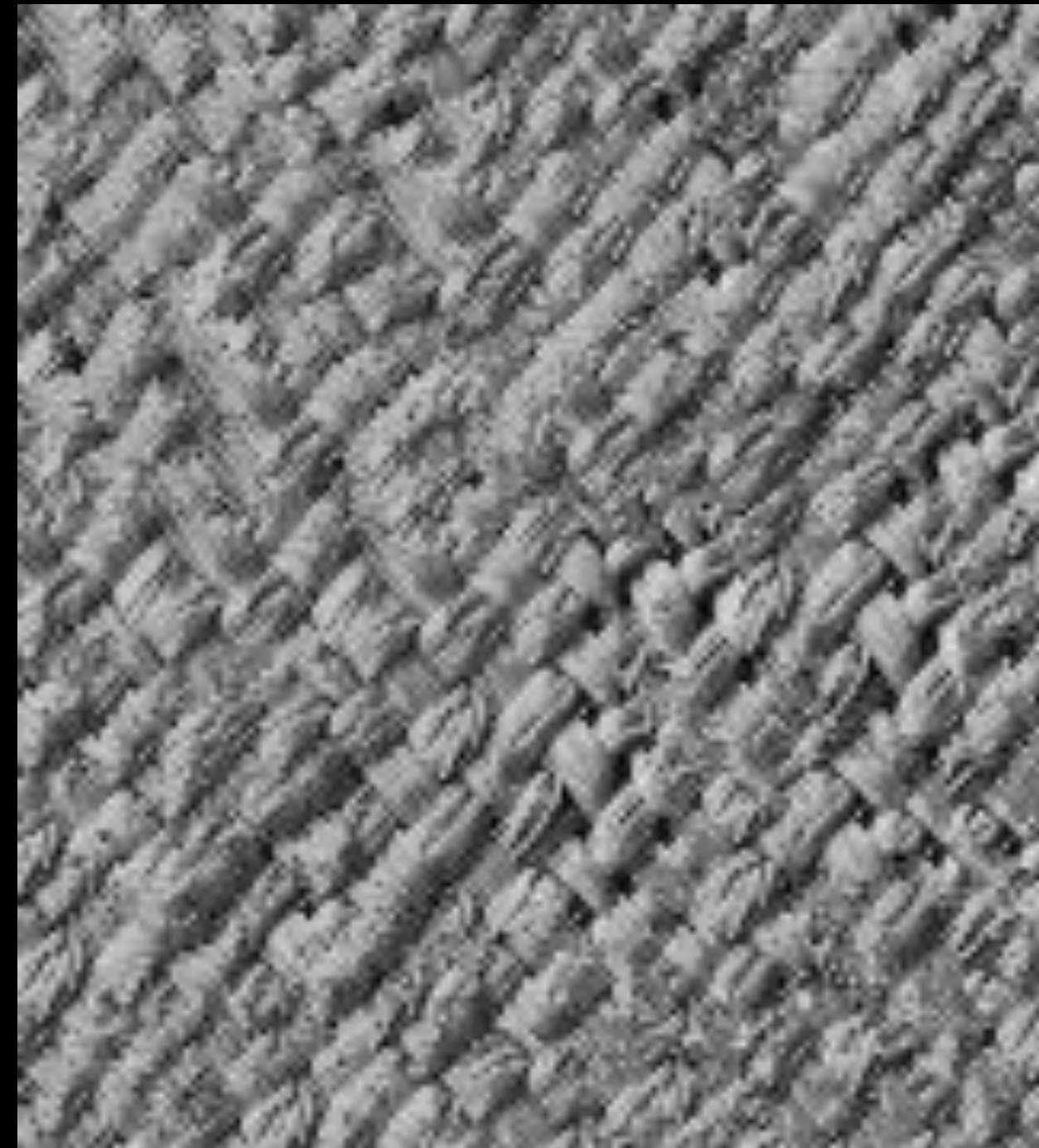


Synthesis Results

french canvas



rafia weave

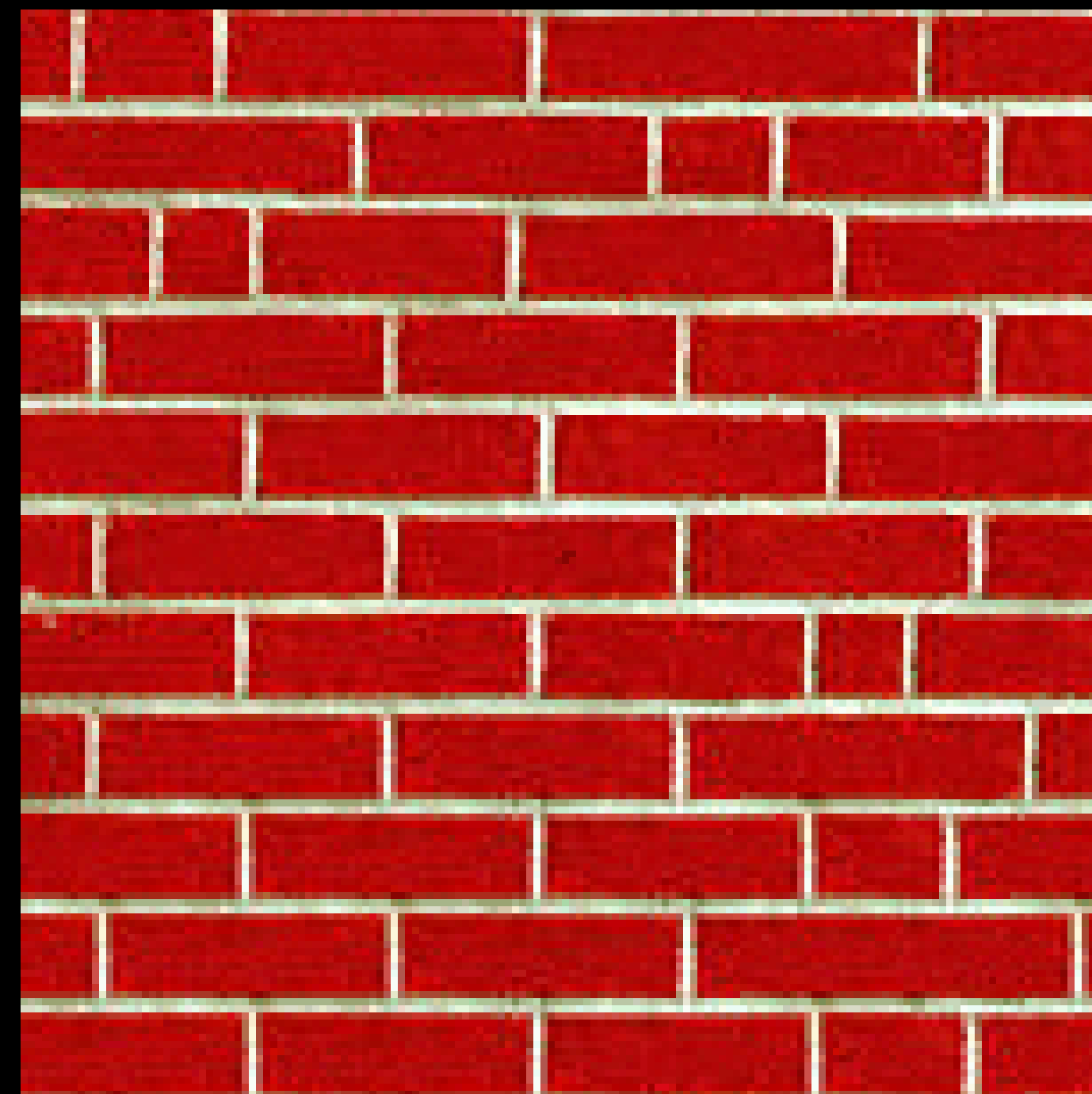
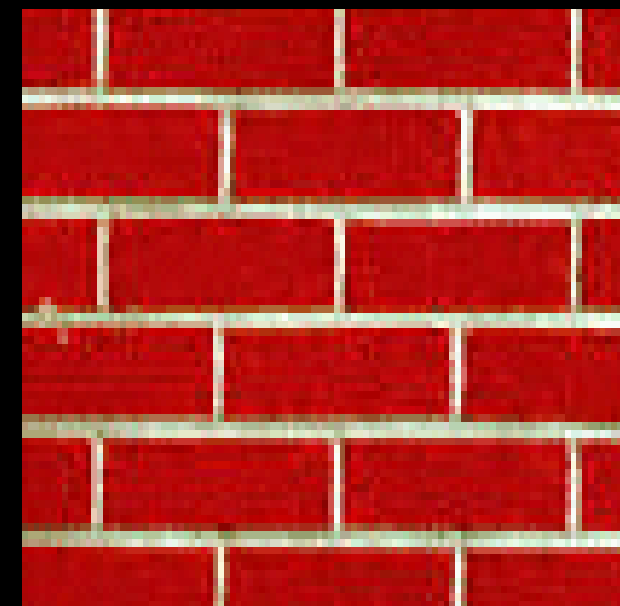


More Results

white bread



brick wall



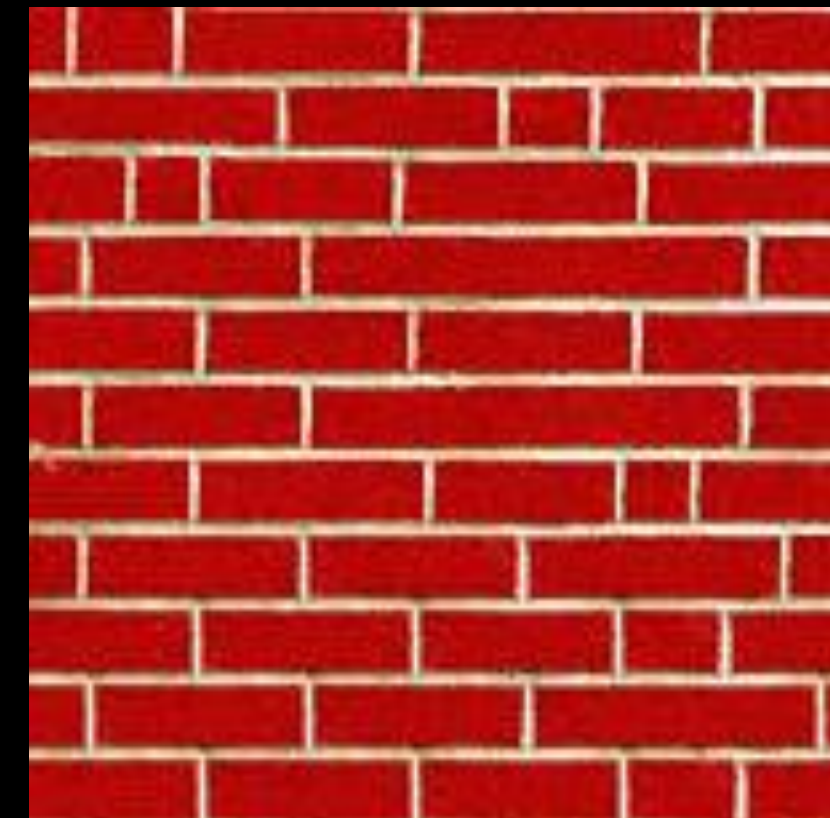
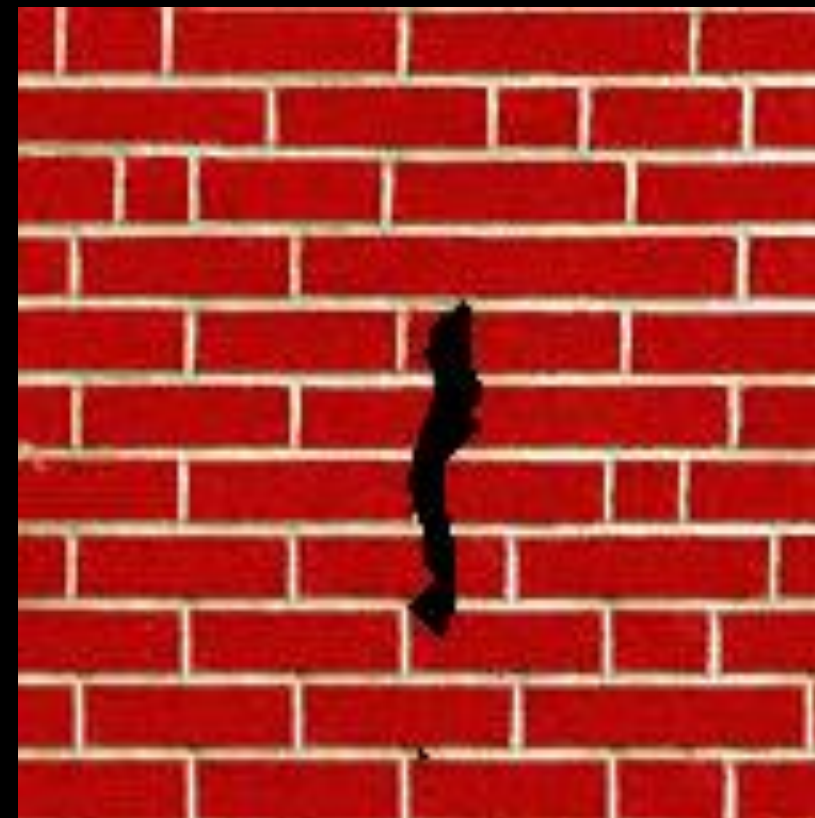
Homage to Shannon

coming in the unsensational
r Dick Gephardt was fai
rful riff on the looming
only asked, "What's your
tions?" A heartfelt sigh
story about the emergen
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Hole Filling



Extrapolation

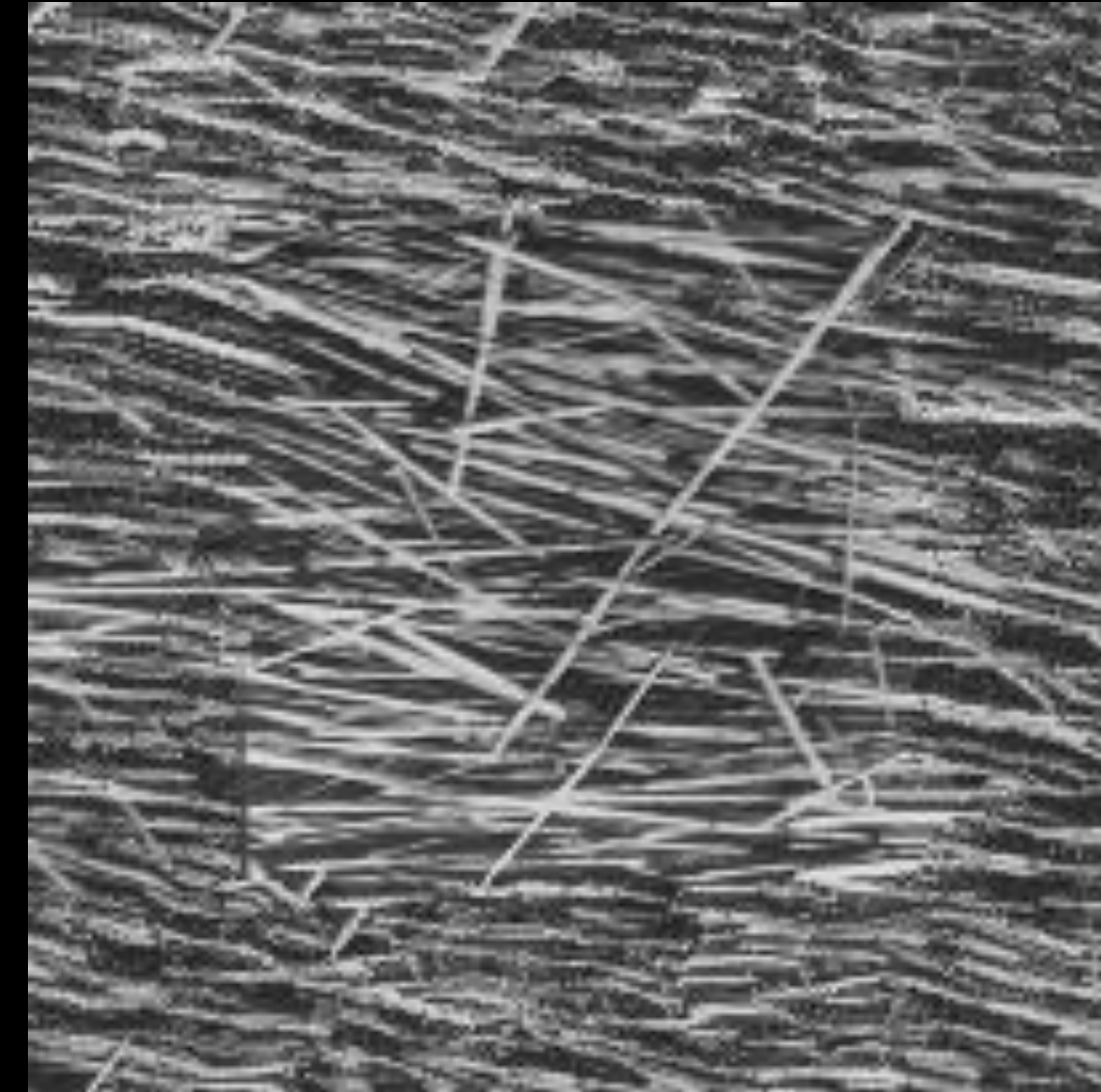
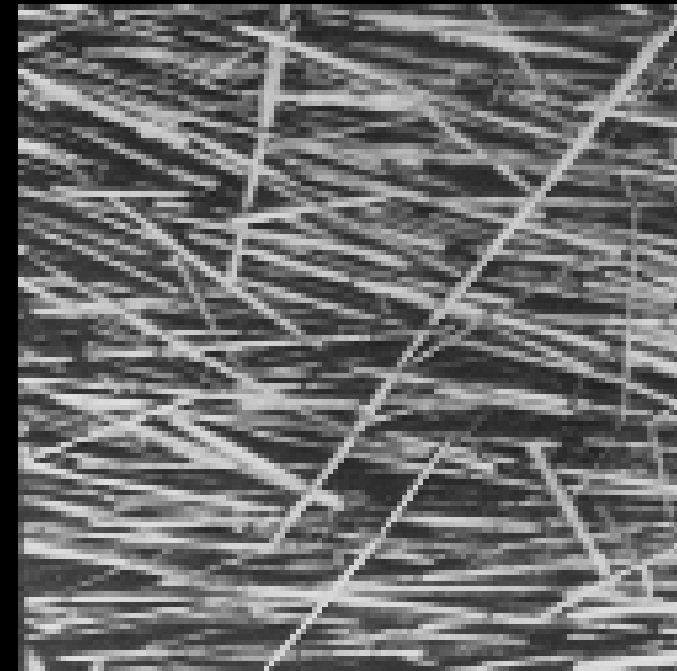


Image Analogies

Aaron Hertzmann^{1,2}

Chuck Jacobs²

Nuria Oliver²

Brian Curless³

David Salesin^{2,3}

¹New York University

²Microsoft Research

³University of Washington

Image Analogies



A



A'



B



B'



Image Analogies

Goal: Process an image by example



A

■
■



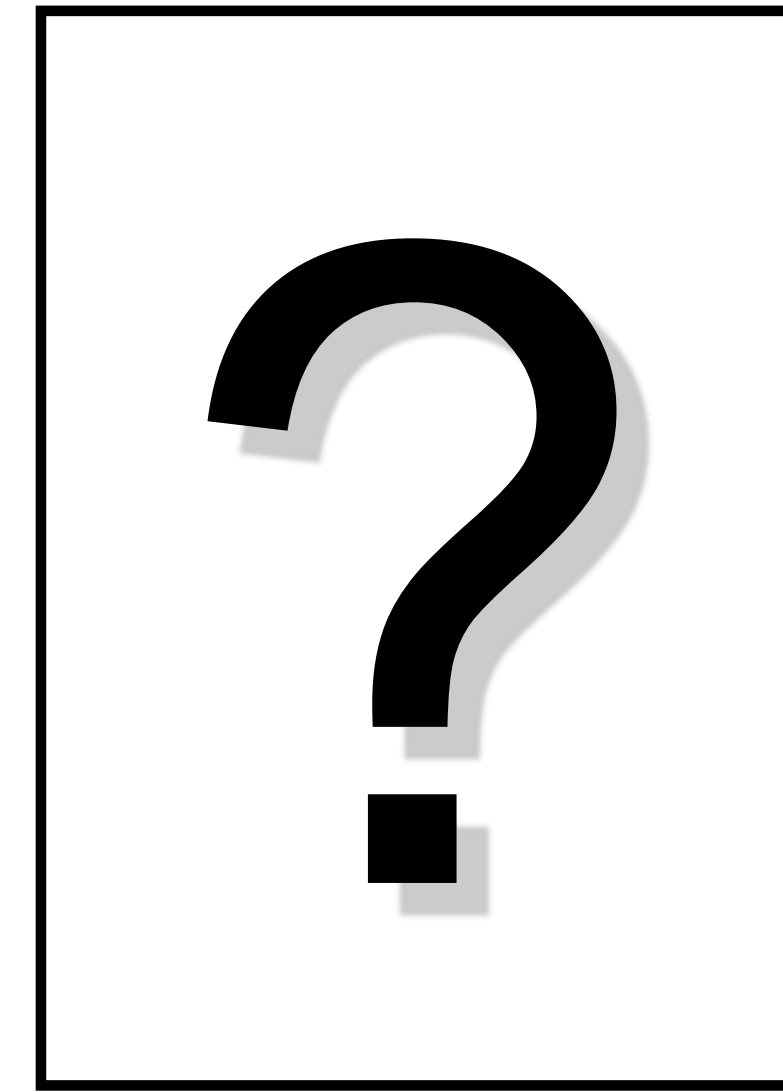
A'

■ ■
■ ■



B

■
■



B'

Non-parametric sampling



A

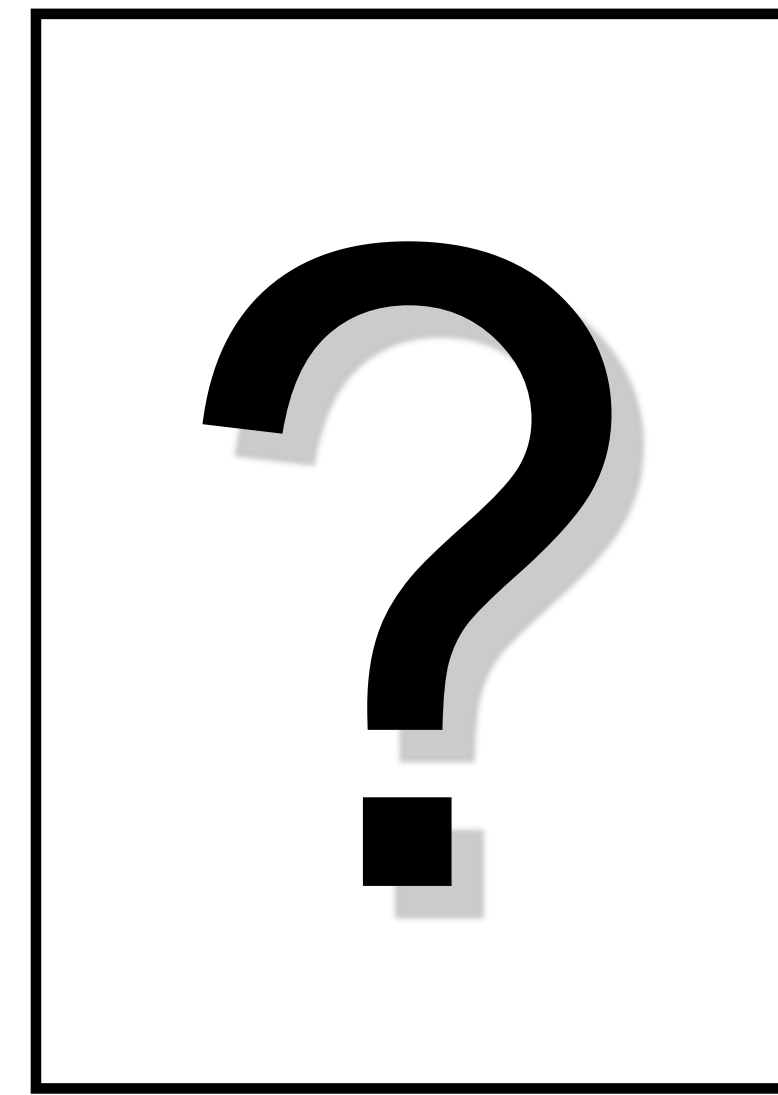


A'

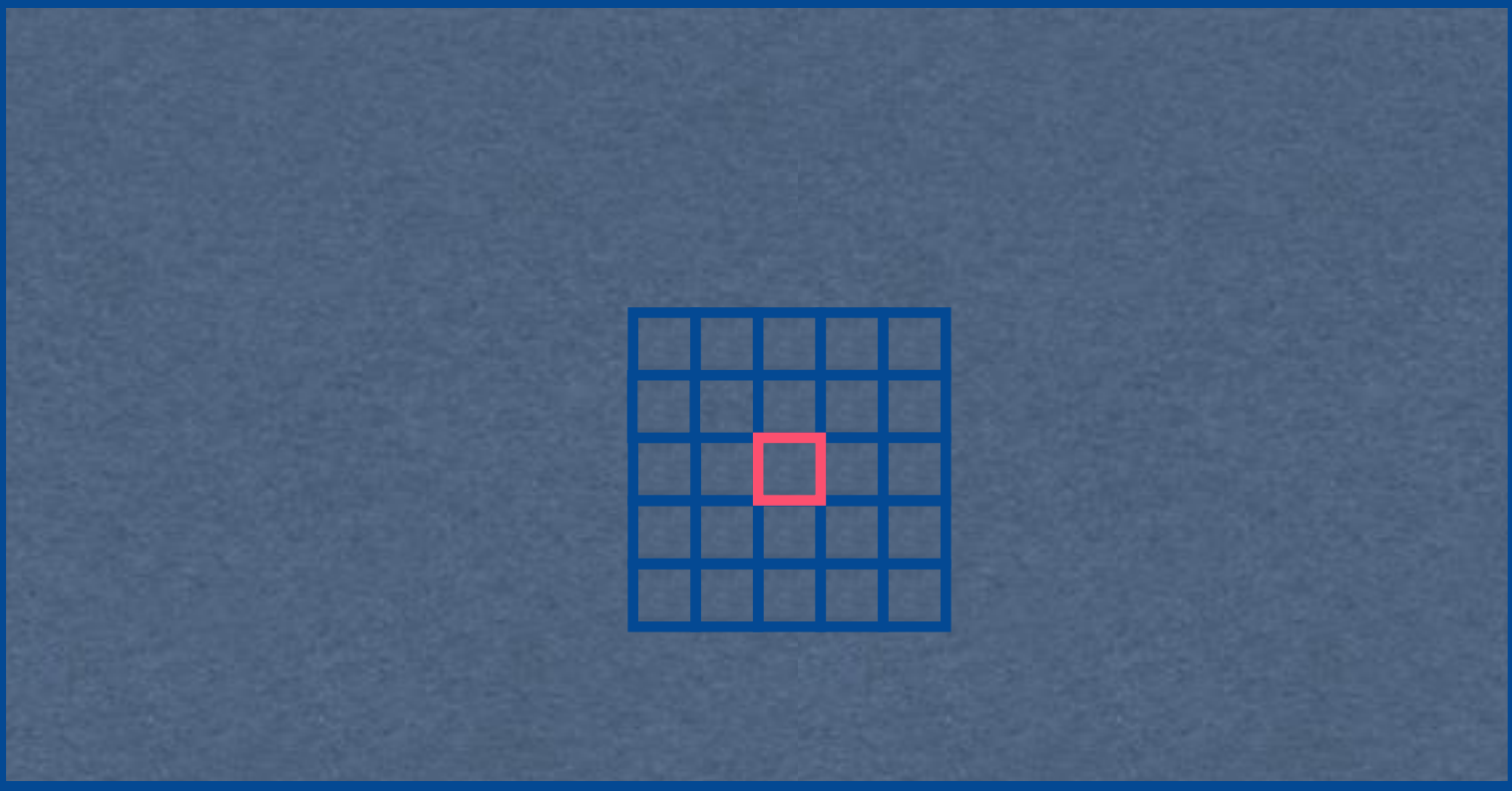
⋮



B

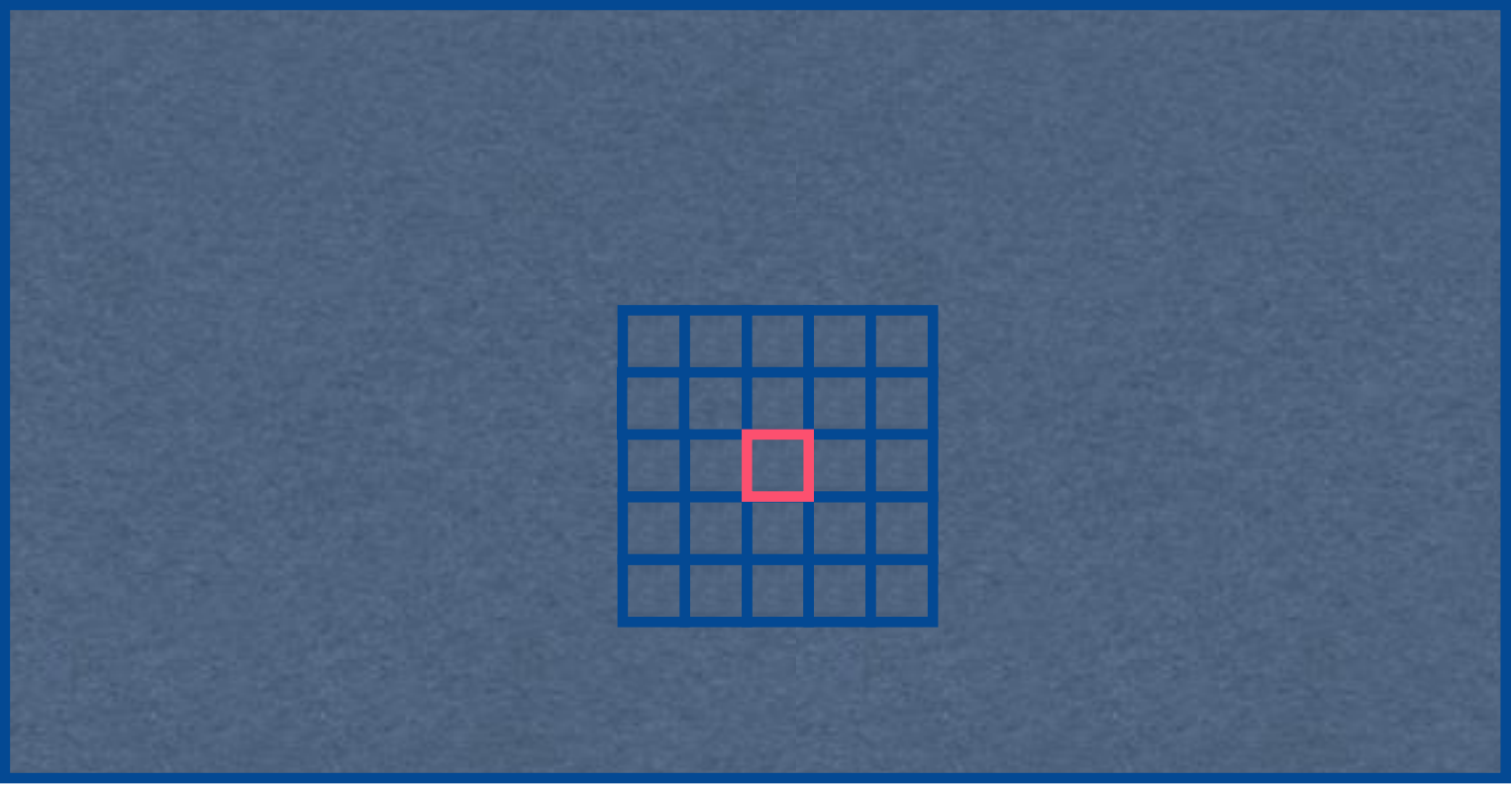


B'



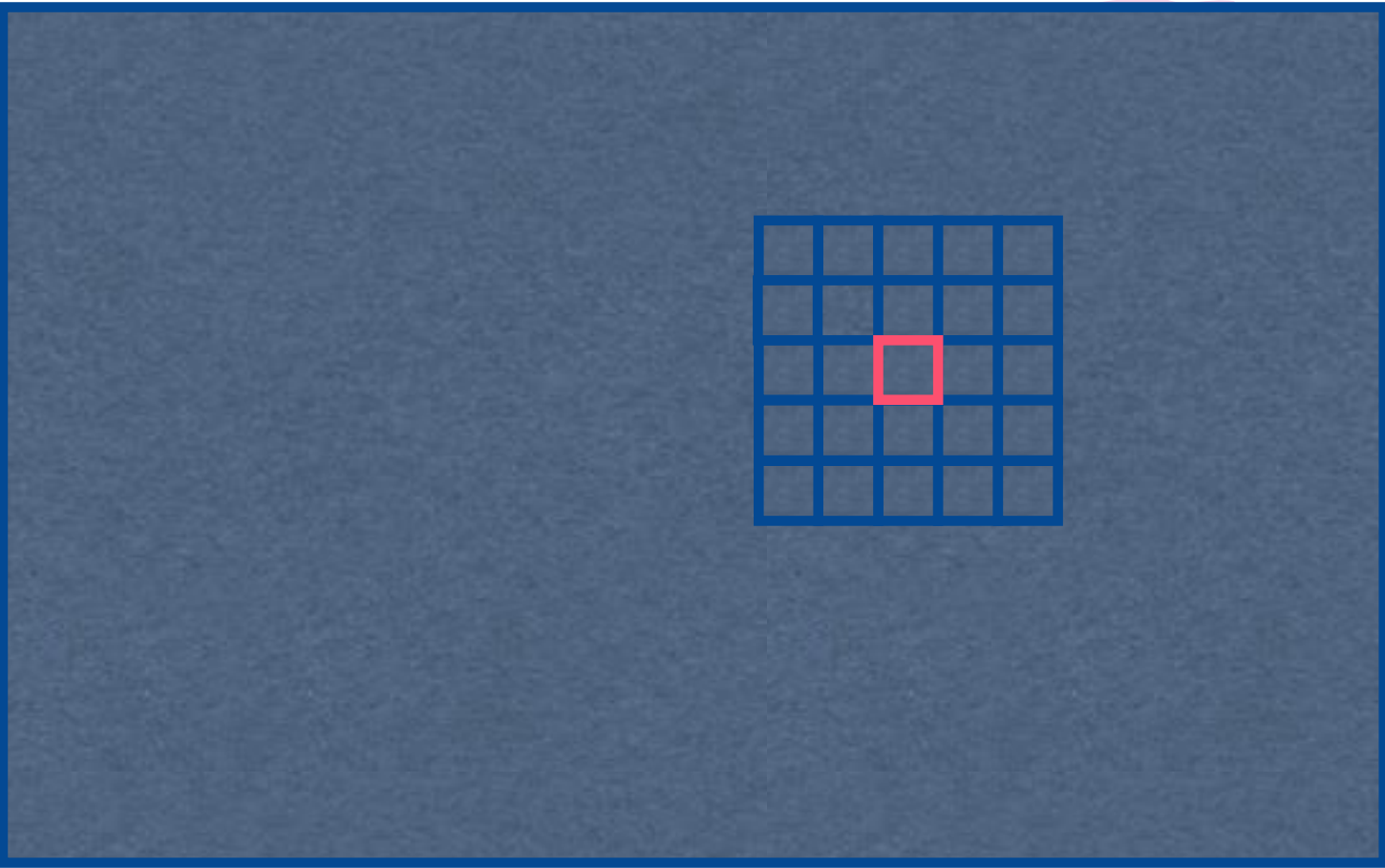
A

⋮



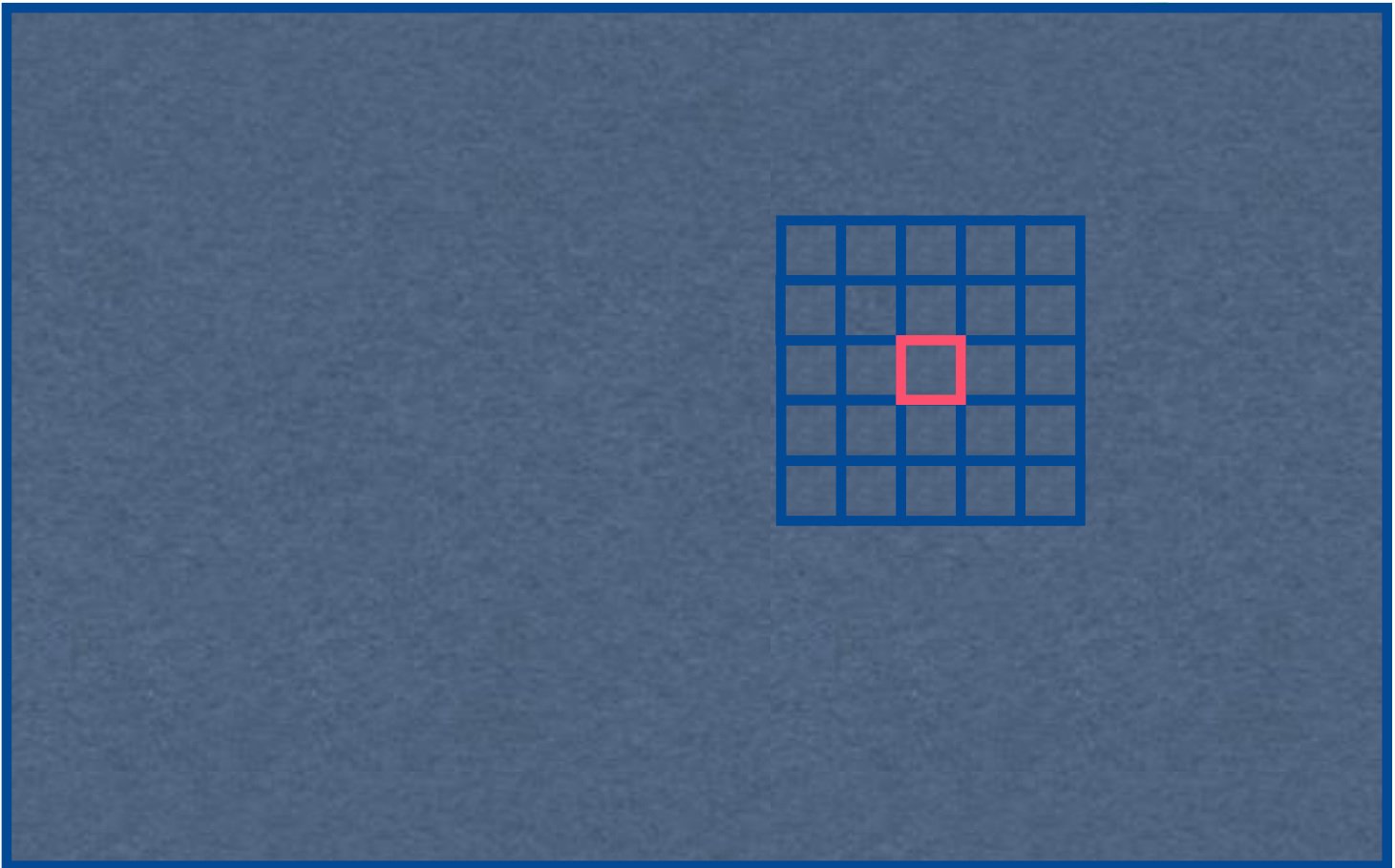
A'

⋮

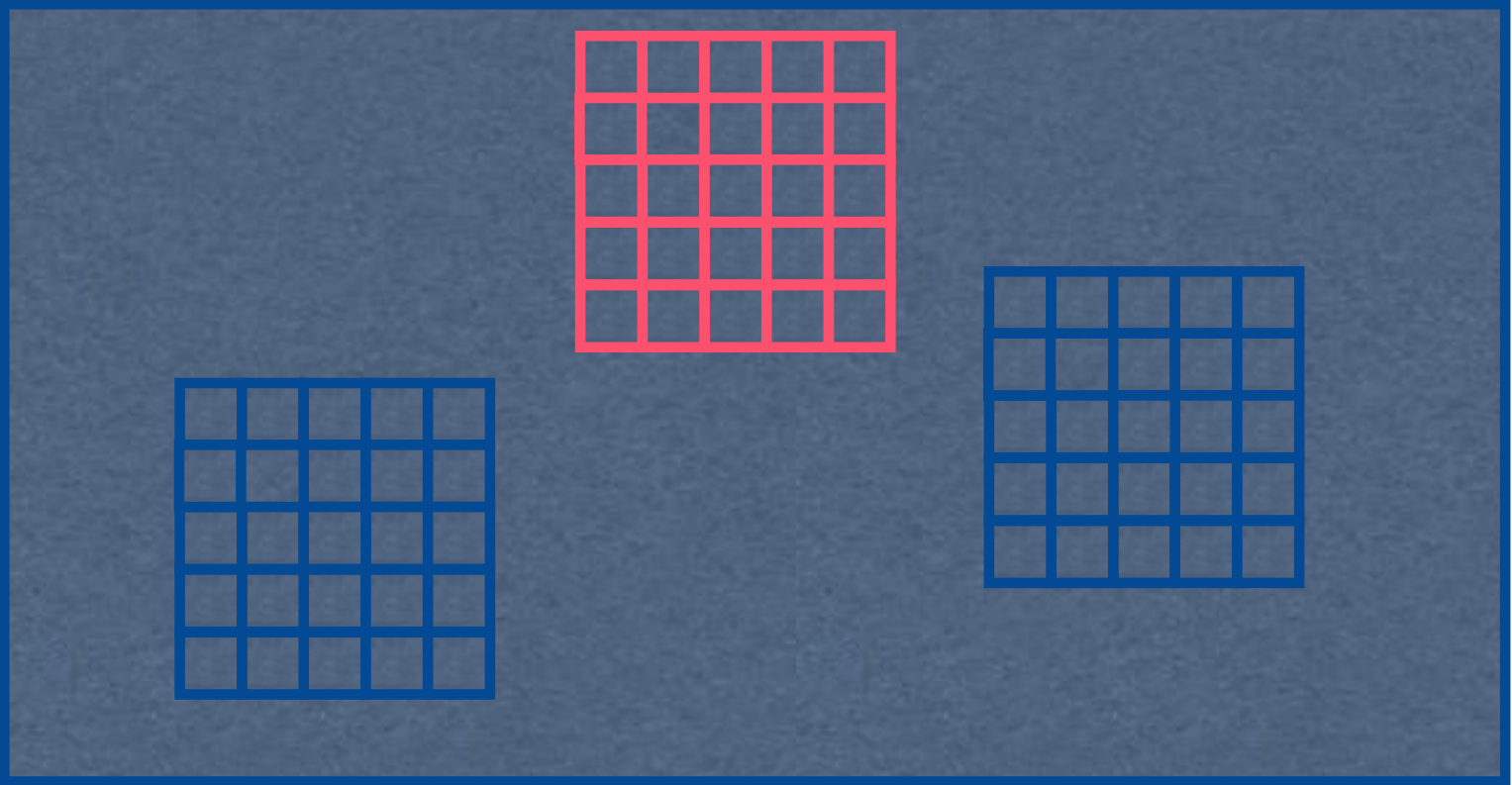


B

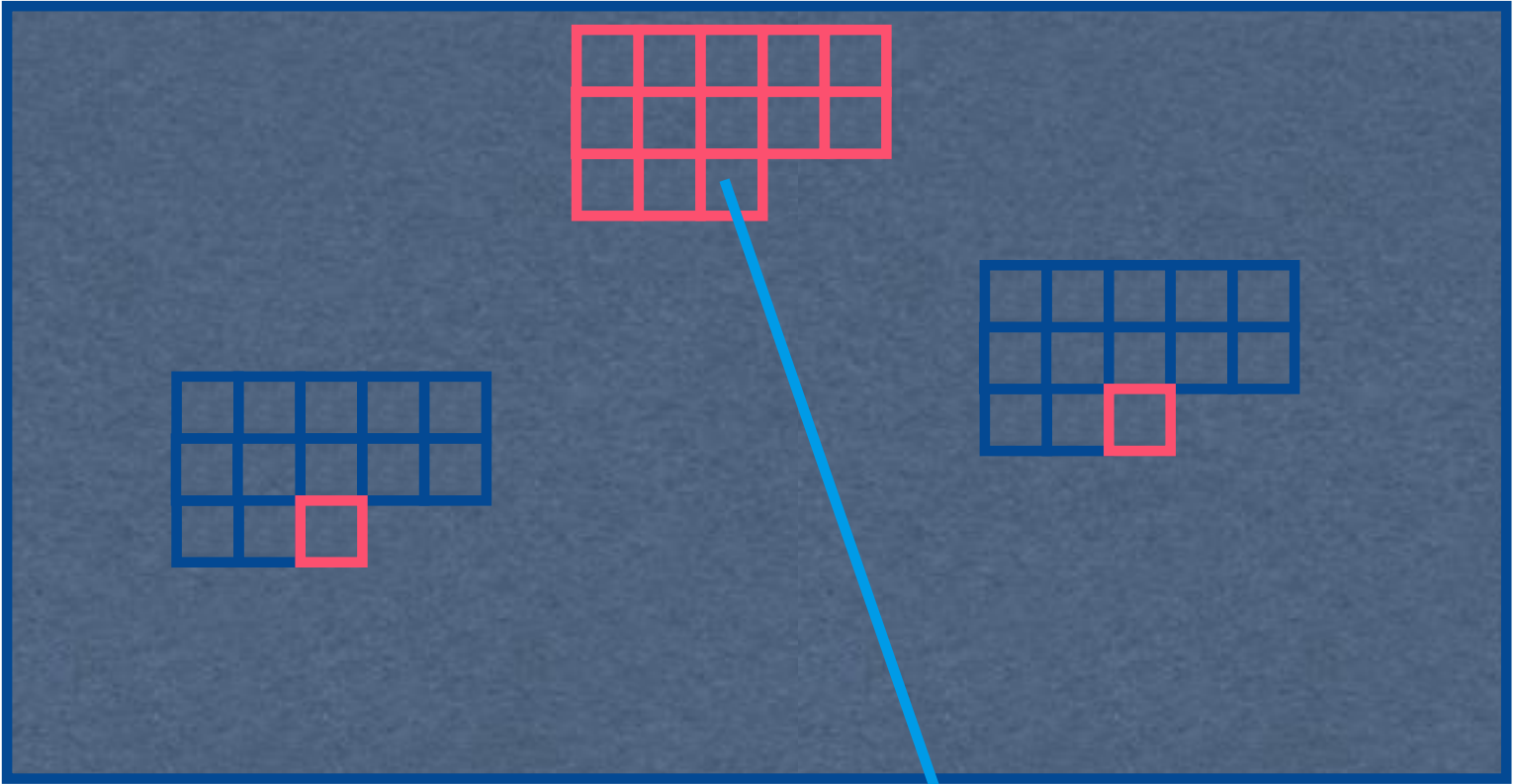
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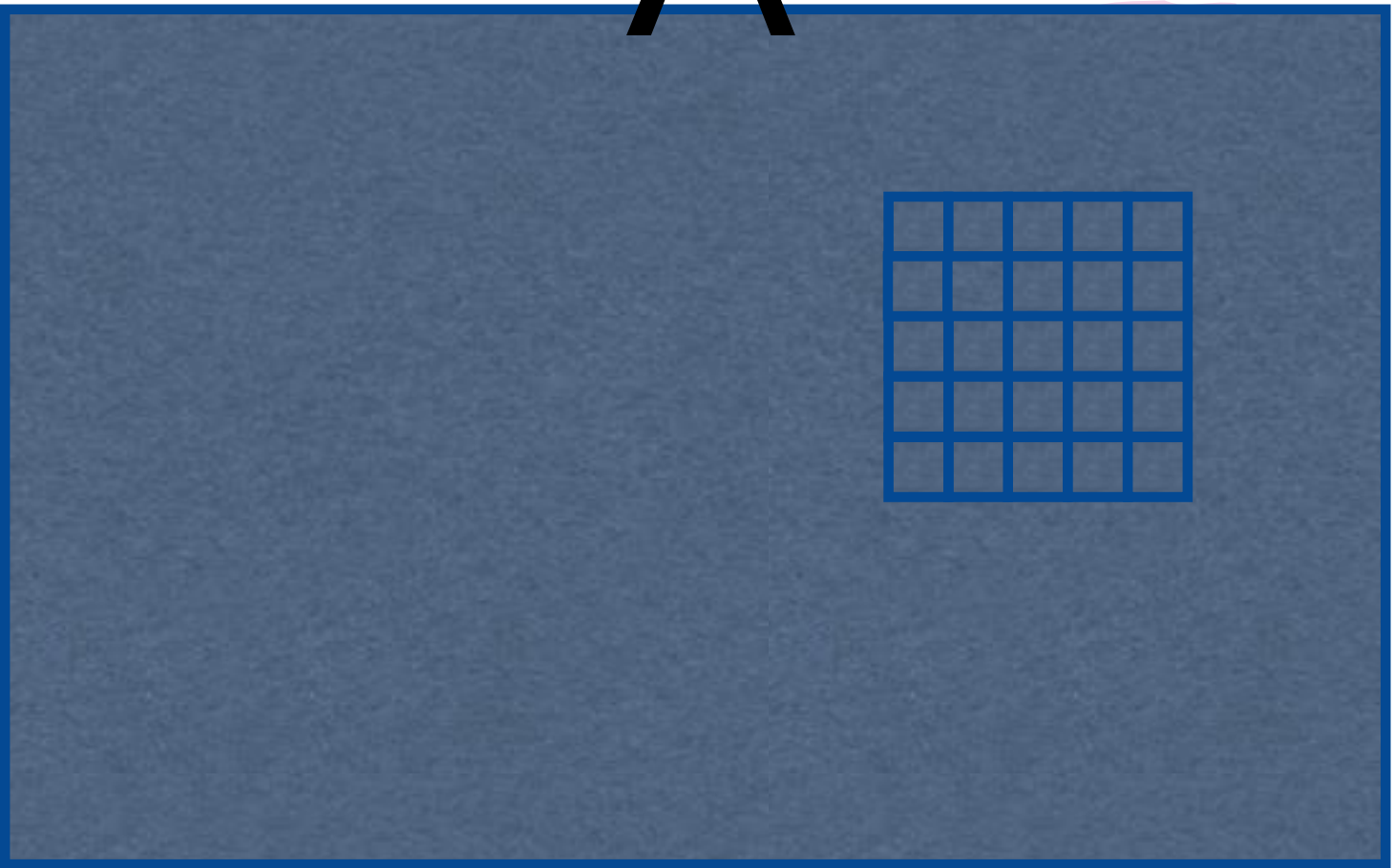
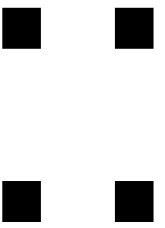
B'



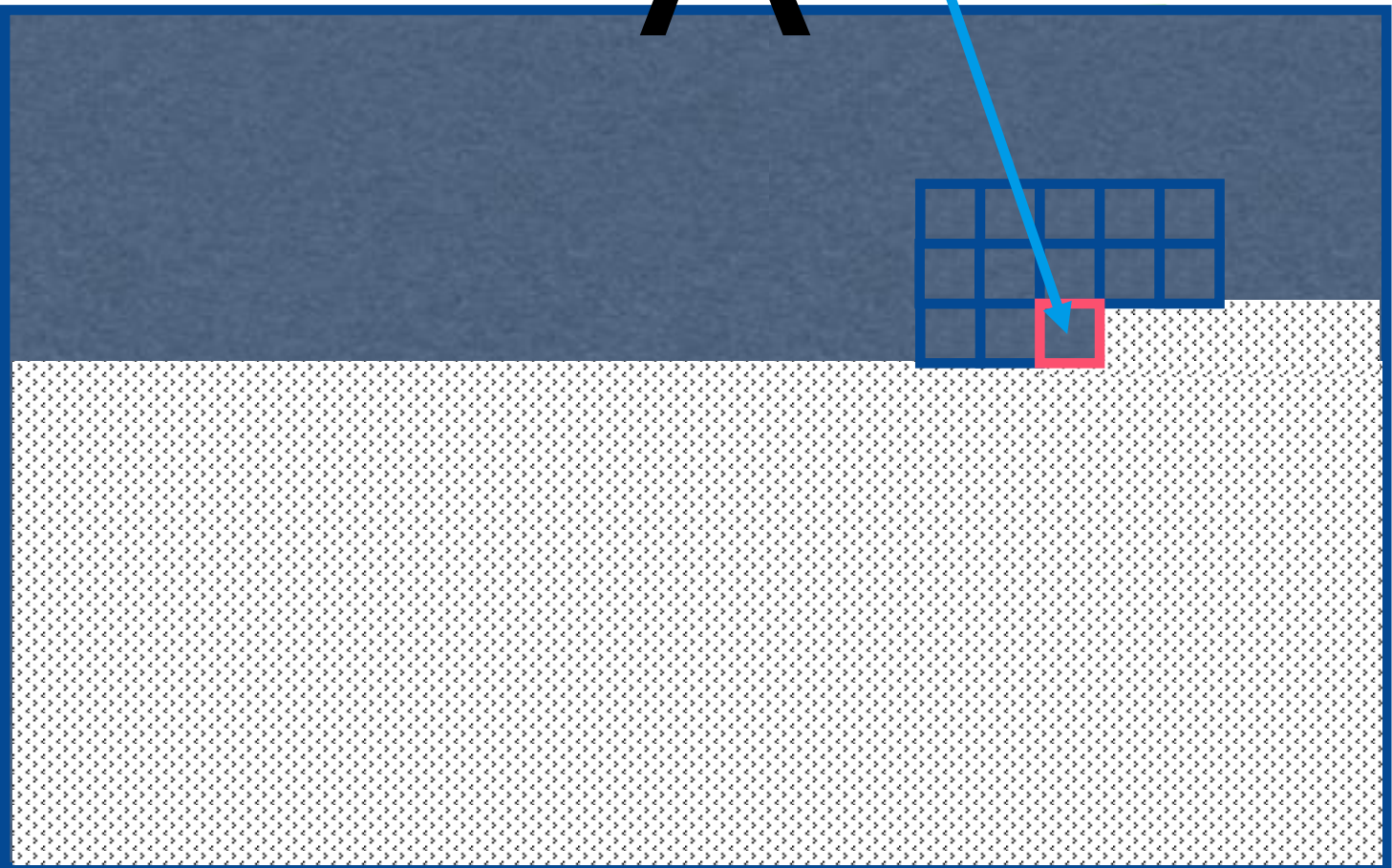
A



A'



B



B'

Blur Filter



Unfiltered source (A)



Filtered source (A')



Unfiltered target (B)

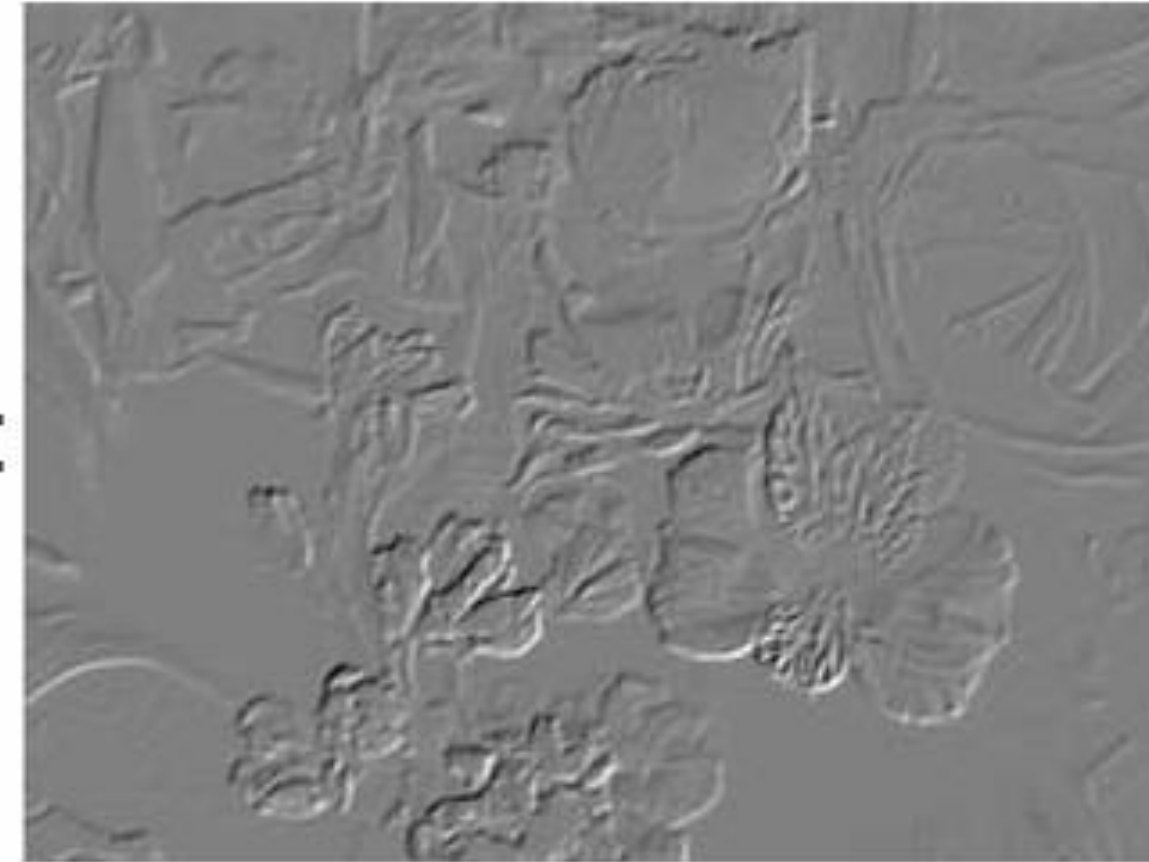


Filtered target (B')

Edge Filter



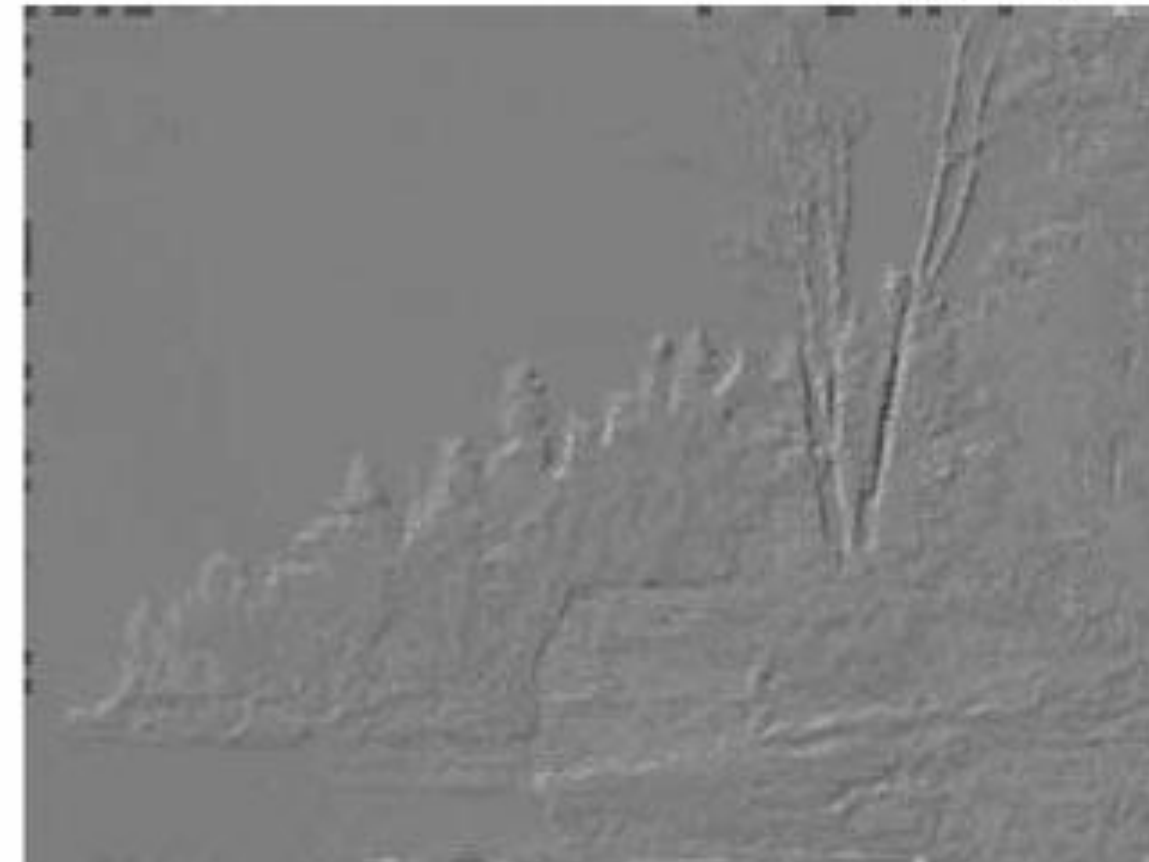
Unfiltered source (A)



Filtered source (A')



Unfiltered target (B)



Filtered target (B')

Artistic Filters



A



A'



B



B'

Colorization



Unfiltered source (A)

▪
▪



Filtered source (A')

▪ ▪
▪ ▪



Unfiltered target (B)

▪
▪



Filtered target (B')

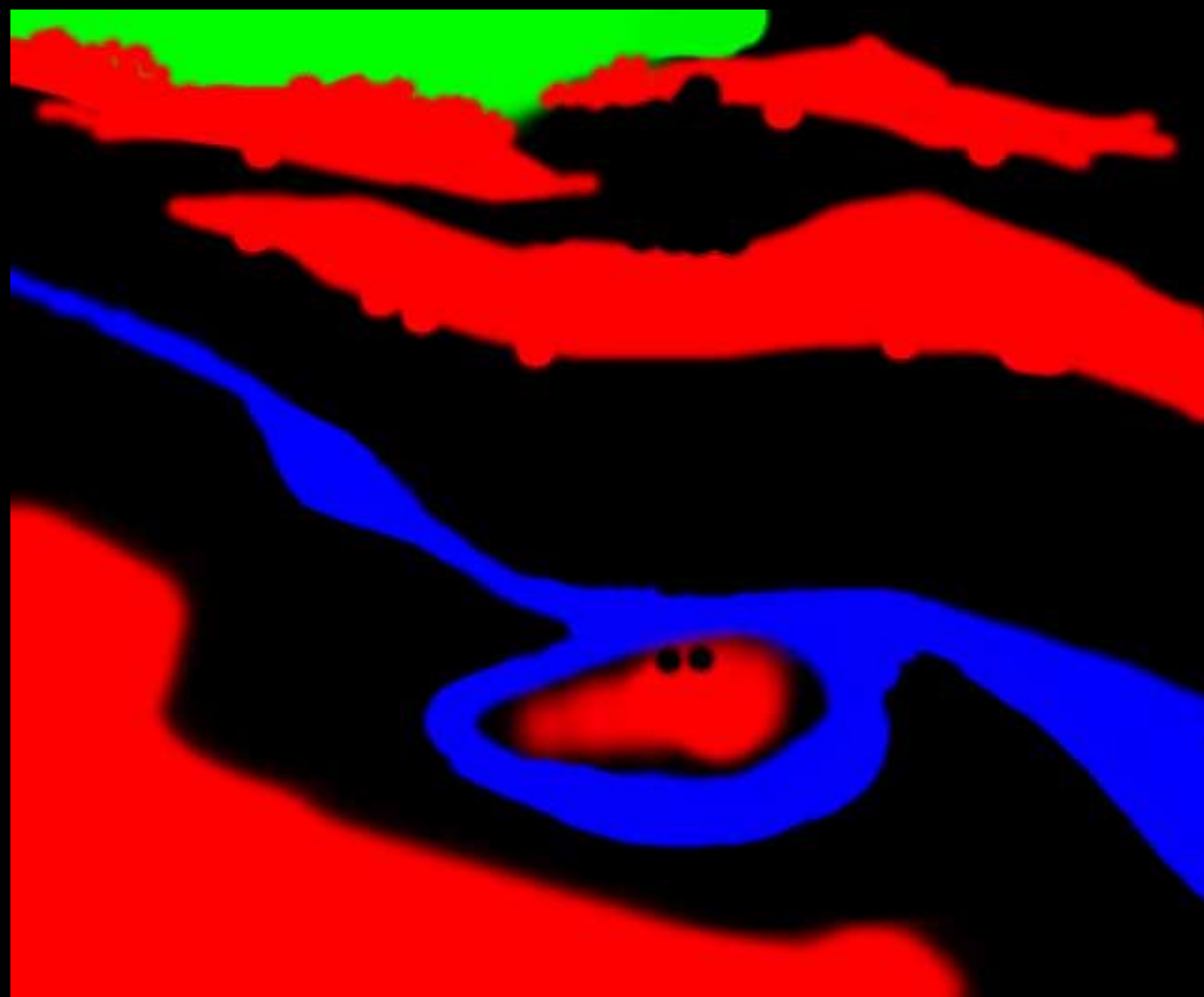
Texture-by-numbers



A



A'



B



B'

Visual Prompting via Image Inpainting

Amir Bar*, Yossi Gandelsman*,
Trevor Darrell, Amir Globerson, Alexei A Efros

NeurIPS 2022



* Equal contribution

Visual Prompting

Task examples

Query

Result

Task e

• •

• •

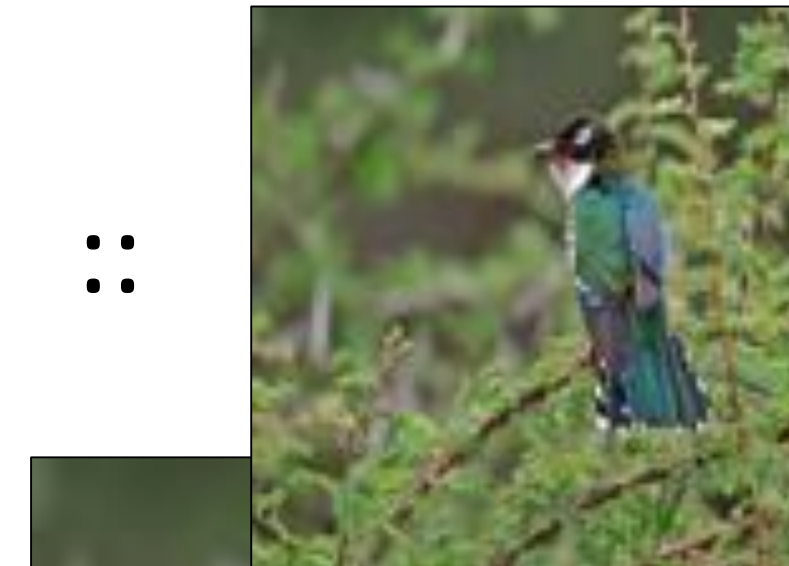
Result



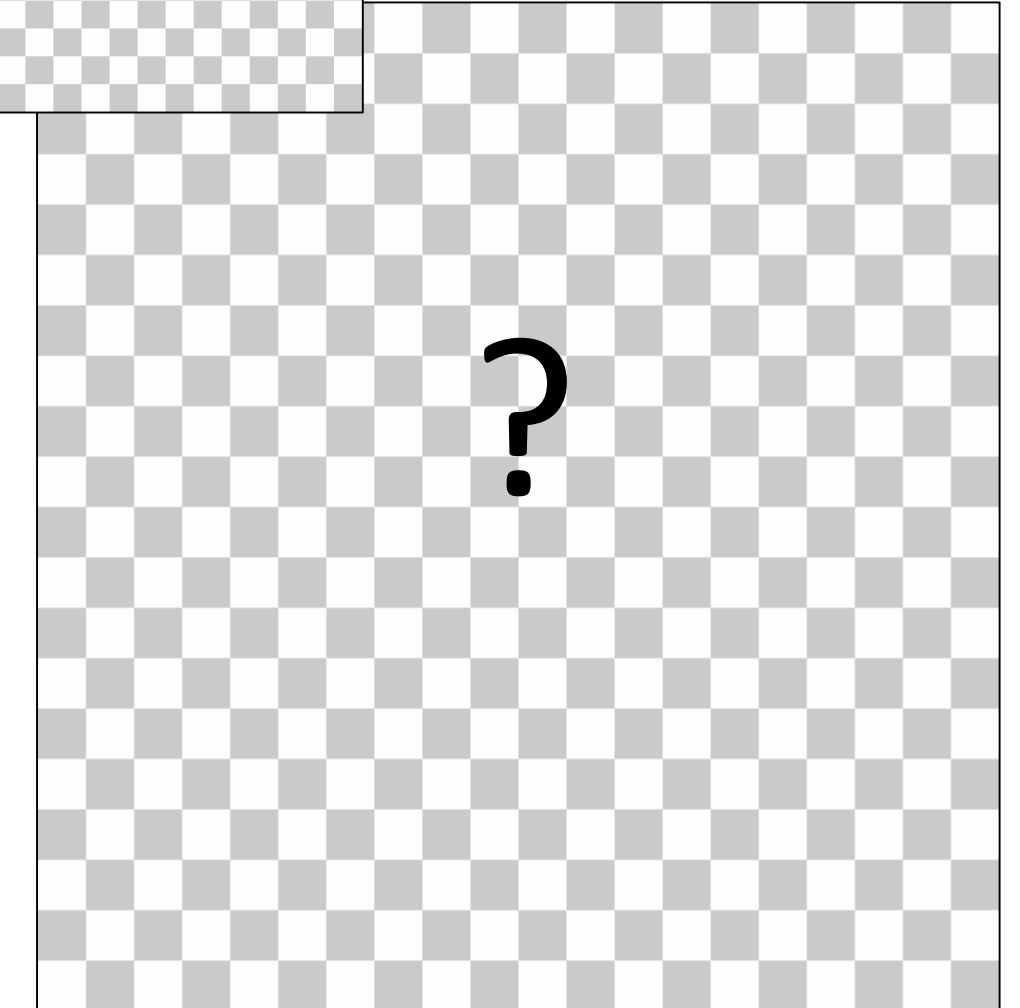
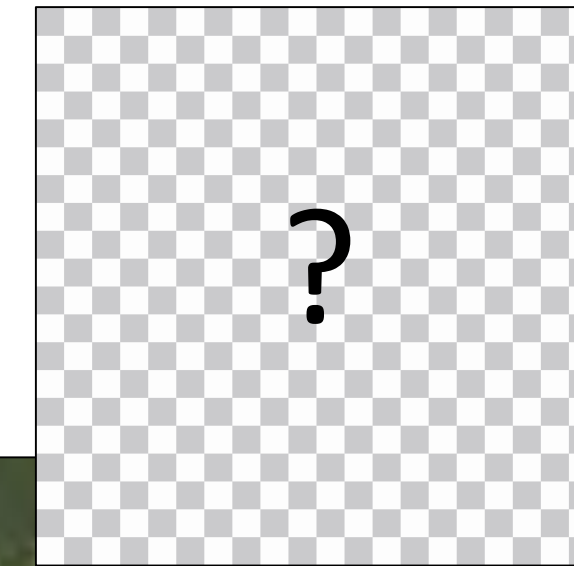
A



A'



B



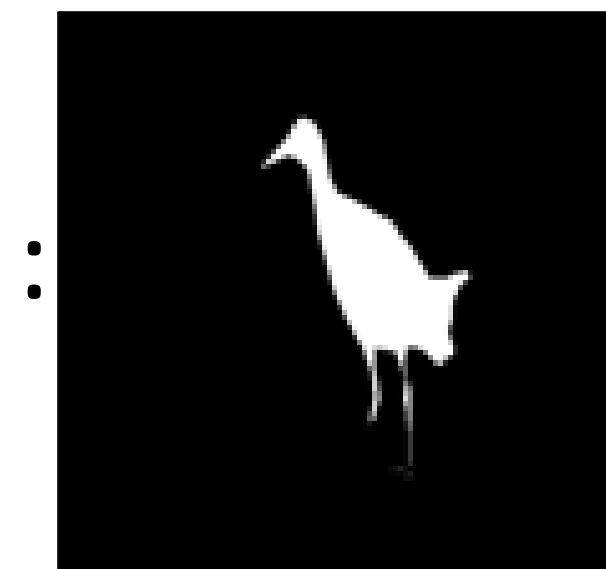
B'

Visual Prompting

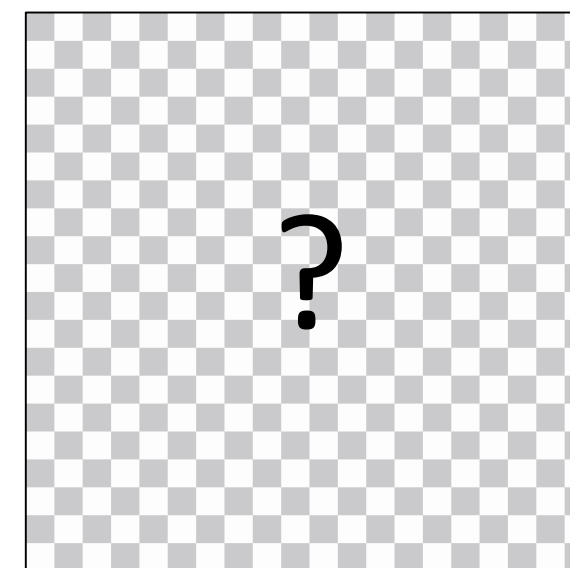
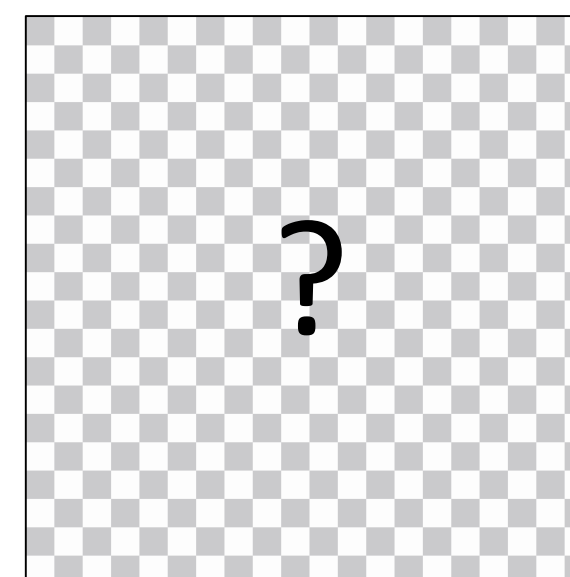
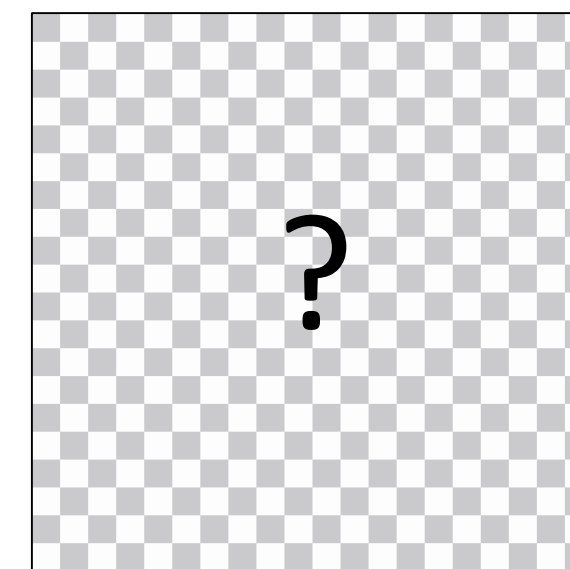
Task examples

Query

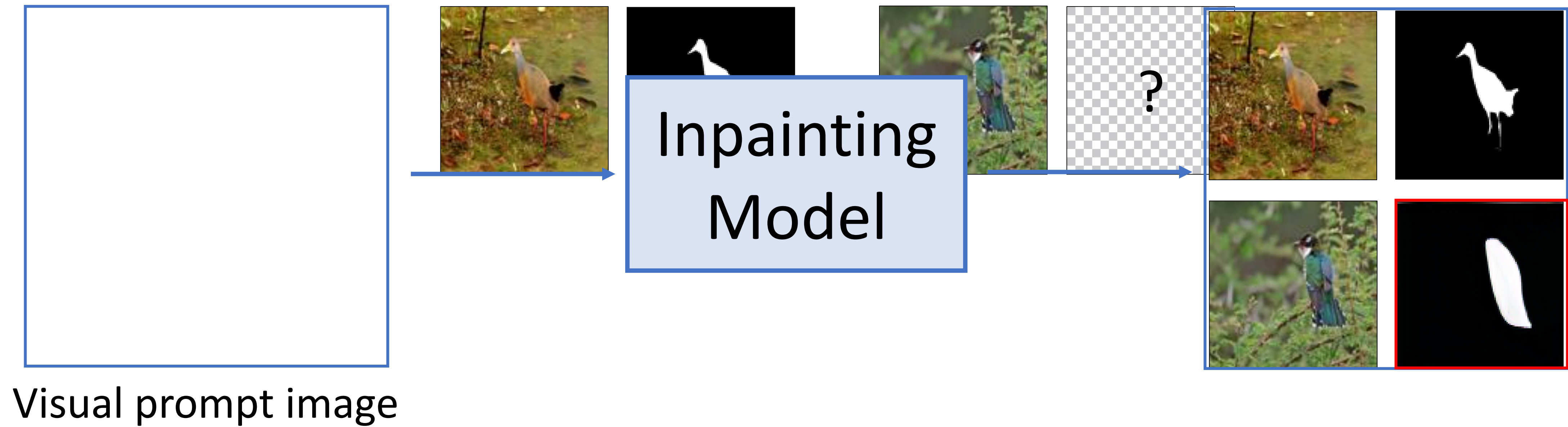
Result



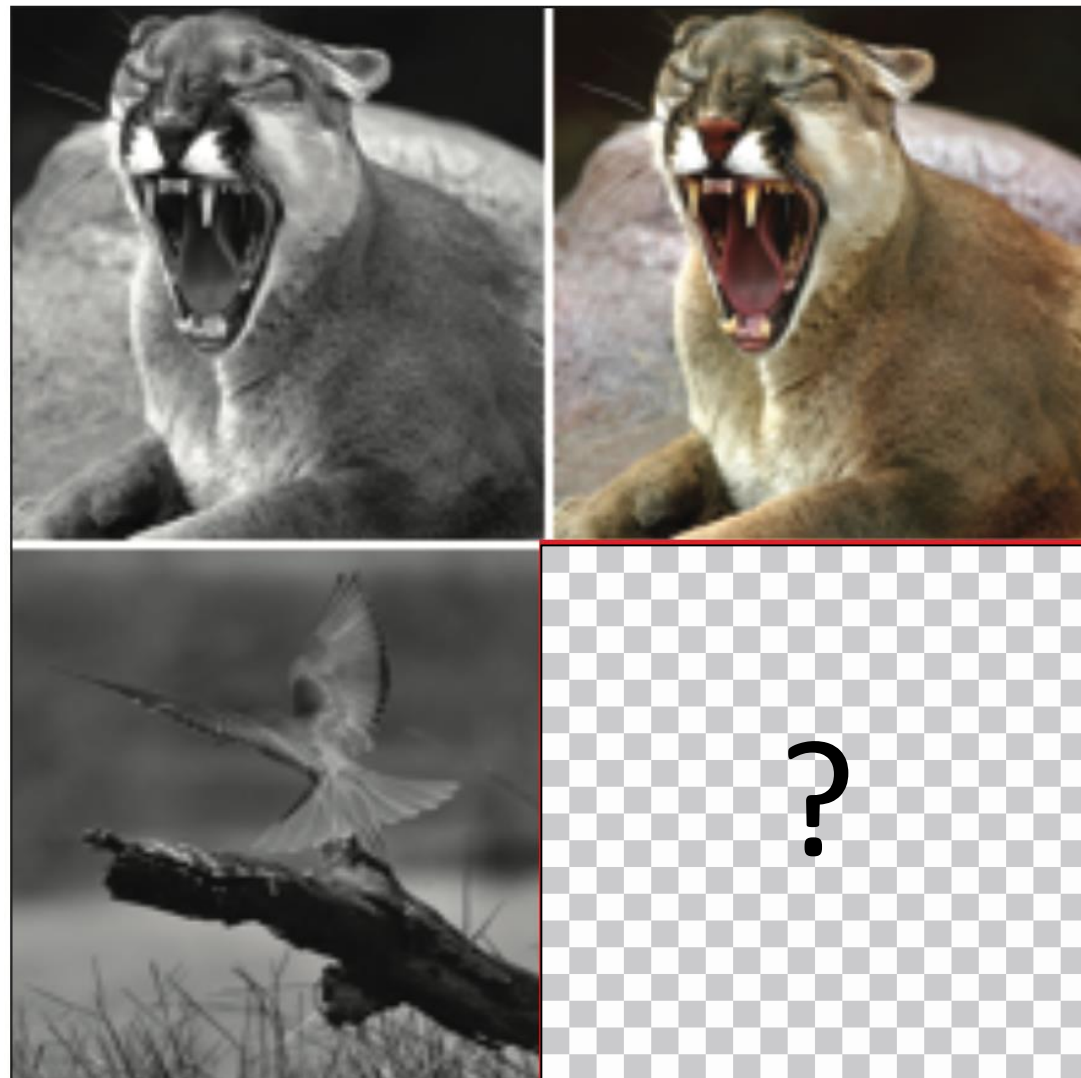
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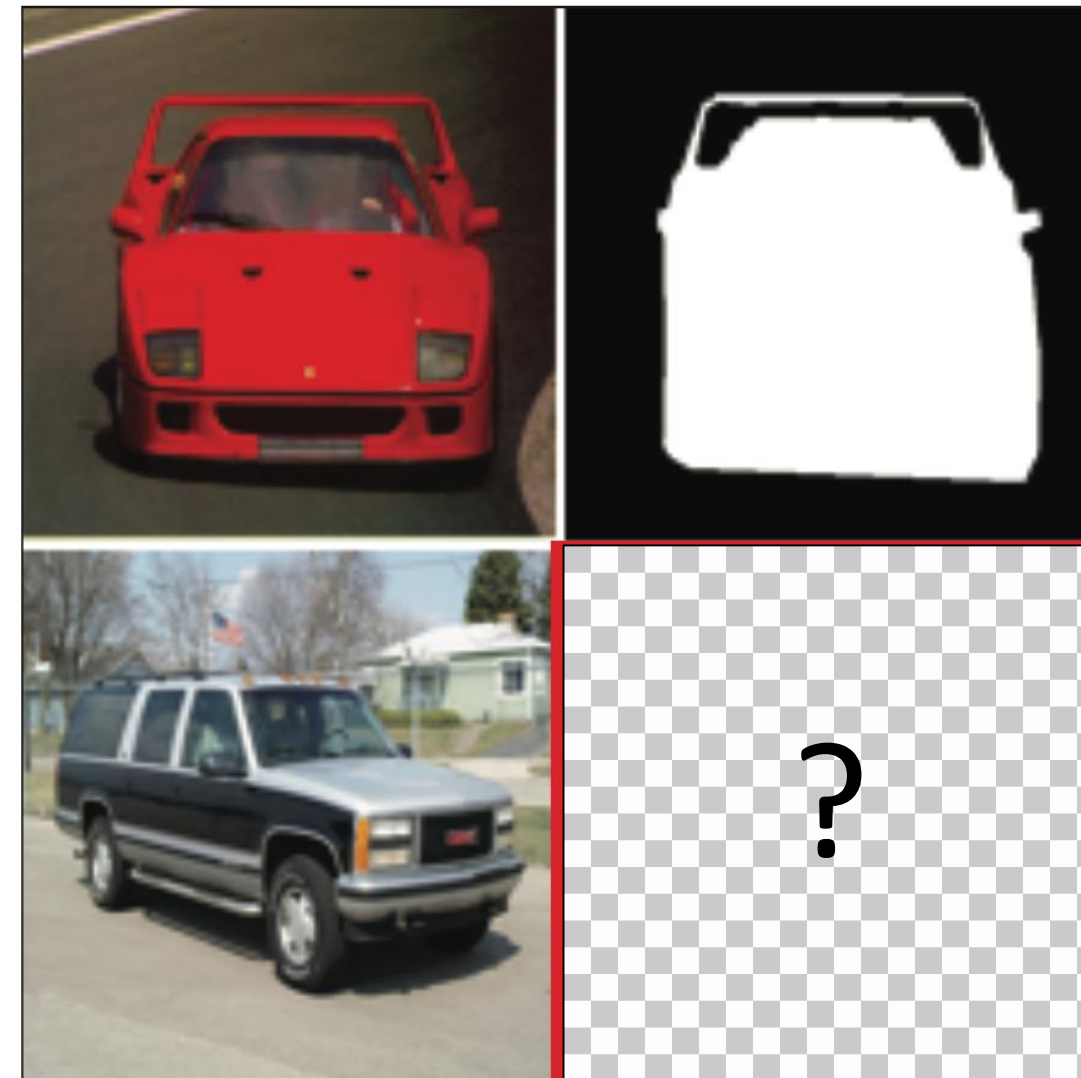
Inpainting models to the rescue!



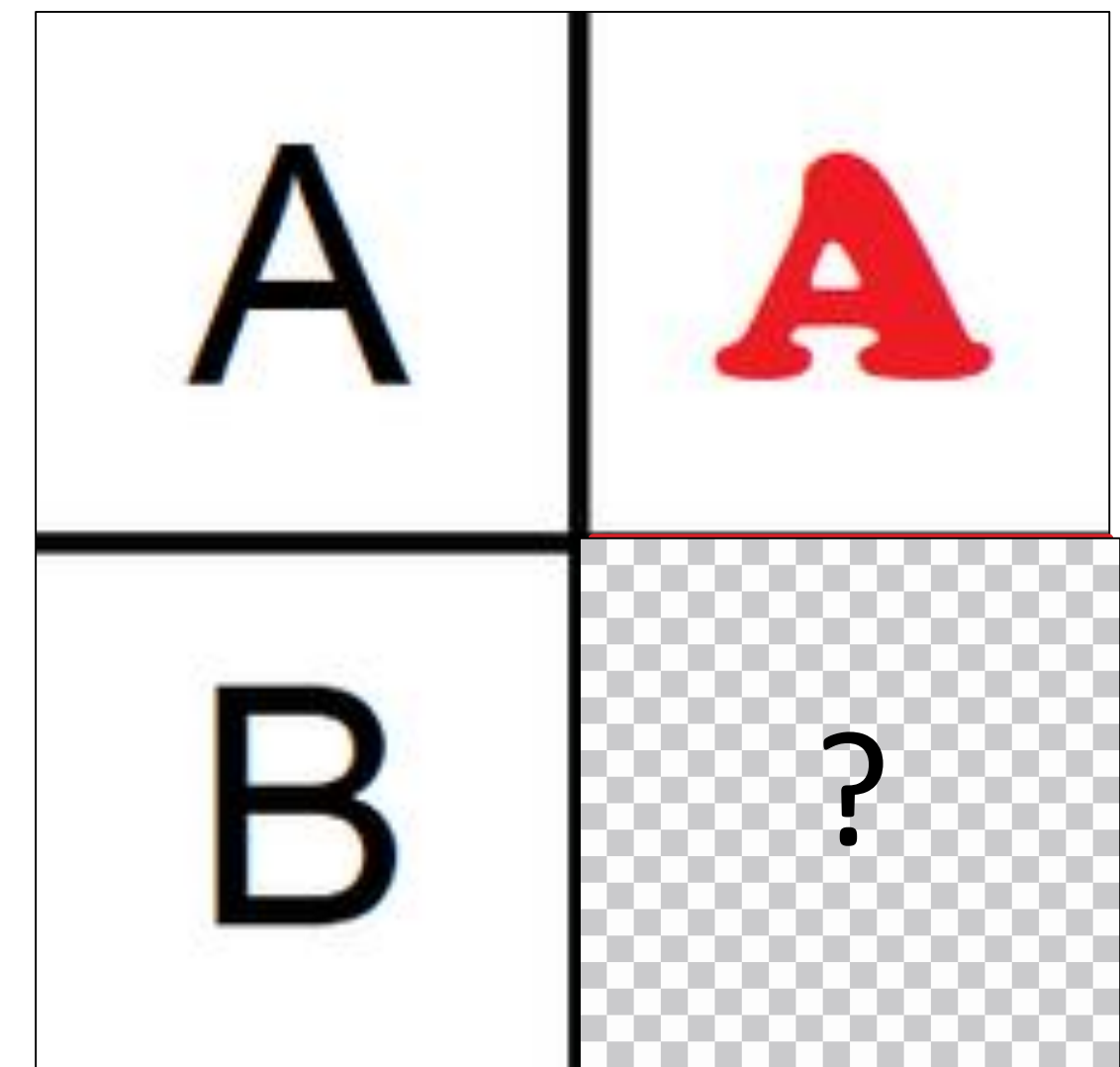
Wide range of tasks



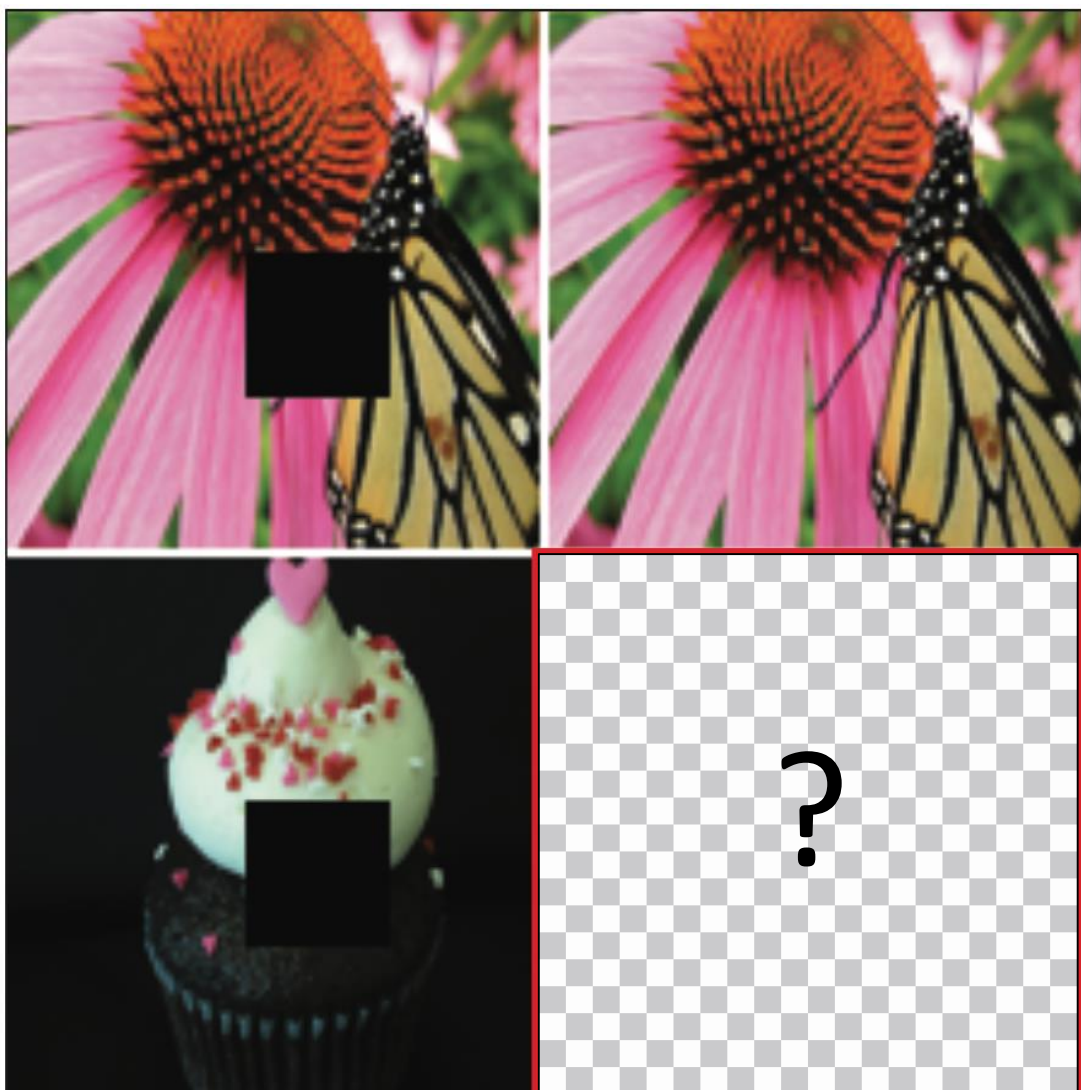
Colorization



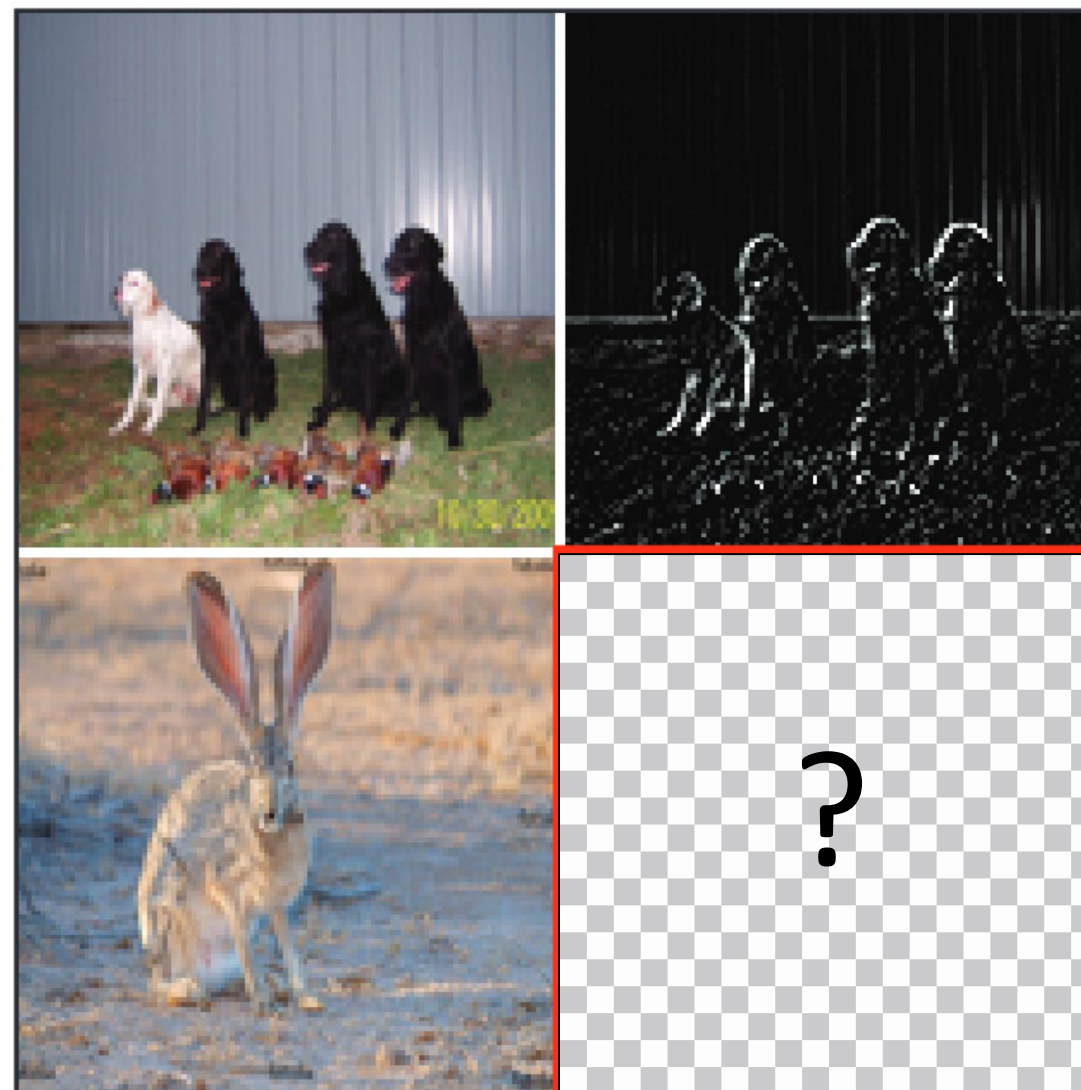
Segmentation



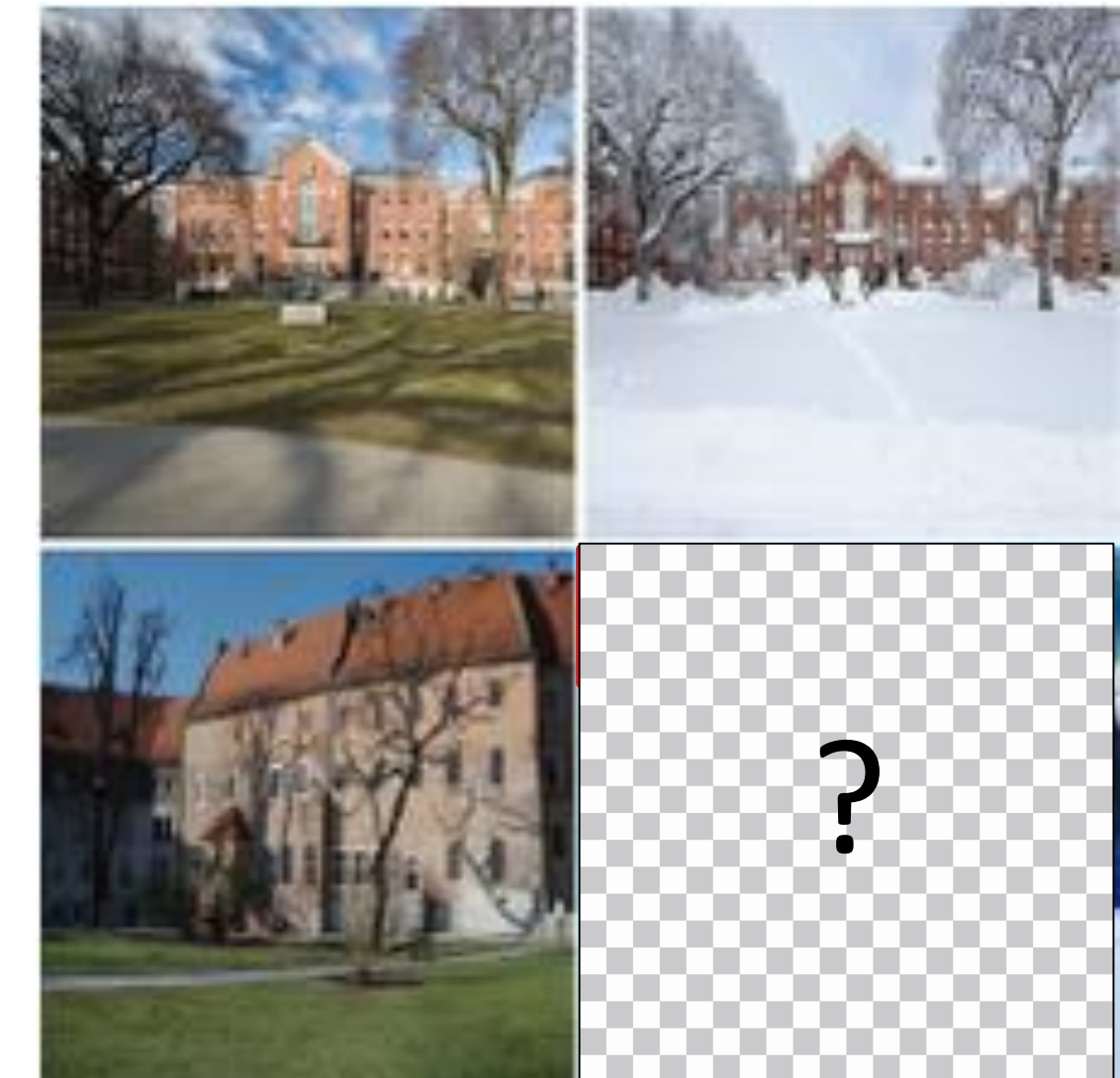
Font Style Transfer



Inpainting



Edge Detection



Style Transfer

Early one morning the sun was shining I was laying in bed
Wondering if she had changed at all if her hair was still red

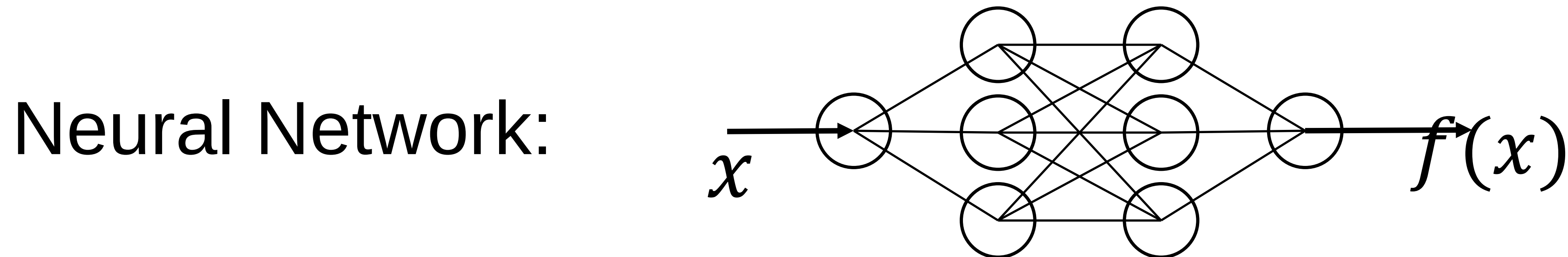
$$P(x_n | x_{n-1}, x_{n-2}, x_{n-3}, x_{n-4}, x_{n-5}, x_{n-6}, x_{n-7}, x_{n-8}, x_{n-9}, x_{n-10}, x_{n-11}, x_{n-12}, x_{n-13})$$

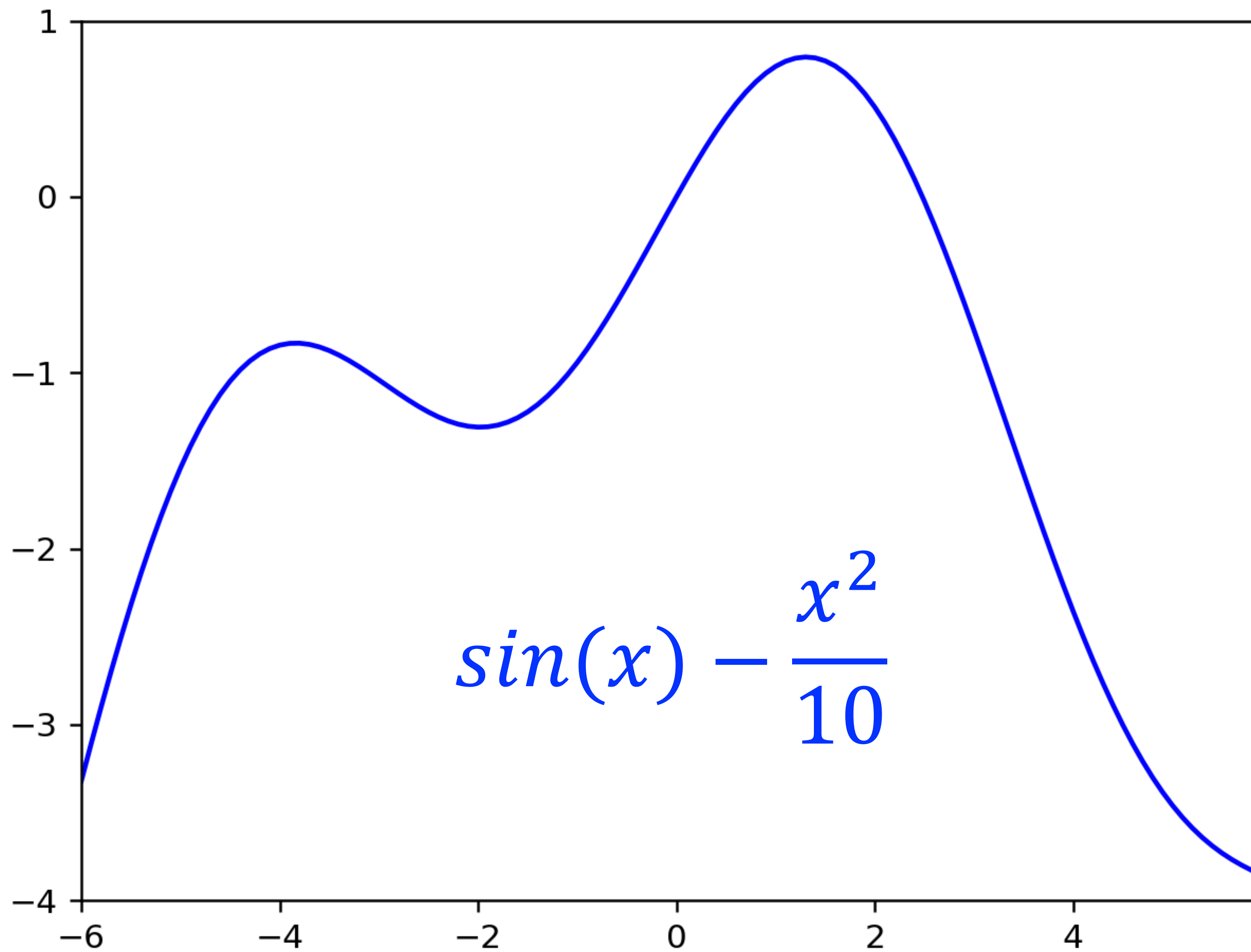
10^{70} combinations

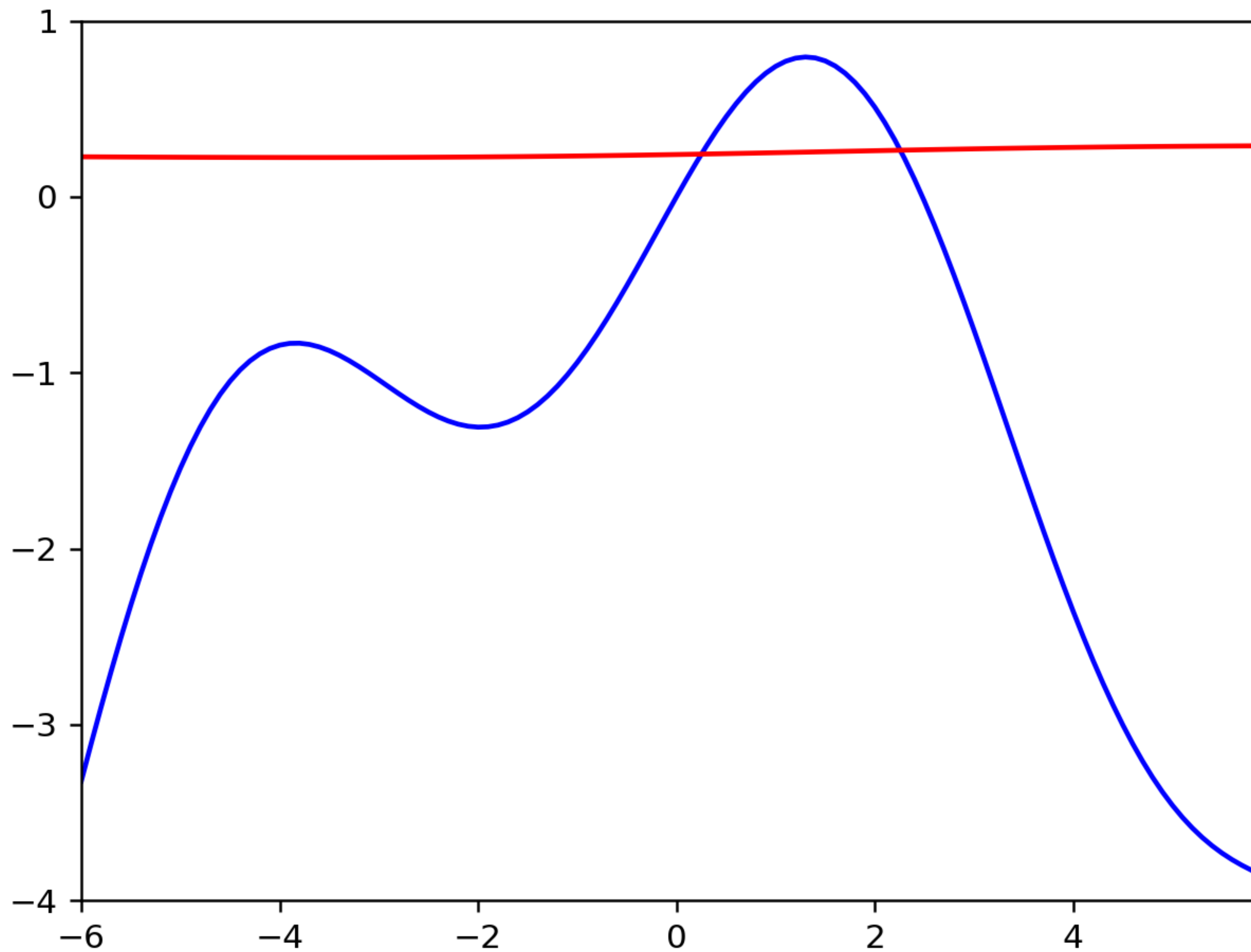
Function Approximation

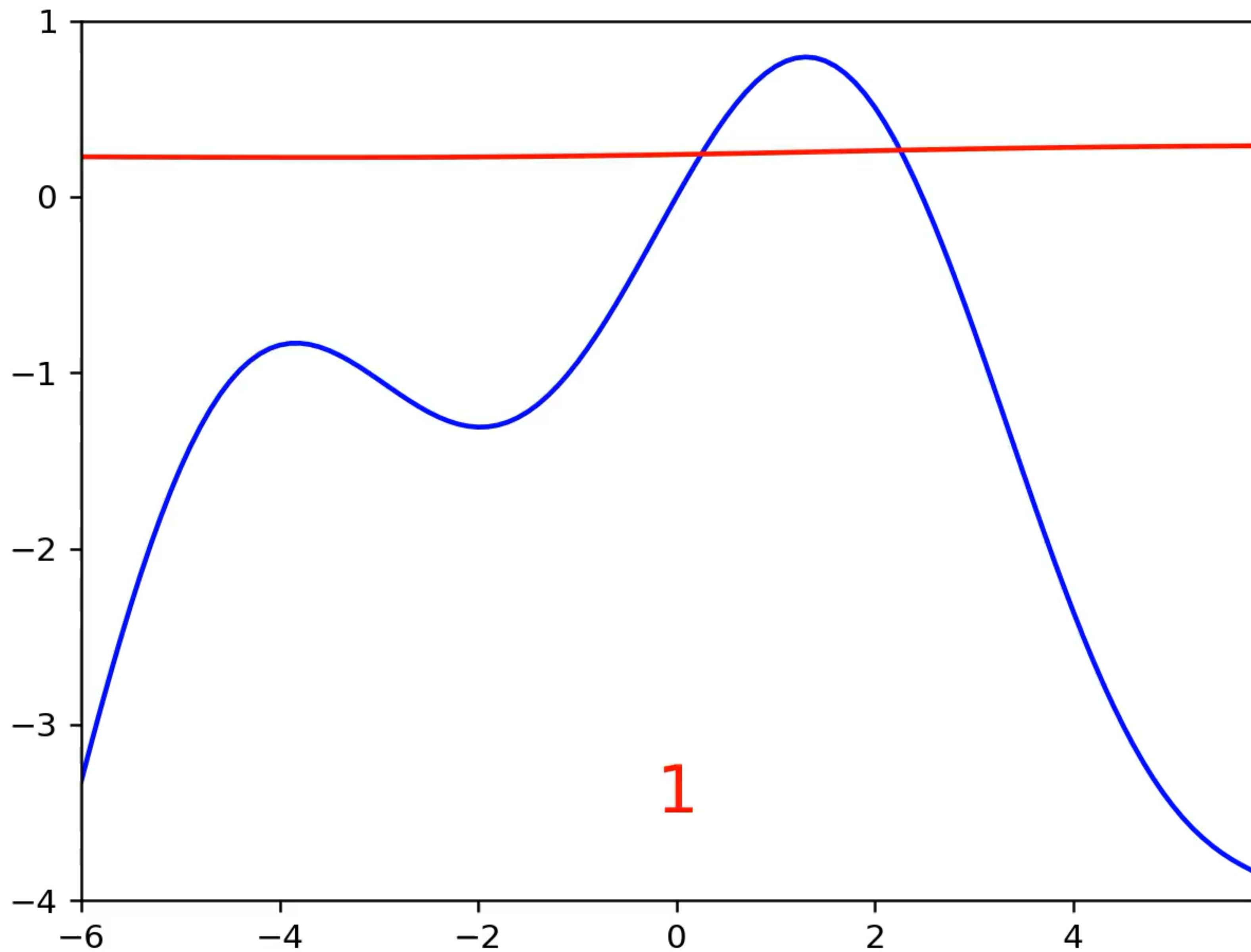
Fourier Series: $f(x) = \text{~} + \text{~} + \text{~} + \text{~} + \dots$

Taylor Series: $f(x) = \text{---} + \text{ / } + \text{ U } + \text{ ~ } + \dots$









$$P(x_n | x_{n-1}, x_{n-2}, x_{n-3}, x_{n-4}, x_{n-5}, x_{n-6}, x_{n-7}, \dots)$$

Early one morning the sun was shining I was laying in bed
Wondering if she had changed at all if her hair was still red

red



Early one morning the sun was shining I was laying in bed
Wondering if she had changed at all if her hair was still

red

neural network

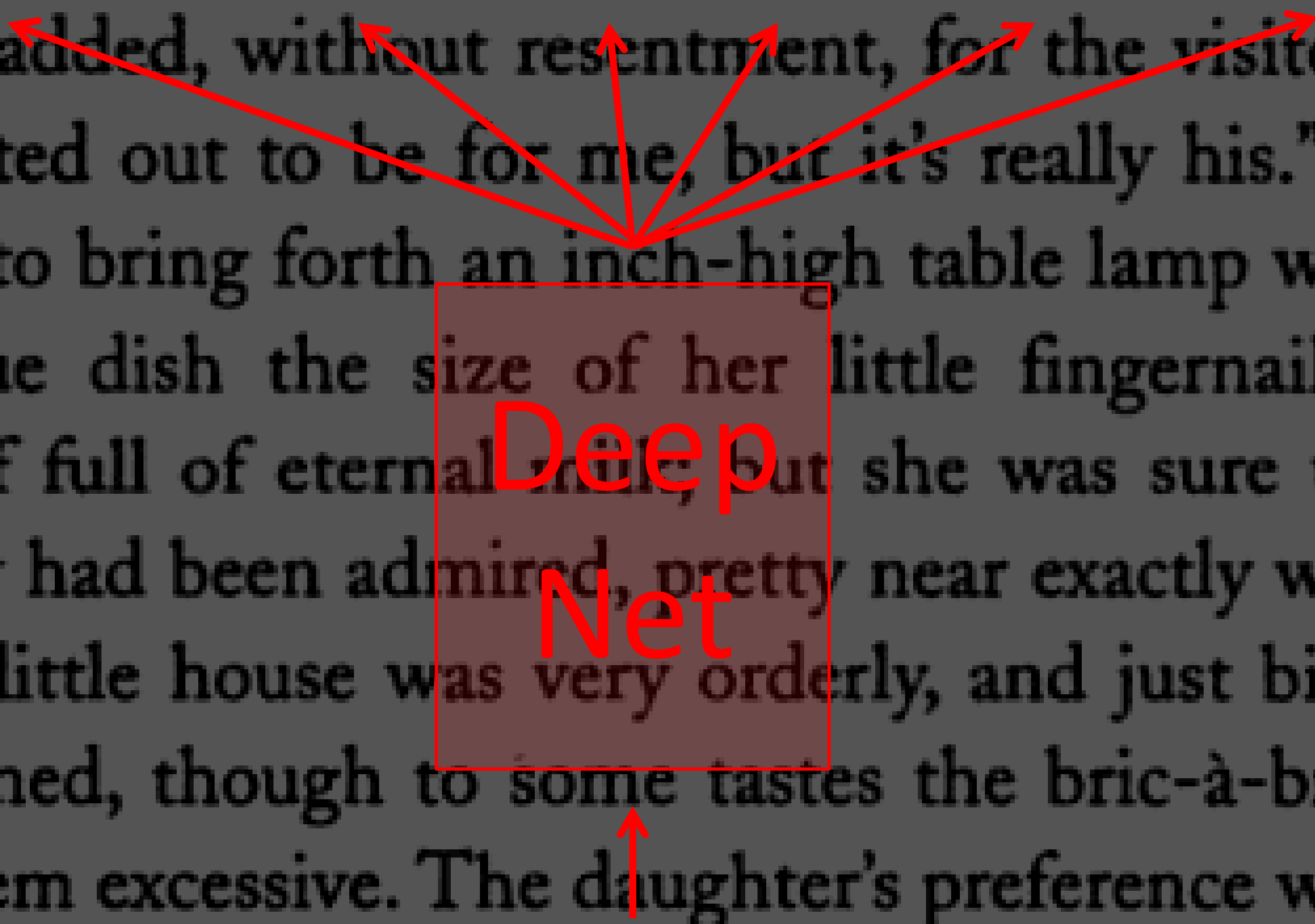
Early one morning the sun was shining I was laying in bed wondering if she had changed at all if her hair was still

0	a
1	able
2	about
	⋮
39	apex
	⋮
56,356	zenith
	⋮
92,487	zygote

word2vec

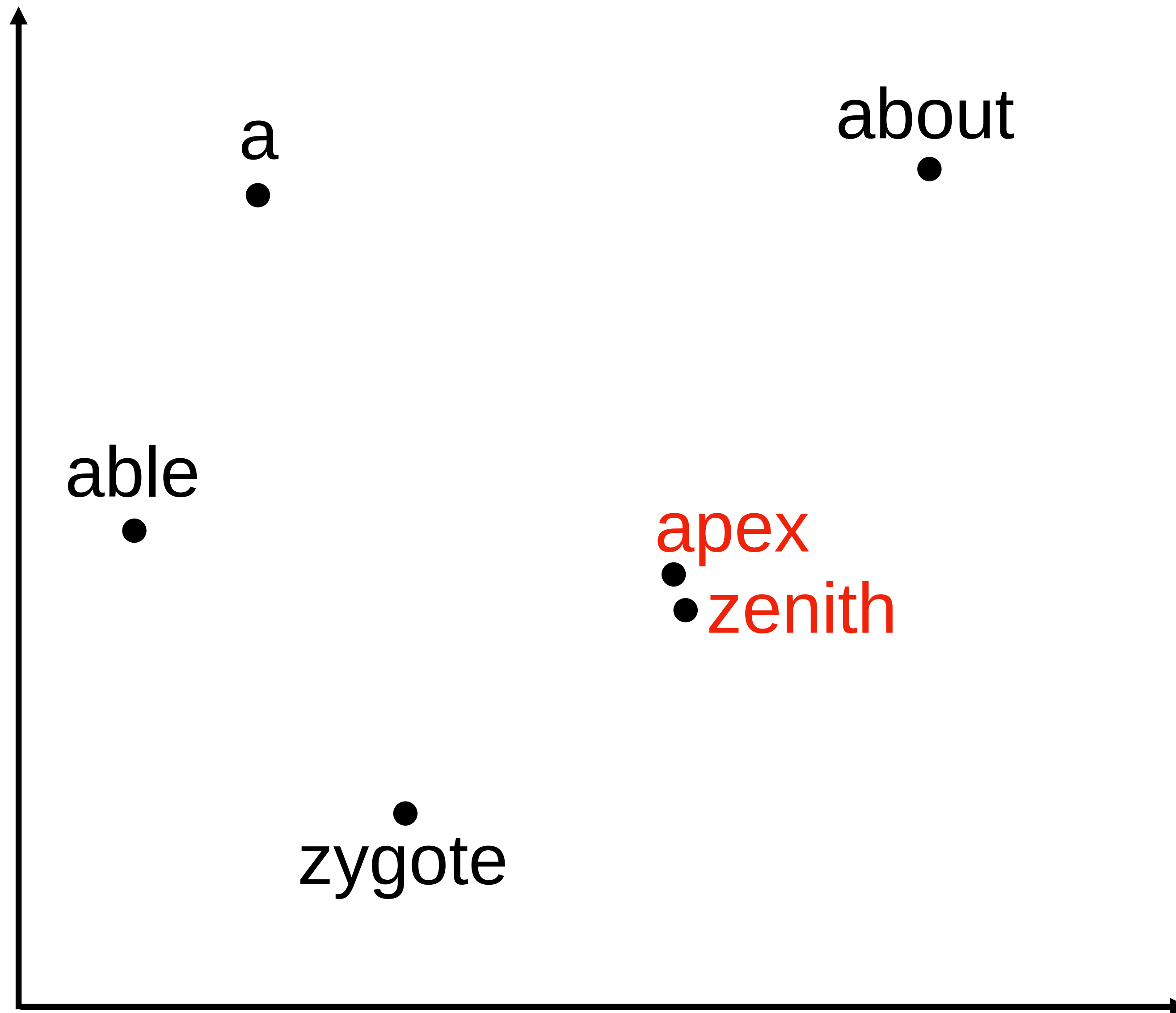
[Collobert & Weston 2008; Mikolov et al. 2013]

house, where the professor lived without his wife and child; or so he said jokingly sometimes: "Here's where I live. My house." His daughter often added, without resentment, for the visitor's information, "It started out to be for me, but it's really his." And she might reach in to bring forth an inch-high table lamp with fluted shade, or a blue dish the size of her little fingernail, marked "Kitty" and half full of eternal milk; but she was sure to replace these, after they had been admired, pretty near exactly where they had been. The little house was very orderly, and just big enough for all it contained, though to some tastes the bric-à-brac in the parlor might seem excessive. The daughter's preference was for the store-bought gimmicks and appliances, the toasters and carpet sweepers of Lilliput, but she knew that most adult visitors would



Deep
Net

The diagram illustrates word embeddings using the words 'Deep' and 'Net' as an example. A semi-transparent red rectangle is centered over the text. The words 'Deep' and 'Net' are written in red inside this rectangle. Five red arrows originate from a single point at the top center of the rectangle and point to the words 'added', 'without', 'resentment', 'for', and 'information' in the text above. A single red arrow originates from the bottom center of the rectangle and points to the word 'daughter's' in the text below. This visualizes how a model can learn the semantic relationship between words by analyzing their context in a sentence.



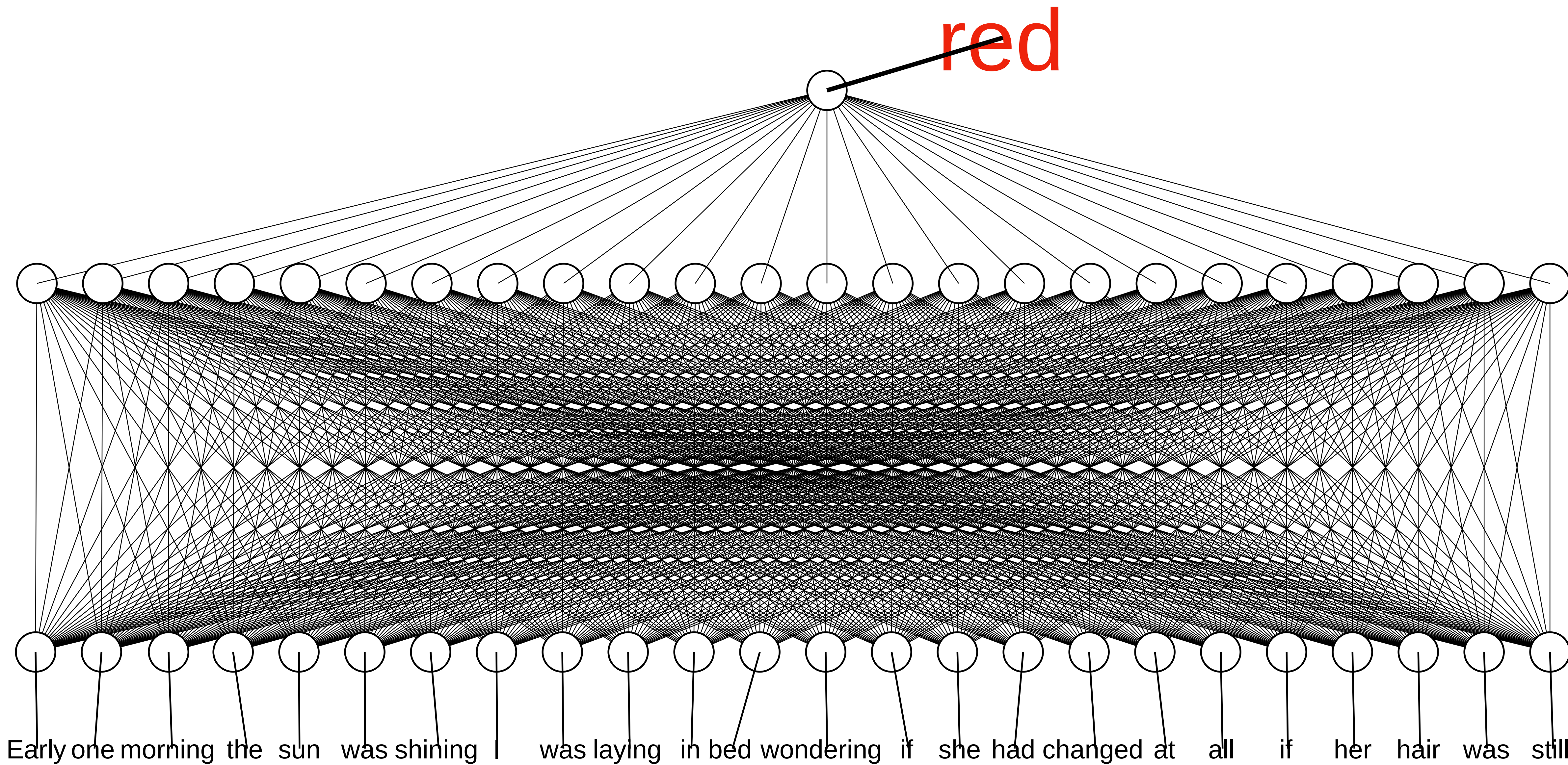
Word Embedding (e.g., word2Vec, GloVe)

slide from Steve Seitz's [video](#)

red

neural network

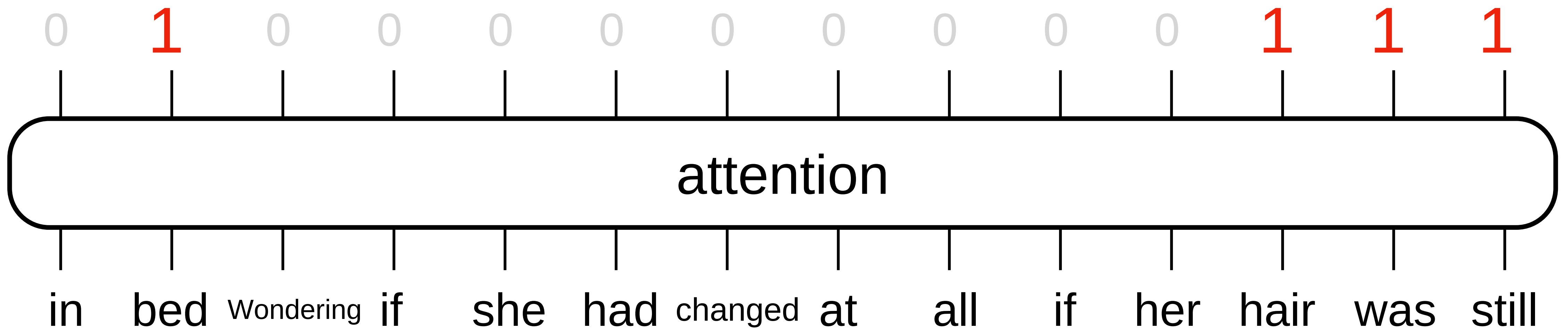
Early one morning the sun was shining I was laying in bed wondering if she had changed at all if her hair was still



Early one morning the sun was shining I was laying in bed
Wondering if she had changed at all if her hair was still ?

bed

hair was still red



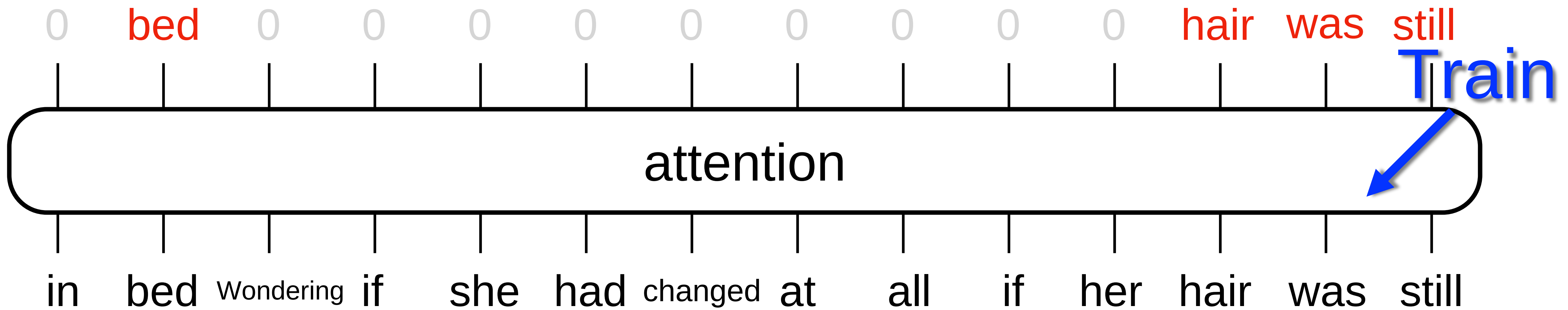
red

next word prediction

0 bed 0 0 0 0 0 0 0 0 0 hair was still

attention

in bed Wondering if she had changed at all if her hair was still



Two roads diverted in a yellow **wood**
And sorry I could not travel **both**
And be one traveler, long I **stood**
And looked down as far as I **could**
To where it bent in the **undergrowth**;

Robert Frost, *Road Not Taken*

slide from Steve Seitz's [video](#)

Train



red

next word prediction

0 bed 0 0 0 0 0 0 0 0 0 hair was still

attention

in bed Wondering if she had changed at all if her hair was still

brown

Train



next word prediction

0 bed 0 0 0 0 0 0 0 0 0 hair was still

attention

in bed Wondering if she had changed at all if her hair was still

brown

Train

next word prediction

0 bed 0 0 0 0 0 0 0 0 0 hair was still

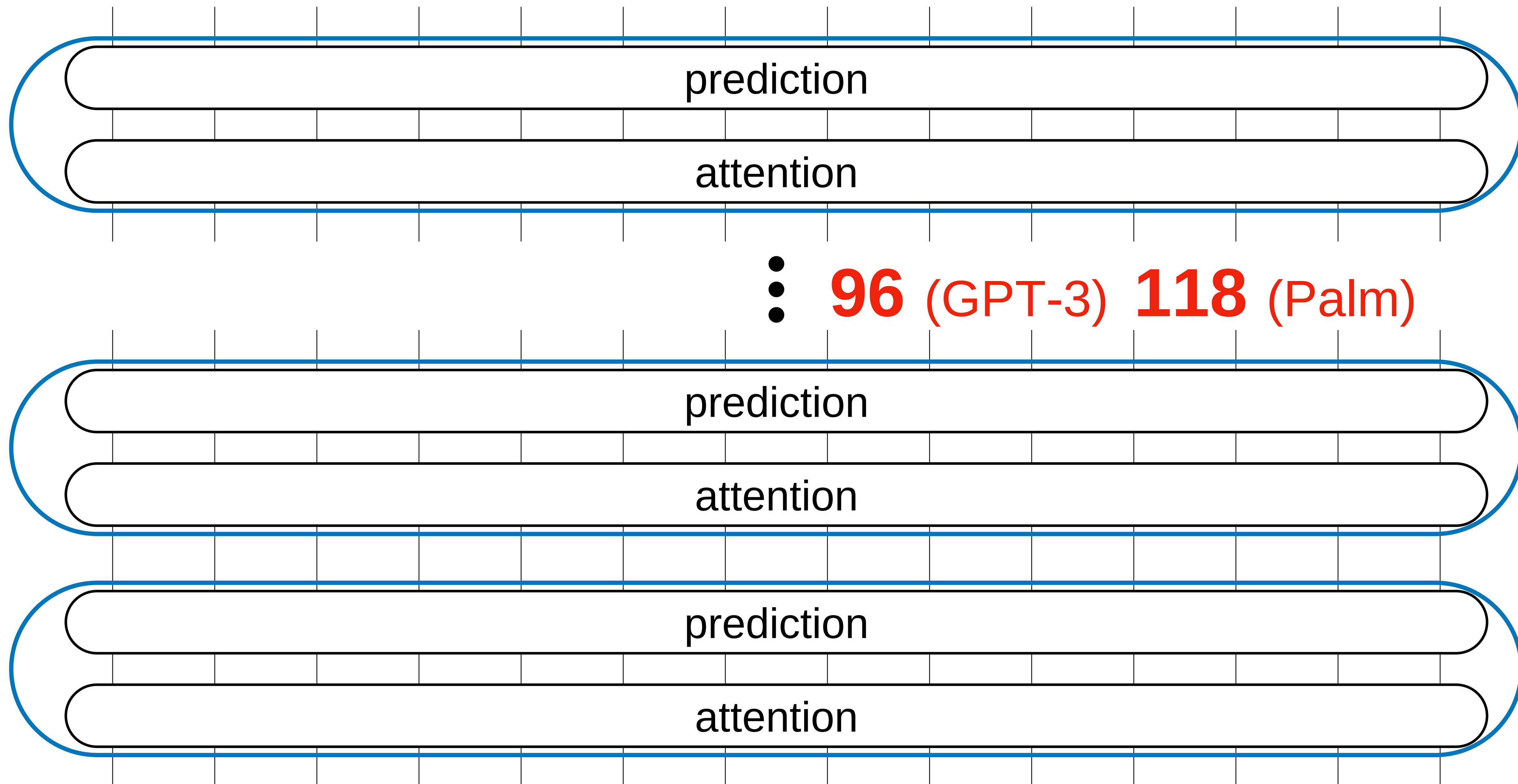
attention

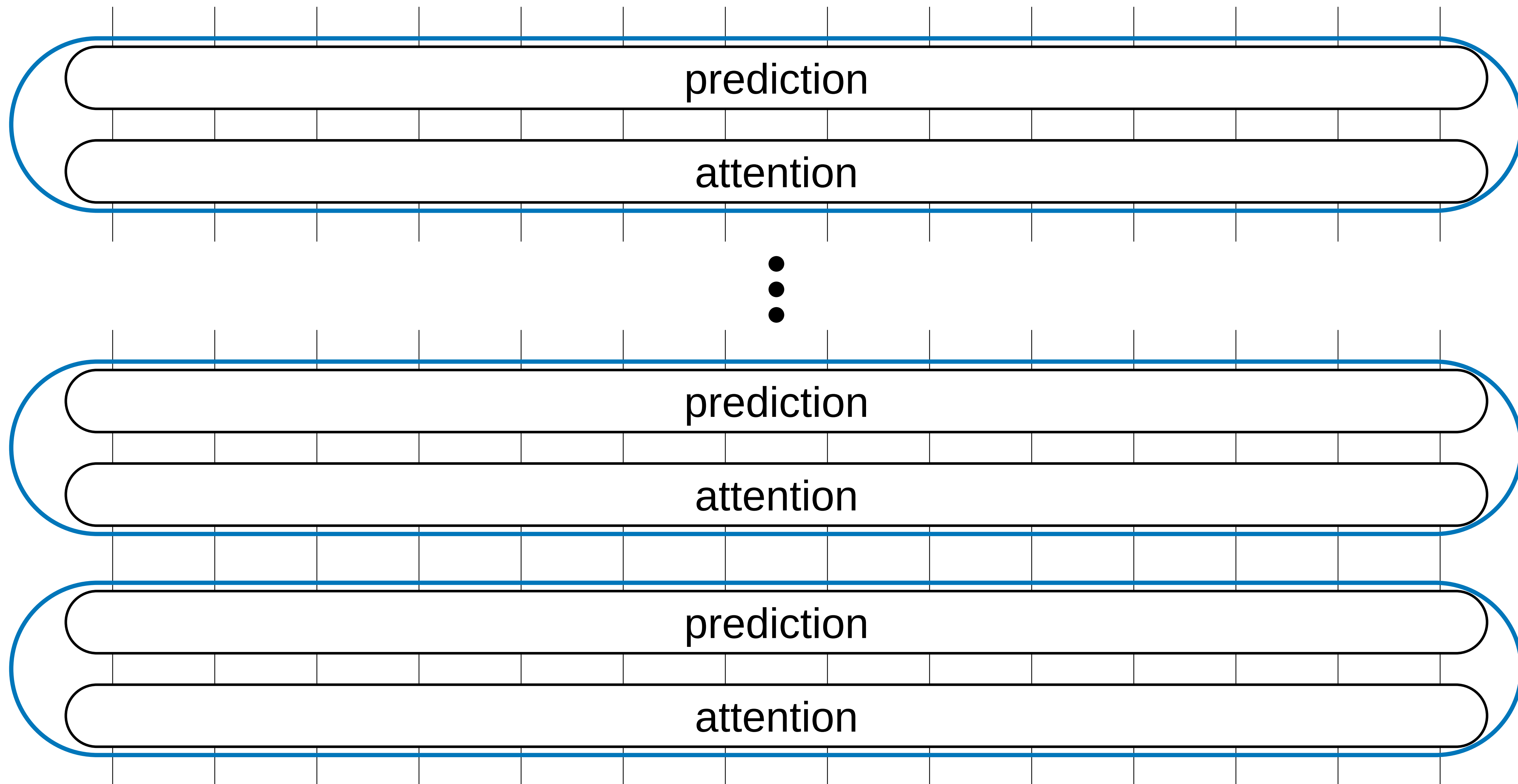
in bed Wondering if she had changed at all if her hair was still

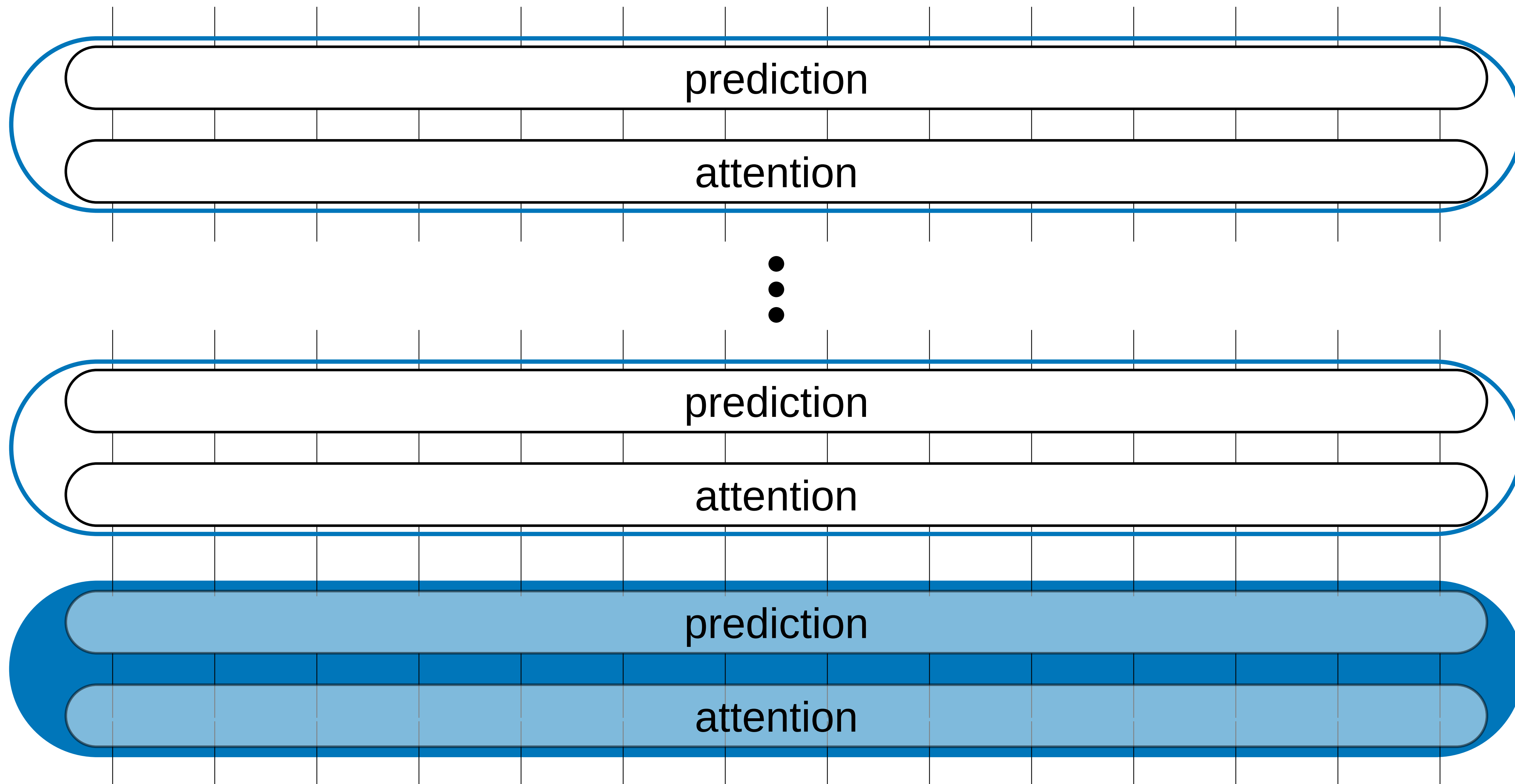
red

Transformer

in bed Wondering if she had changed at all if her hair was still







Syntax

slide from Steve Seitz's [video](#)

Semantics

prediction

attention

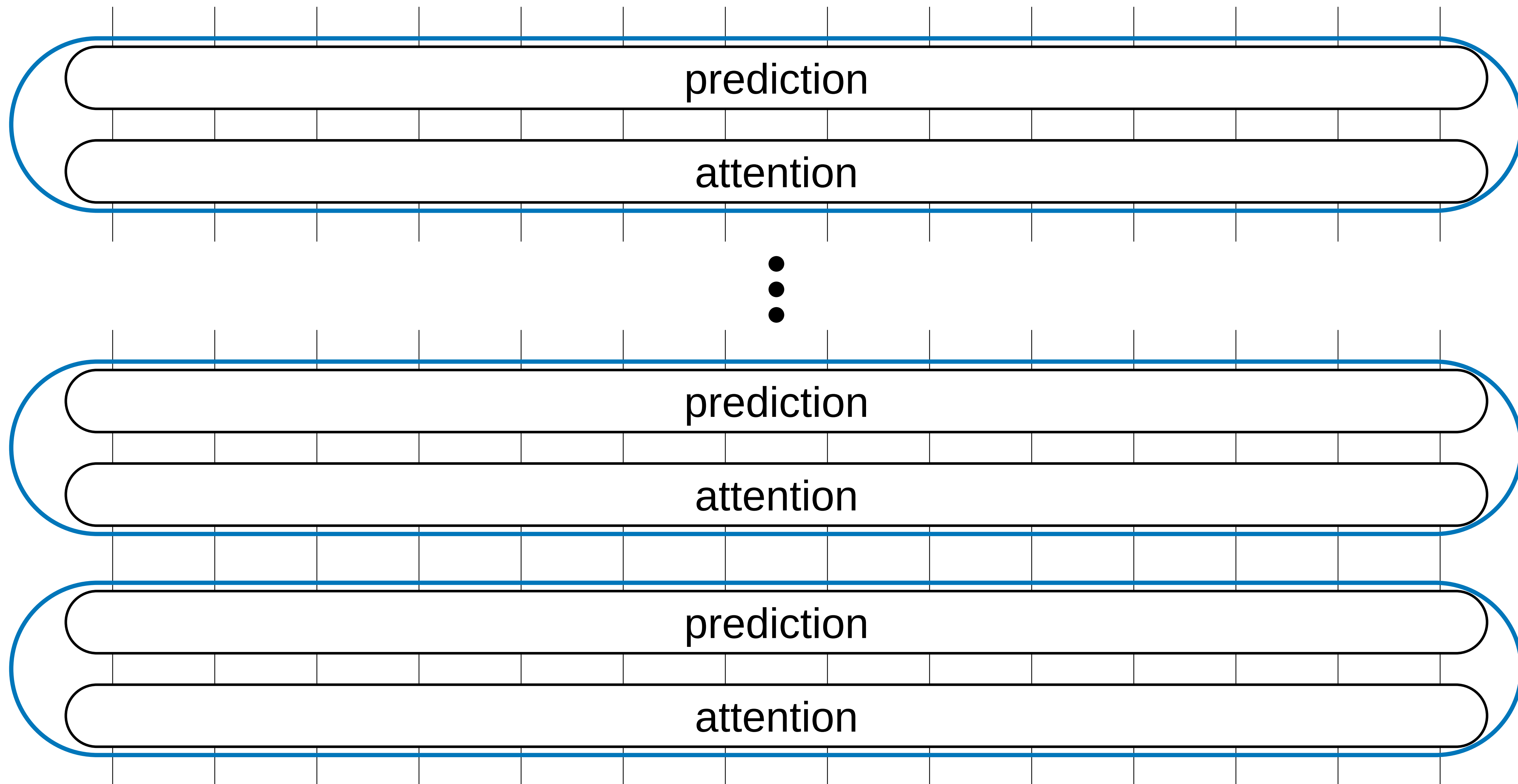
⋮

prediction

attention

prediction

attention



How much data
to train?

All of it...

355 years

a month

The 16th President was

The capital of Zimbabwe is

Frank Zappa's middle name is

Napoleon was born on this date

The prime factorization of 19456721434 is

Queen Victoria's maiden name was

US per-capita income in 1957 was

The lat long coordinates of Rome are

The 16th President was Abraham Lincoln

The capital of Zimbabwe is Harare

Frank Zappa's middle name is Vincent

Napoleon was born on this date 1769

The prime factorization of 19456721434 is $2 \times 3 \times 3 \times 17$

Queen Victoria's maiden name was Alexandrina Victoria

US per-capita income in 1957 was \$2,974

The lat long coordinates of Rome are 41.894722, 12.48

a pattern of characters that looks like a star

• ○ •
○ ○ ○
• ○ •

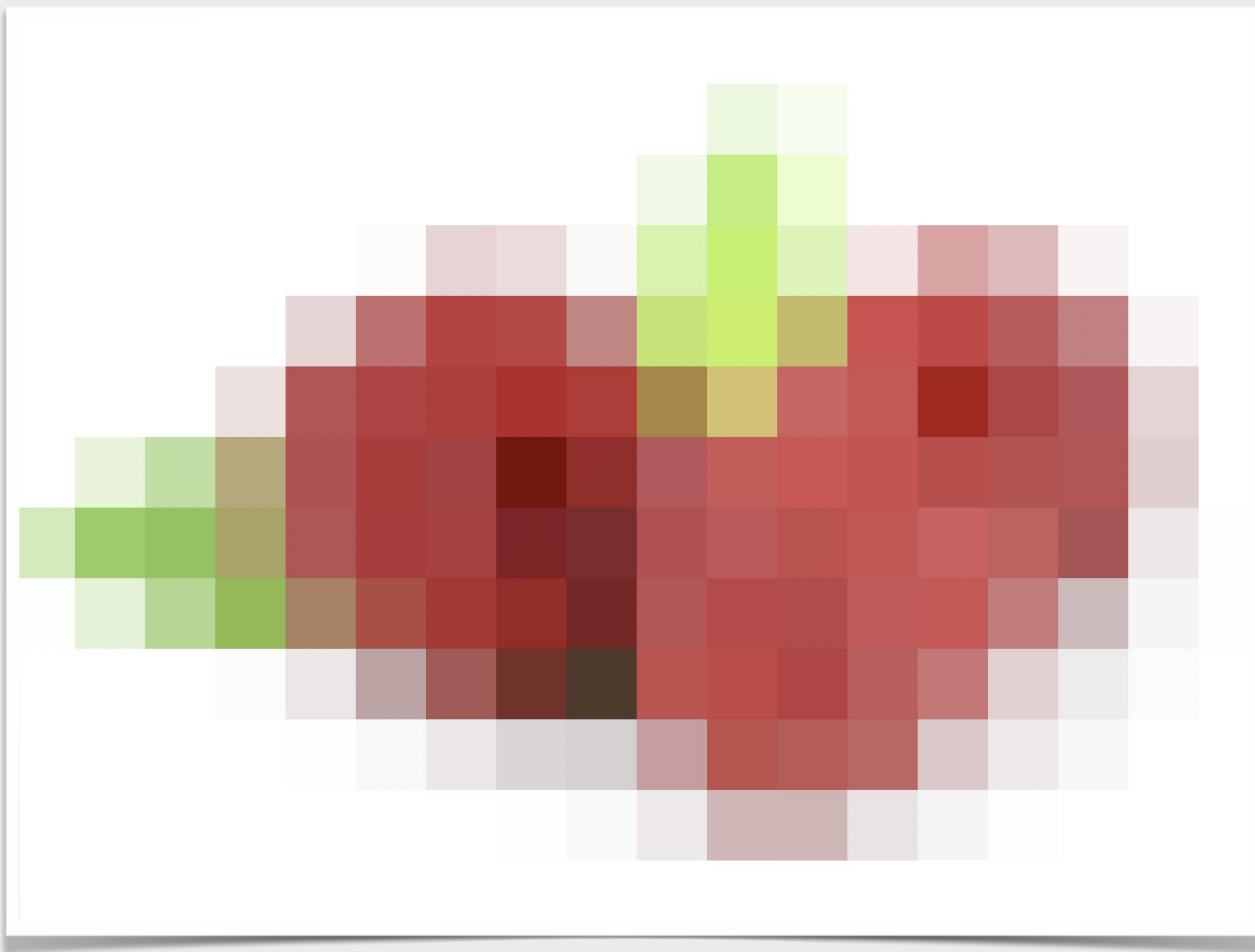
a pattern of characters that looks like a vertical line

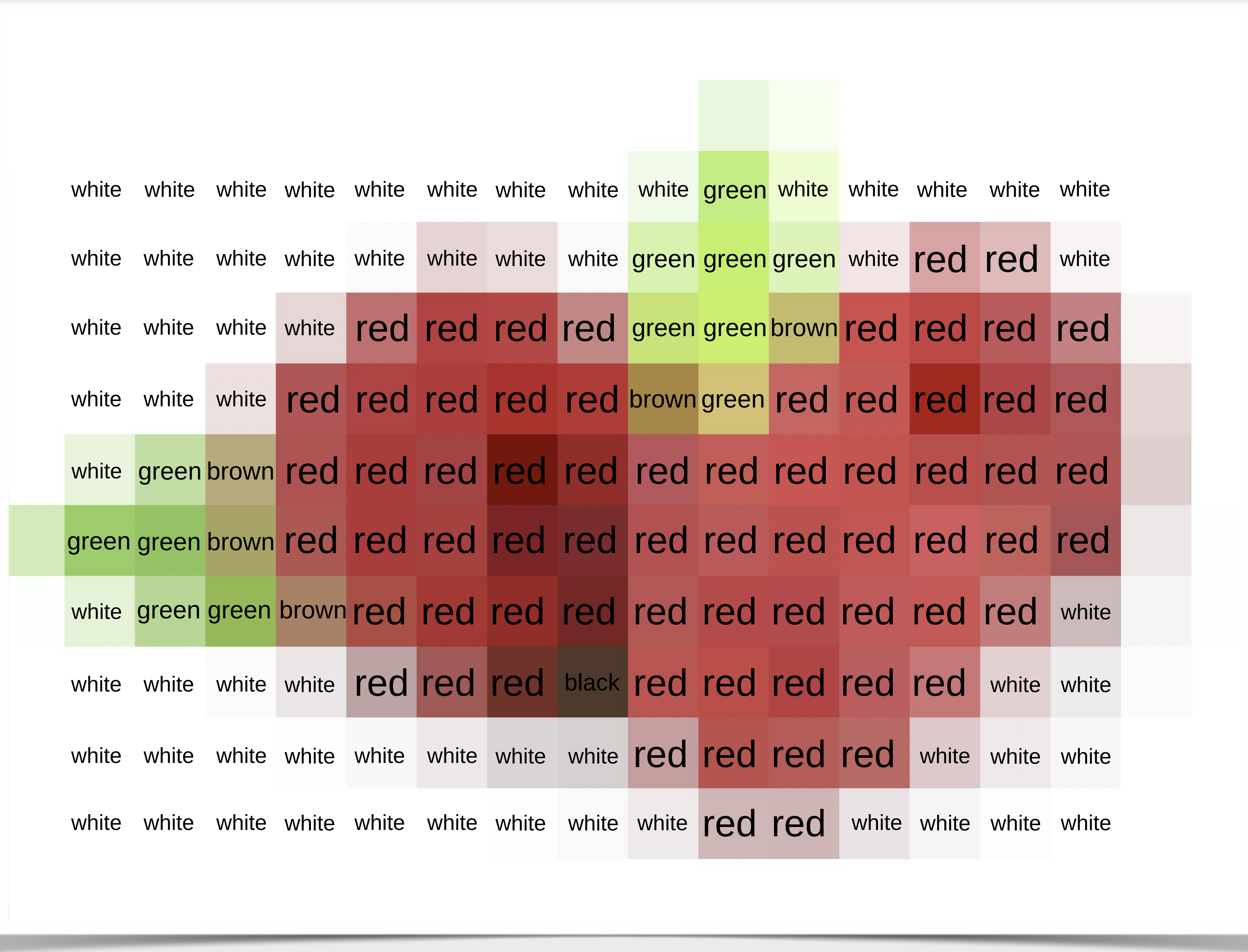
• ○ •
• ○ •
• ○ •
• ○ •
• ○ •

a pattern of characters that looks like a triangle

. ○ .
. ○ ○ .
. ○ ○ ○ .
. ○ ○ ○ ○ .
. ○ ○ ○ ○ ○ .







white white white white white white white white white green white white white white
white white white white white white white white green green green white red red white
white white white white red red red red green green brown red red red red
white white white red red red red red brown green red red red red red red
white green brown red red red red red red red red red red red red red
green green brown red red red red red red red red red red red red red
white green green brown red red red red red red red red red red red white
white white white white red red red black red red red red red white white
white white white white white white white white red red red red white white white
white white white white white white white white white white red red white white white white

(255,0,0)

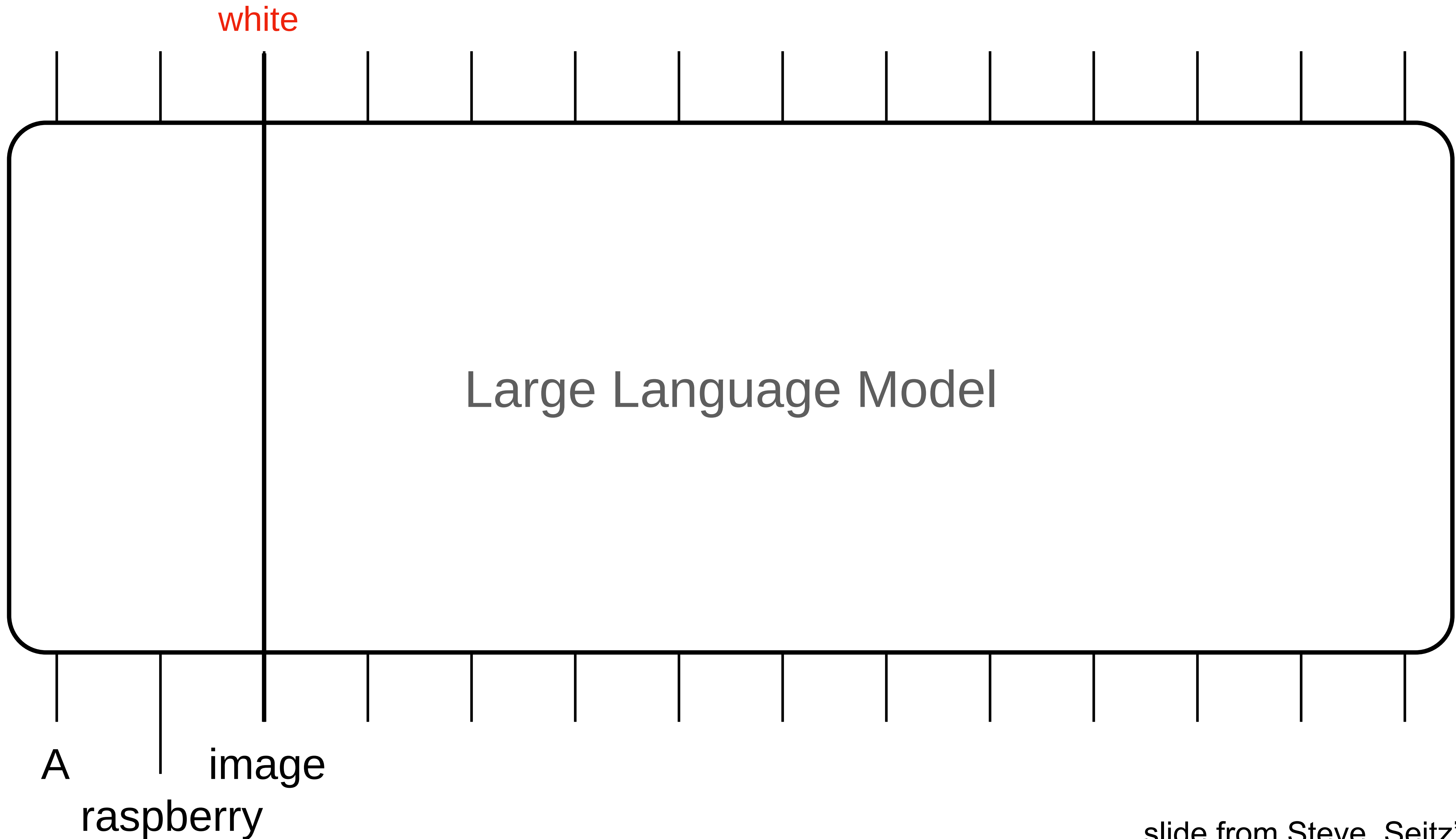
white white white white white white white white white green white white white white white
white white white white white white white white green green green white red red white
white white white white red red red red green green brown red red red red
white white white red red red red red brown green red red red red red red
white green brown red red red red red red red red red red red red red
green green brown red red red red red red red red red red red red red
white green green brown red red red red red red red red red red red white
white white white white red red red black red red red red red white white
white white white white white white white white red red red red white white white
white white white white white white white white white white red red white white white white

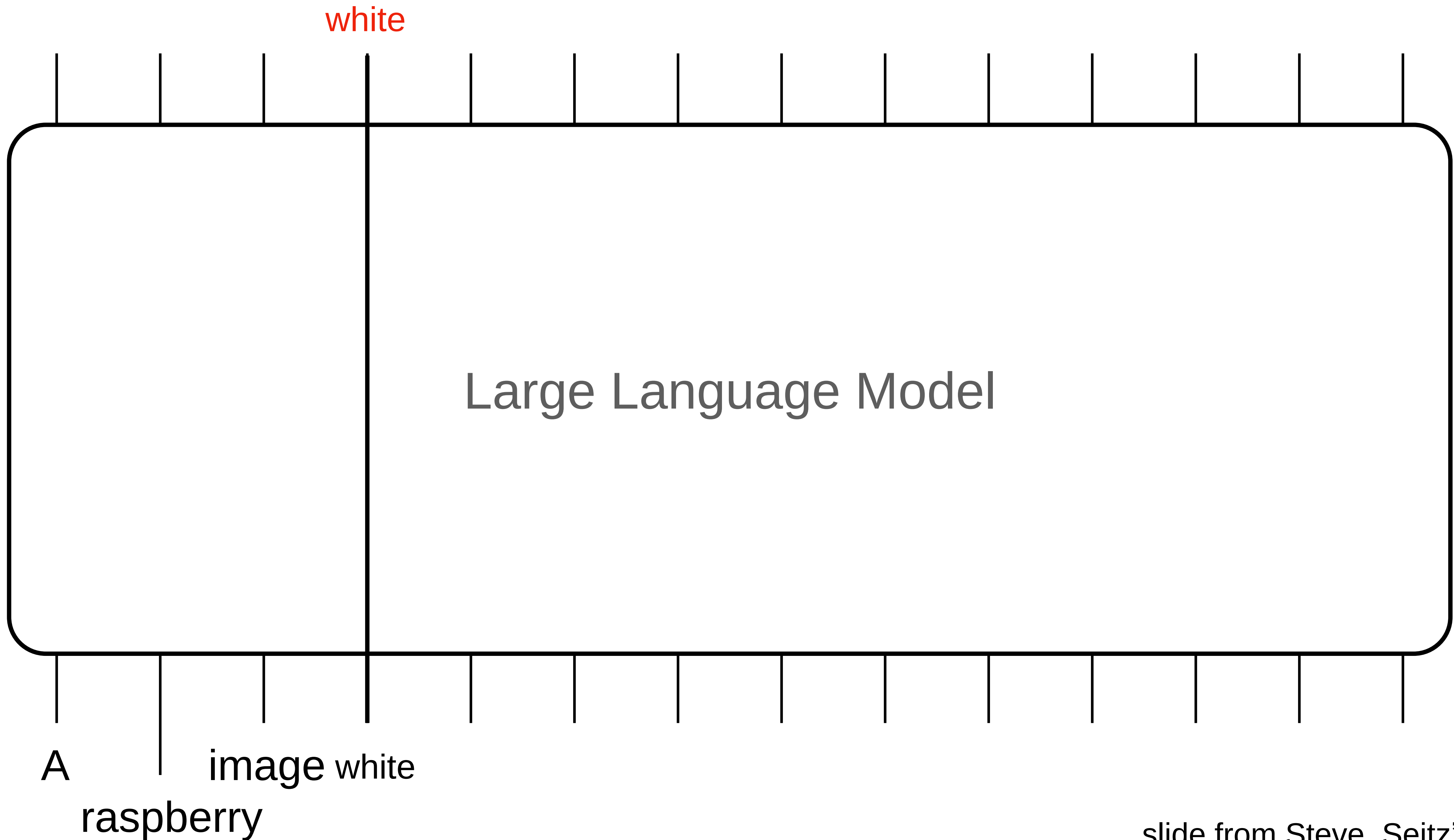


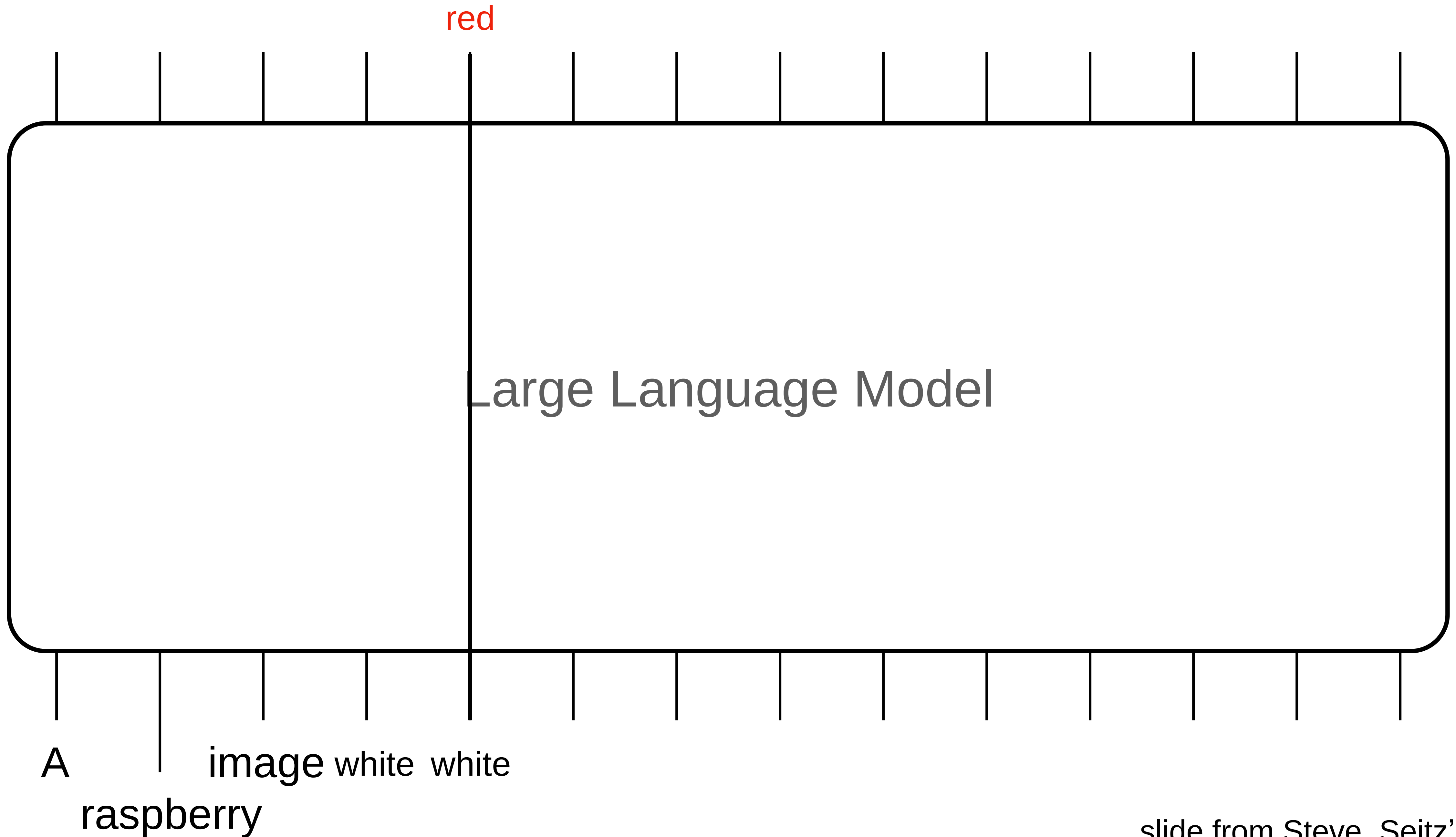
raspberries pancakes sunsets

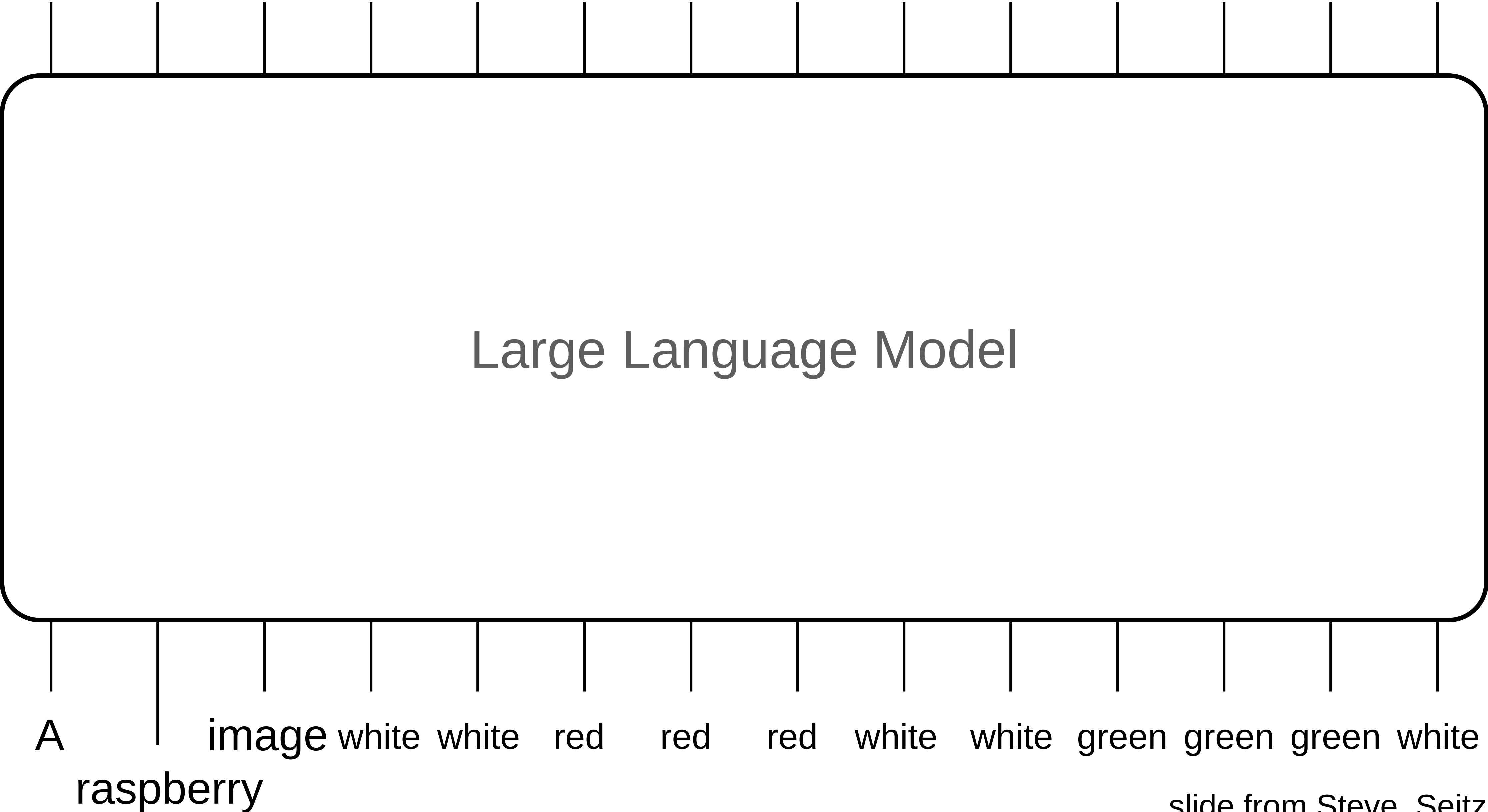
1 Billion

slide from Steve Seitz's [video](#)









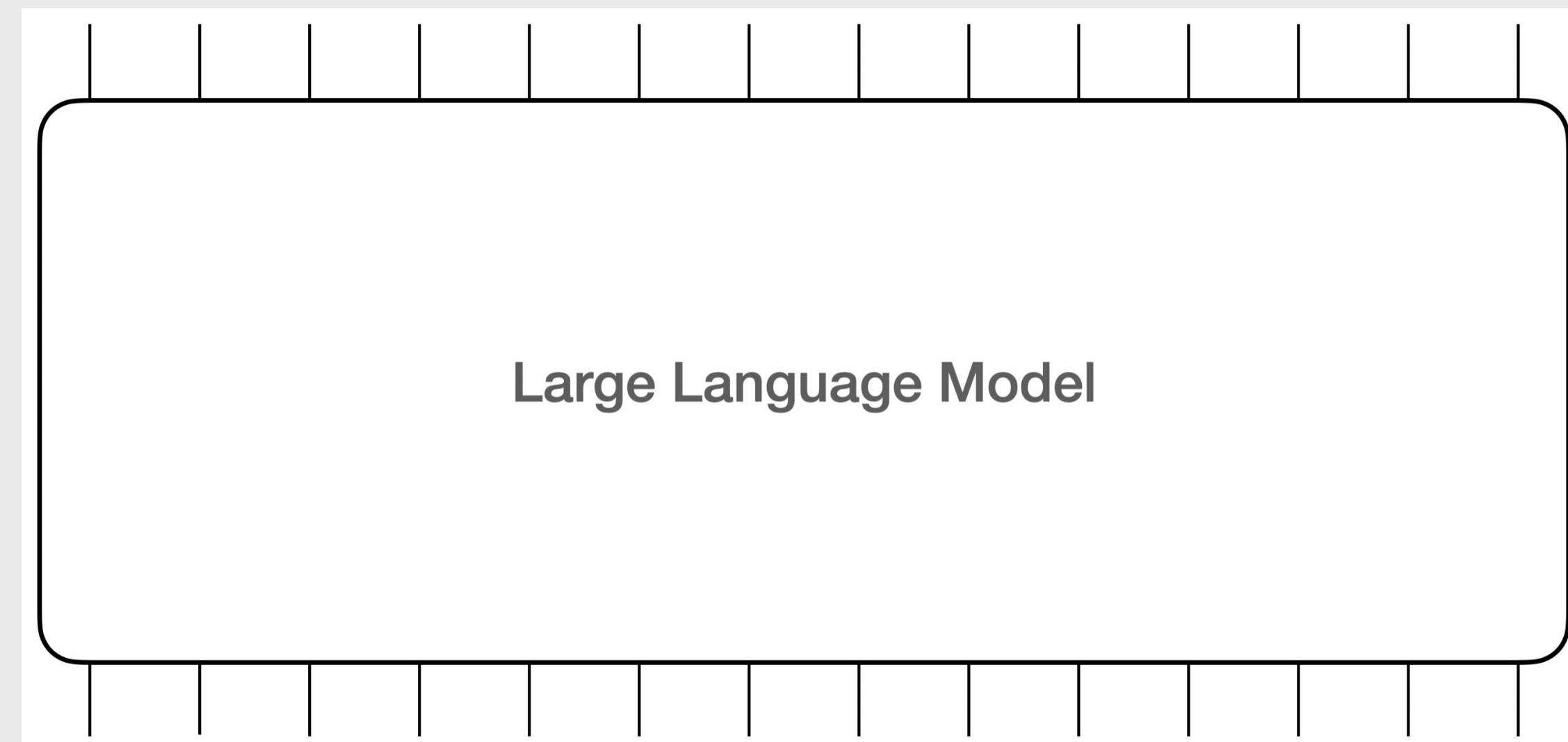
Large Language Model

A image white white red red red white white green green green white
raspberry

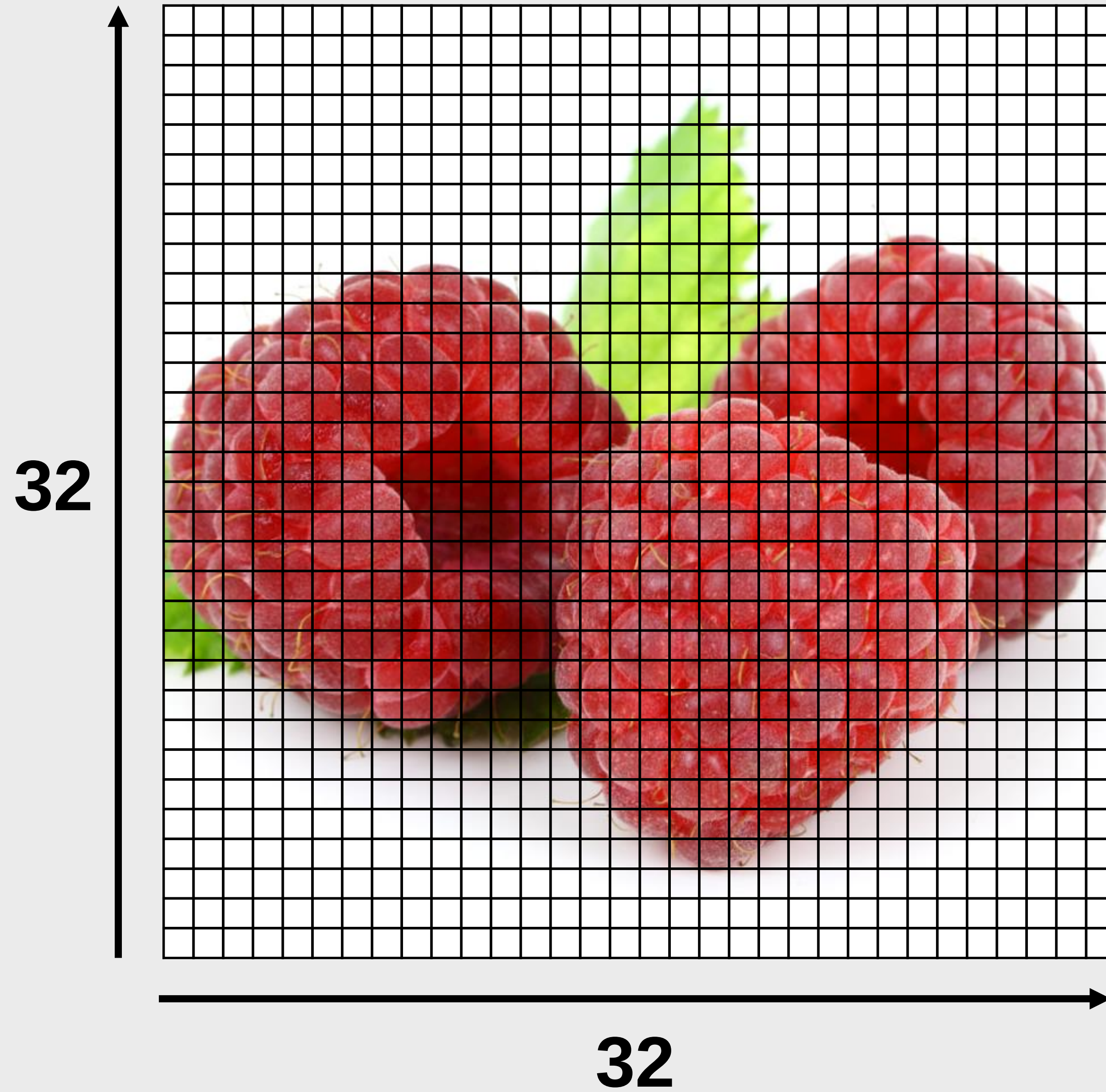
slide from Steve Seitz's [video](#)



1,000,000s of pixels



1,000s of words



$$32 \times 32 = 1024$$

Visual words

[illegible]

squirrel reaching for a nut

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 6 6 6 1 1 6 6 4 4 4 9 9 9 9 9 9
1 1 1 1 1 7 7 1 1 1 1 1 1 1 1 1

squirrel reaching for a nut

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	1	1	6	6	4	4	4	9	9	9	9	9	9
1	1	1	1	1	7	7	1	1	1	1	1	1	1	1	1																

squirrel reaching for a nut

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	1	1	6	6	4	4	4	9	9	9	9	9	9
1	1	1	1	1	7	7	1	1	1	1	1	1	1	1	1																

squirrel reaching for a nut

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	1	1	6	6	4	4	4	9	9	9	9	9	9
1	1	1	1	1	7	7	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	

squirrel reaching for a nut

1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	1	1	6	6	4	4	4	9	9	9	9	9	9
1	1	1	1	1	7	7	1	1	1	1	1	1	1	1	6														

squirrel reaching for a nut

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	1	1	6	6	4	4	4	9	9	9	9	9	9
1	1	1	1	1	7	7	1	1	1	1	1	1	1	1	1	6															

squirrel reaching for a nut

1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	6	6	6	1	1	6	6	4	4	4	9	9	9	9	9	9
1	1	1	1	1	7	7	1	1	1	1	1	1	1	1	1	6															

squirrel reaching for a nut

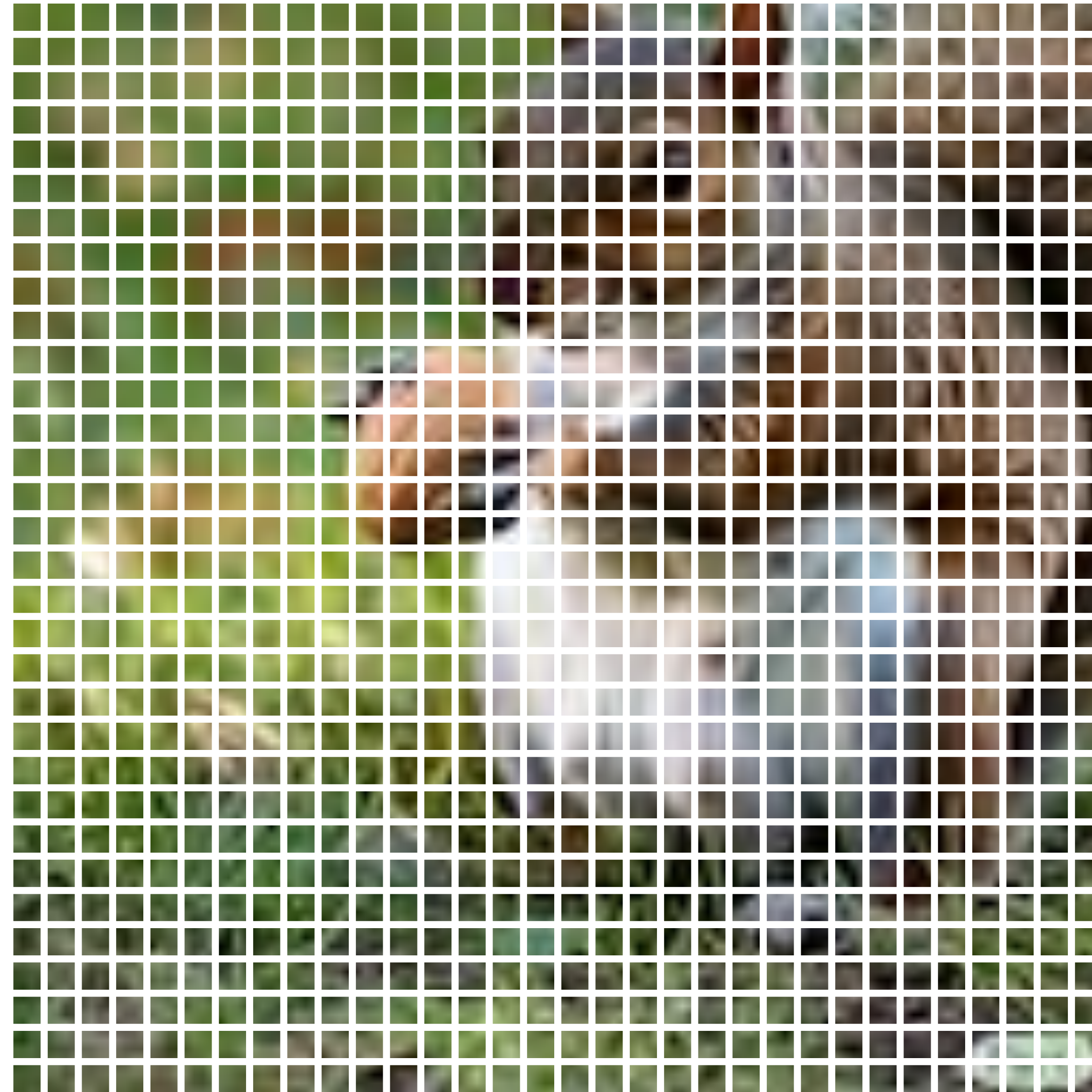
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 6 6 6 1 1 6 6 4 4 4 9 9 9 9 9 9
1 1 1 1 1 7 7 1 1 1 1 1 1 1 1 1 6

squirrel reaching for a nut

```
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 6 6 6 1 1 6 6 4 4 4 9 9 9 9 9 9
1 1 1 1 1 7 7 1 1 1 1 1 1 1 1 6 6 6 6 6 6 6 4 1 4 9 9 9 9 9 9
1 1 7 1 1 1 7 1 1 1 1 1 1 1 6 6 6 6 6 6 6 4 1 4 9 9 9 9 9 9
1 1 7 1 1 1 1 1 1 1 1 1 1 2 6 2 6 6 2 2 6 2 9 9 9 9 9 9 9 9
1 1 1 7 7 1 1 1 1 1 1 1 1 6 6 6 6 2 5 2 2 4 9 9 9 9 9 9 9 9
1 1 1 7 1 1 1 1 1 1 1 1 2 6 6 6 2 5 2 2 0 9 9 9 9 9 9 9 9 9
1 1 1 1 1 1 2 2 2 2 2 2 1 2 6 6 6 2 6 2 6 2 0 9 9 9 9 9 9 9
2 1 1 1 1 2 2 1 2 2 2 2 1 2 6 6 6 6 6 2 6 4 9 9 9 9 9 9 8 8 8
2 2 1 1 1 1 1 1 1 1 1 1 1 6 6 2 6 6 6 4 4 9 9 9 9 9 9 9 8 8
1 1 1 1 1 1 1 1 1 1 1 1 1 6 6 0 0 0 0 0 4 9 9 9 9 9 9 9 9 8
1 1 1 1 1 1 1 1 1 7 1 5 2 2 2 0 0 0 0 0 4 9 9 9 9 9 9 9 9 8
1 1 1 1 1 1 1 1 1 1 5 2 2 2 2 0 0 0 0 4 4 6 9 9 9 9 9 9 9 9
1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 5 2 0 5 4 6 9 9 9 9 9 9 9 9 9
1 1 1 1 7 1 1 1 1 1 2 2 6 6 5 5 2 9 9 9 9 9 9 9 9 9 9 9 9 9
1 1 1 1 7 2 2 2 1 1 2 2 6 5 5 0 2 9 9 9 9 9 9 9 9 9 9 9 9 9
1 1 1 7 2 2 2 2 1 1 2 6 6 5 5 0 0 4 4 4 9 9 0 0 0 0 9 9 9 9 9
1 1 0 1 2 1 2 1 1 1 1 1 1 0 0 0 0 4 0 0 0 4 0 0 0 0 9 9 9 9 8
1 1 1 1 1 1 1 1 1 1 1 7 1 1 0 0 0 0 0 0 0 0 0 0 0 0 9 9 9 8
1 1 1 1 1 1 1 1 1 7 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 9 9 9 8 8
1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 4 4 9 9 9 8 9
1 1 1 1 1 7 7 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 4 4 9 9 5 5 9
1 1 1 1 1 7 7 7 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 4 4 9 9 5 4 3
1 1 1 1 1 1 7 7 3 3 3 3 3 3 4 0 4 0 0 4 0 0 0 0 4 4 9 9 5 4 4
3 3 3 3 3 3 3 3 3 3 3 3 3 3 4 5 0 0 0 0 0 0 0 0 4 9 9 9 4 3 3
3 3 3 3 3 3 3 3 3 3 4 4 4 3 3 3 5 3 0 4 4 4 4 4 4 9 9 9 4 3 8
3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 8 3 4 4 4 4 4 9 3 9 4 8 8
3 3 8 3 3 3 3 3 3 3 3 3 3 8 3 8 8 3 3 8 3 8 4 4 8 4 8 3 8 3 3 3
8 8 3 8 8 3 8 3 3 3 8 3 8 3 3 3 3 3 8 3 8 4 8 8 8 3 3 3 3 3 3
3 3 8 3 3 3 3 3 3 3 8 8 8 8 3 3 8 8 3 3 8 8 8 8 8 8 3 8 3 8
3 3 8 8 3 3 3 8 3 3 3 3 3 3 3 8 8 3 3 8 3 3 3 8 3 8 8 8 3 8
3 3 8 8 3 3 3 8 8 3 8 3 3 3 3 3 3 3 3 3 3 3 3 8 3 8 8 8 8 8
3 3 8 3 3 3 3 8 3 8 8 3 3 3 3 3 3 3 3 3 3 3 3 3 8 8 8 8 8 8
```

squirrel reaching for a nut

```
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 6 6 6 1 1 6 6 4 4 4 9 9 9 9 9 9
1 1 1 1 1 7 7 1 1 1 1 1 1 1 1 6 6 6 6 6 6 6 4 1 4 9 9 9 9 9 9
1 1 7 1 1 1 7 1 1 1 1 1 1 1 6 6 6 6 6 6 6 4 1 4 9 9 9 9 9 9
1 1 7 1 1 1 1 1 1 1 1 1 1 2 6 2 6 6 2 2 6 2 9 9 9 9 9 9 9 9
1 1 1 7 7 1 1 1 1 1 1 1 1 6 6 6 6 2 5 2 2 4 9 9 9 9 9 9 9 9
1 1 1 7 1 1 1 1 1 1 1 1 2 6 6 6 2 5 2 2 0 9 9 9 9 9 9 9 9 9
1 1 1 1 1 1 2 2 2 2 2 2 1 2 6 6 6 2 6 2 6 2 0 9 9 9 9 9 9 9
2 1 1 1 1 2 2 1 2 2 2 2 1 2 6 6 6 6 6 2 6 4 9 9 9 9 9 9 8 8 8
2 2 1 1 1 1 1 1 1 1 1 1 1 6 6 2 6 6 6 4 4 9 9 9 9 9 9 9 8 8
1 1 1 1 1 1 1 1 1 1 1 1 1 6 6 0 0 0 0 0 4 9 9 9 9 9 9 9 9 8
1 1 1 1 1 1 1 1 1 7 1 5 2 2 2 0 0 0 0 0 4 9 9 9 9 9 9 9 9 8
1 1 1 1 1 1 1 1 1 1 5 2 2 2 2 0 0 0 0 4 4 6 9 9 9 9 9 9 9 9
1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 5 2 0 5 4 6 9 9 9 9 9 9 9 9 9
1 1 1 1 7 1 1 1 1 1 2 2 6 6 5 5 2 9 9 9 9 9 9 9 9 9 9 9 9 9
1 1 1 1 7 2 2 2 1 1 2 2 6 5 5 0 2 9 9 9 9 9 9 9 9 9 9 9 9 9
1 1 1 7 2 2 2 2 1 1 2 6 6 5 5 0 0 4 4 4 9 9 0 0 0 0 9 9 9 9 9
1 1 0 1 2 1 2 1 1 1 1 1 1 0 0 0 0 4 0 0 0 4 0 0 0 0 9 9 9 9 8
1 1 1 1 1 1 1 1 1 1 1 7 1 1 0 0 0 0 0 0 0 0 0 0 0 0 9 9 9 8
1 1 1 1 1 1 1 1 1 7 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 9 9 9 8 8
1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 4 4 9 9 9 8 9
1 1 1 1 1 7 7 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 4 4 9 9 5 5 9
1 1 1 1 1 7 7 7 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 4 4 9 9 5 4 3
1 1 1 1 1 1 7 7 3 3 3 3 3 3 4 0 4 0 0 4 0 0 0 0 4 4 9 9 5 4 4
3 3 3 3 3 3 3 3 3 3 3 3 3 3 4 5 0 0 0 0 0 0 0 0 4 9 9 9 4 3 3
3 3 3 3 3 3 3 3 3 3 4 4 4 3 3 3 5 3 0 4 4 4 4 4 4 9 9 9 4 3 8
3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 8 3 4 4 4 4 4 9 3 9 4 8 8
3 3 8 3 3 3 3 3 3 3 3 3 3 8 3 8 8 3 3 8 3 8 4 4 8 4 8 3 8 3 3 3
8 8 3 8 8 3 8 3 3 3 8 3 8 3 3 3 3 3 8 3 8 4 8 8 8 3 3 3 3 3 3
3 3 8 3 3 3 3 3 3 3 8 8 8 8 3 3 8 8 3 3 8 8 8 8 8 8 3 8 3 8
3 3 8 8 3 3 3 8 3 3 3 3 3 3 3 8 8 3 3 8 3 3 3 8 3 8 8 8 3 8
3 3 8 8 3 3 3 8 8 3 8 3 3 3 3 3 3 3 3 3 3 3 3 8 3 8 8 8 8 8
3 3 8 3 3 3 3 8 3 8 8 3 3 3 3 3 3 3 3 3 3 3 3 3 3 8 8 8 8 8
```

squirrel reaching for a nut



squirrel reaching for a nut



**Up-sampled
4x**

squirrel reaching for a nut

slide from Steve Seitz's [video](#)



squirrel reaching for a nut

Parti, <https://parti.research.google/>



squirrel reaching for a nut underwater

slide from Steve Seitz's [video](#)



fossil of a squirrel reaching for a nut

slide from Steve Seitz's [video](#)



squirrel made of toothpicks wearing sunglasses reaching for a nut
slide from Steve Seitz's [video](#)



DLSR photograph of a whimsical fantasy house shaped like a squirrel
with windows and a door, in the forest

slide from Steve Seitz's [video](#)



Squirrel reaching for a nut. by Leonardo da Vinci

slide from Steve Seitz's [video](#)

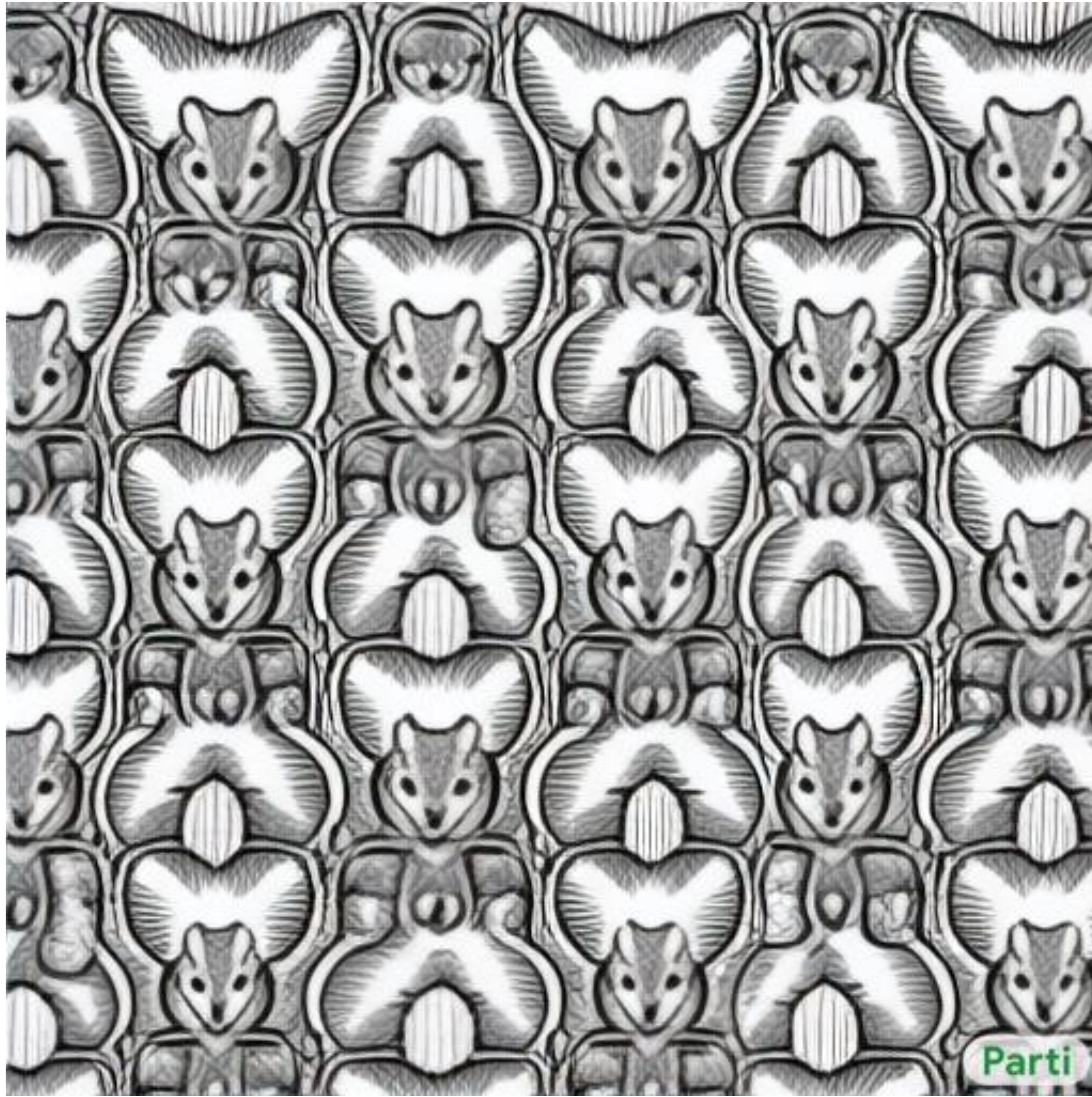


Squirrel reaching for a nut. Van Gogh painting

slide from Steve Seitz's [video](#)



Intricately carved cathedral door of a squirrel reaching for a nut
slide from Steve Seitz's [video](#)



Squirrel reaching for a nut. Woodcut tessellation pattern by M.C. Escher
slide from Steve Seitz's [video](#)



Squirrel reaching for a nut. Latte art

slide from Steve Seitz's [video](#)

Algorithm vs. Data

Diffusion-based

Auto-regressive

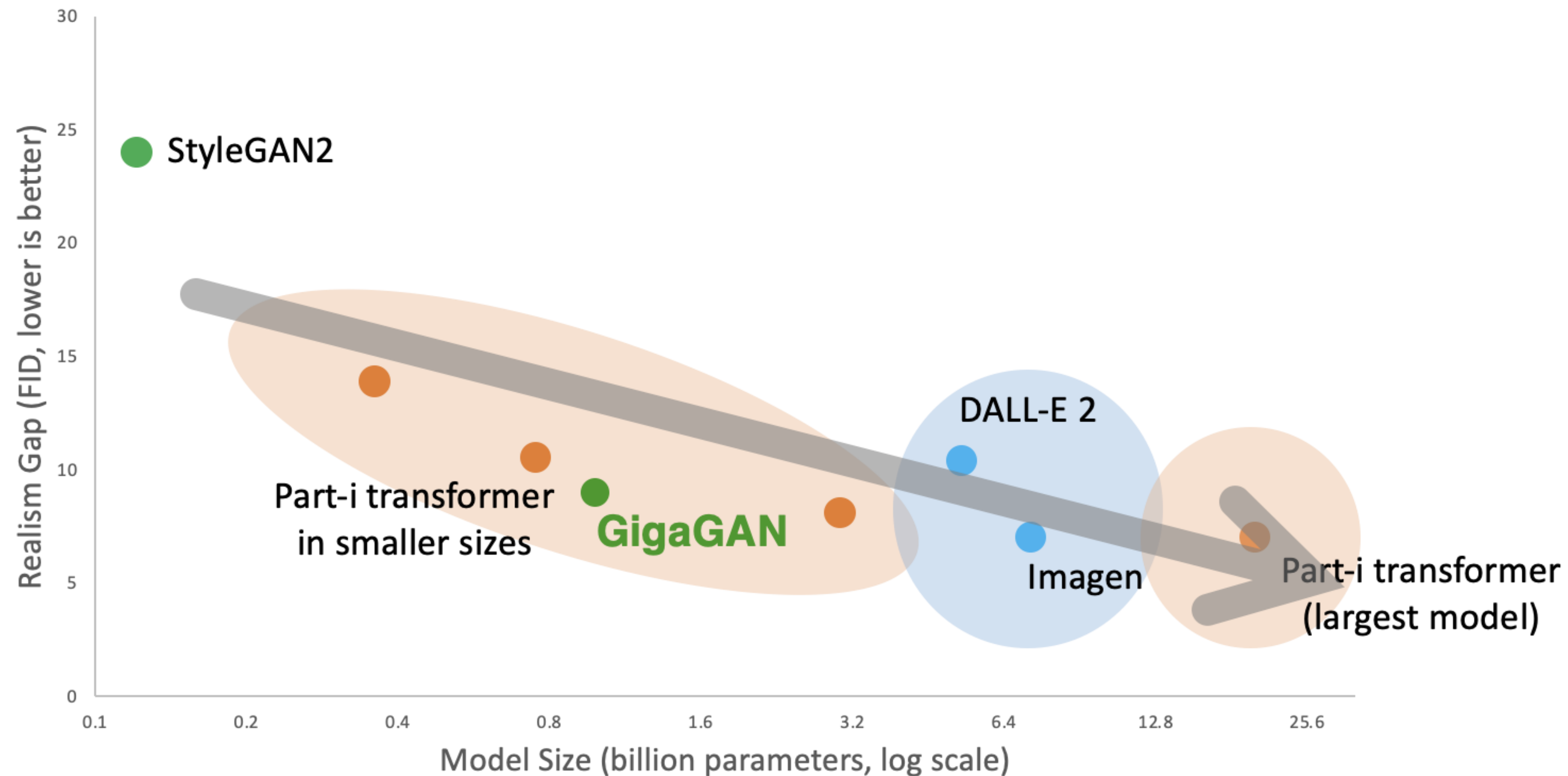
GAN-based



Prompt: *“squirrel reaching for a nut”*

data capacity vs. image quality

Larger Models Attain Better Realism



Graph by Taesung Park

Generative Magic

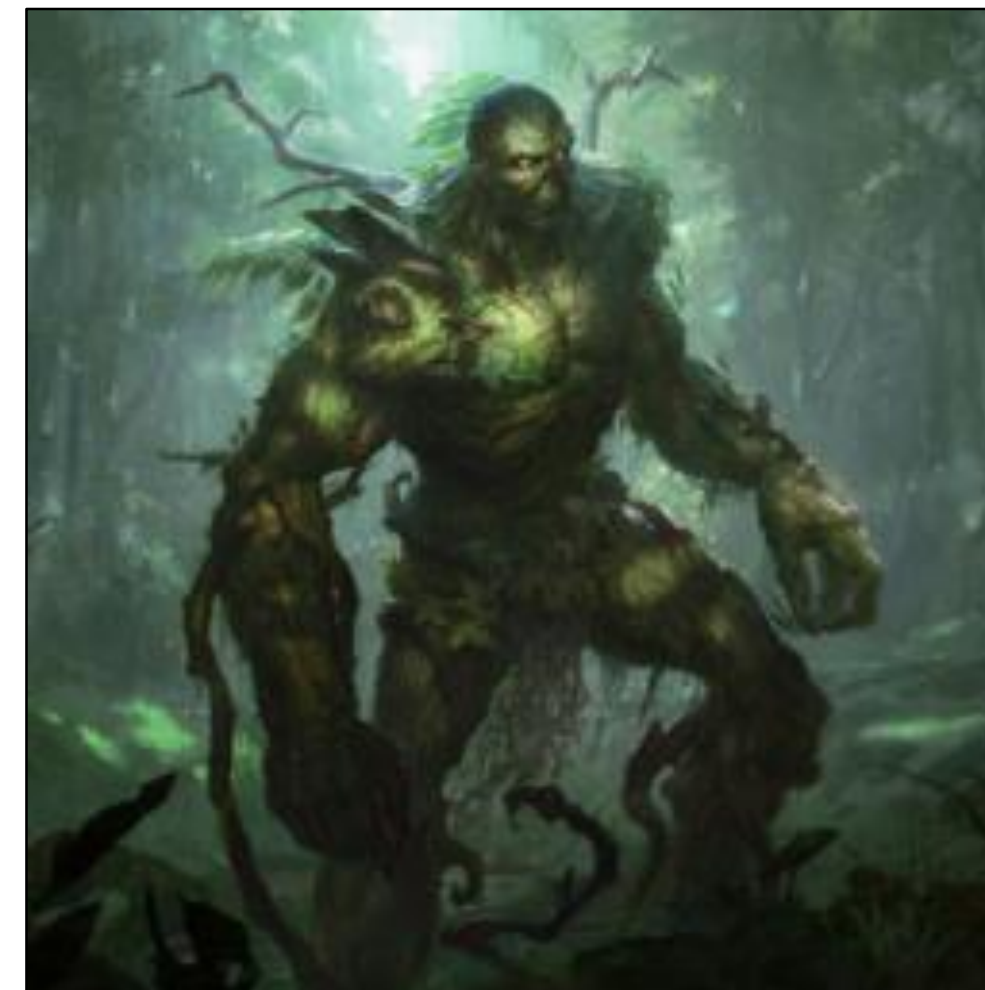


Prompt: “a green creature made of leaves and vines bursting out of the ground ready to attack; detailed, best on artstation, raymond swanland, magic the gathering, epic, stunning, masterpiece”

“influences” from the training data



generated image



to paraphrase Arthur C. Clarke:

*Interpolation in sufficiently
high-dimensional space is
indistinguishable from magic*